# A Full DR11 Strong-Lens Search in the DESI Legacy Surveys: ClaudeNet `v3blend8` Finding with LensJudge v3 HSC Cross-Validation

Greg Benson*  Xiaosheng Huang†

June 2026

*Internal Research Report*

**Abstract**

We deploy the contaminant-aware ClaudeNet `v3blend8` ensemble and the LensJudge v3 vetting cascade on the final DESI Legacy Surveys data release (DR11). The **DR11-south** (DECam, native *griz*) sweep scores all $5.38 \times 10^7$ galaxies with the five-member v2-lean stage and re-ranks the survivors with the eight-member `v3blend8` ensemble; a DR11-native recalibration is required because the deeper DR11 reductions have fatter random-galaxy score tails than DR9, tightening the matched-FPR thresholds (known-lens grade-A recall at the $10^{-4}$ operating point falls 62% → 41% relative to DR10 — a genuine DR9→DR11 domain shift, not a threshold artifact, so the DR11 survivor list under-recovers grade-A lenses and is not complete). Group-conformal selection remains *power-limited* at $m \approx 5.4 \times 10^7$ (zero certified at $\alpha \leq 0.25$ with an 8M-negative calibration), so the candidate list is the `v3blend8`-ranked top-$N$, as in the prior campaigns. The decisive step is LensJudge v3's new **HSC PDR3 tier-2**: vetting the top-500 new candidates as a cascade (cheap detection triage → agentic mimic adjudication → high-resolution escalation, \$40 total) moves **26** HSC-covered candidates from DESI-ambiguous ($p_{\text{lens}} \sim 0.03$–$0.6$) to **24 HSC tier-2 grade-A/B** under LensJudge's automated grader ($p_{\text{lens}}$ up to 0.97; a single automated grader, not a human committee or spectroscopy). An independent SuGOHI cross-match calibrates the engine: `v3blend8` recovers **79** SuGOHI HSC committee lenses into the survivors at 2.7× the survivor-median score, and **15 of the 24** graded A/B are SuGOHI lenses (a sensitivity check — the automated tier-2 grader's precision on the genuinely-new candidates is not established here). The remaining **9** (8 distinct systems) are new to the DESI lens-finder catalogs and SuGOHI; a post-hoc literature crosscheck (Huang+2020, omitted from the campaign crossmatch, + NED/SIMBAD; App. A) then shows **five of these eight systems are previously-published lenses** (mostly Huang+2020; one also SOGRAS), leaving **3** genuinely new — automated-grader candidates pending follow-up. Beyond the HSC-overlap set, the honest-residual cascade also grades **108** candidates A/B at DESI resolution *alone* — with no HSC-SSP or Euclid Q1 coverage (App. B); a literature crosscheck finds **69 are previously-published lenses** (mostly DES, whose footprint this DECaLS-south region heavily overlaps) and **39 are new** (36 grade B and 3 grade A). These cannot be validated beyond DECaLS 1″ — an unknown and likely large fraction are lrg+companion mimics — and the set is residual- and nondeterminism-sensitive (larger under the honest residual, but hotter rather than more complete), so it is a follow-up target list, not

*`benson@cs.usfca.edu`; Department of Computer Science, University of San Francisco, CA 94117, USA

†Department of Physics & Astronomy, University of San Francisco; and Physics Division, Lawrence Berkeley National Laboratory

a catalogue. The honest frame is unchanged from the v3 program: the model recovers known lenses and ranks new candidates the higher-resolution HSC grader rates highly, but *no net-new lens is established from DECaLS pixels alone* (the new candidates depend on HSC resolution and remain unconfirmed), and net-new discovery beyond existing high-resolution catalogs is resolution-bounded — the value of the *resolution lever* (now HSC over $\sim$1,300 deg$^2$, far wider than Euclid Q1) is to *convert* overlap candidates, not to manufacture discoveries. The **DR11-north** (BASS/MzLS) arm — which needs instrument-native retraining (v3 is DECam-specific) — is fully staged but deferred by a 7-day Perlmutter maintenance.

# 1 Premise

The ClaudeNet v3 program concluded that for DECaLS-resolution lens finding the binding constraint is *lens-vs-mimic separation* and *vetting resolution*, not architecture: `v3blend8` (the SHIP-gated eight-member ensemble: `effnet_B`, `zoobot_N`, and the v2-hard + v3-mimic-blend versions of `effnet_S2`, `effnet_B3`, `resnet46_C`) rejects common contaminants far better than v2 but is statistically tied with the originals at the hardest real-vs-mimic frontier, which is resolution-bounded (Inchausti Reyes et al., 2025). The productive mode is therefore *DESI pre-screen → high-resolution confirmation in the overlap.* This campaign operationalizes that on the final release (DR11) and exploits the new lever in LensJudge v3: an **HSC PDR3** tier-2 grader (Aihara et al., 2022) ($0.168''$/px, $\sim 0.6''$ seeing) whose footprint (HSC-SSP Wide, $\sim$1,300 deg$^2$) is $\sim 20\times$ the Euclid Q1 area used previously, vastly widening the confirmable overlap.

# 2 The DR11-south sweep

DR11-south is the deepest DECam reduction to date: 1,600 sweep files (1.7 TB) yielding a parent of **53,809,040** galaxies after the published Inchausti cuts (TYPE$\in${SER,EXP,DEV,REX}, NOBS$_{grz} \geq 3$, $m_z < 20$), 51.4M with native $i$. We shard into 32 footprint-pure, brick-disjoint parts, extract 101-px $grz$ cutouts (100% success, 6.0 TB), and score every galaxy with the five v2-lean members.

**DR11-native recalibration (required).** The DR9-fit per-member $10^{-4}$-FPR thresholds, when transferred to DR11 score-space, are *too tight*: DR11's deeper imaging produces a fatter high-score tail of random galaxies (e.g. the `resnet46_C` isotonic score saturates, its $10^{-4}$ threshold moving $0.81 \to 1.00$, effectively reducing the union to four members). We recalibrate from the scored population itself (a 8 M-galaxy random NegEval), yielding **95,104** survivors. Re-measuring recall of the published Inchausti 811 against these survivors gives grade-A 41% (37/90), versus 62% for the DR10-recalibrated DR10 sweep — a real DR9→DR11 model domain shift (the DR9-trained members under-score a fraction of DR11 grade-A lenses, *and* the random tail is fatter), not a threshold artifact (the denominator includes north and parent-cut lenses, so 41% is a conservative floor).

**Selection.** Scoring the survivors with all eight `v3blend8` members and ranking by the mean calibrated score, the top-300 contain **102 known** lenses (recovered into the top of the list — a consistency check) and **198 new** candidates (top-300 `v3blend8` score 0.812–0.939). The discovery set is the **top-500 new** (`v3blend8` score 0.734–0.933); the broader survivor pool has 1,071 survivors above `v3blend8` 0.7 (726 new + 345 already-known).

**Certified-FDR (power-limited, reported honestly).** Group-conformal selection (Jin and Candès, 2023) with the 8 M NegEval certifies **zero** candidates at $\alpha = 0.05/0.10/0.25$: the per-group

**Table 1:** DR11-south cascade outcome (500 new candidates, `pass_frac`=0.7).

| quantity | value | note |
|---|---|---|
| graded (stage-1 triage) | 500 | \$0.012 each |
| escalated (agentic stage-2) | 350 | v2 mimic rubric |
| reached HSC tier-2 | 26 | HSC-SSP Wide coverage |
| grade A / B / C / D | 27 / 76 / 125 / 272 | final (DESI- or HSC-grade) |
| **HSC tier-2 grade-A/B** | **24** (19 A, 5 B) | automated; 8 more A are DESI-only |
| total cost | \$40.38 | |

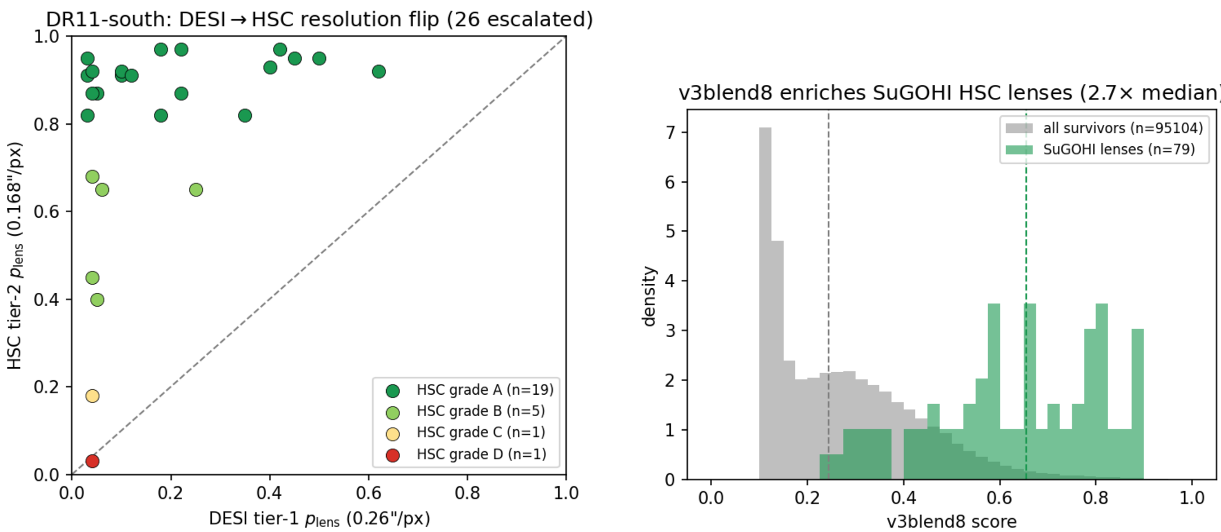



**Figure 1:** Left: the DESI→HSC resolution flip — the 26 HSC-covered candidates move from DESI-ambiguous (tier-1 $p_{\rm lens}$) to the HSC tier-2 grade (automated grader), 24 landing A/B above the $y=x$ line. Right: `v3blend8` enriches the independent SuGOHI HSC lenses to 2.7× the survivor-median score (median lines dashed).

$p$-value floor $1/(n_{\rm cal}+1) \approx 2.5 \times 10^{-7}$ is $\sim$50× short of what full-$m$ Benjamini-Hochberg needs at $m = 5.38 \times 10^7$ to select even one. This is the same power-floor diagnosed at DR9/DR10 scale — rigorous full-population FDR needs an infeasibly large calibration set — so the candidate list is the `v3blend8`-ranked top-$N$, not an FDR-certified set. The arithmetic self-test (14/14) and the byte-level scoring-path check pass.

## 3 LensJudge v3 cascade vetting + HSC cross-validation

The DESI viewer exposes no public DR11 layer (and is unreachable from our host), so we stage the 500 candidates' *grz* cubes from Perlmutter as on-disk FITS, ensuring LensJudge grades the exact pixels the CNN scored. We then run the B7 deploy cascade: a loop-free `direct` detection triage on all 500 ($\sim$\$0.012 each), escalating the top 70% by rank to the agentic v2 mimic-adjudication grader, which itself escalates to **HSC PDR3 tier-2** wherever HSC covers the position. Total cost **\$40.4** (campaign total \$43 of a \$250 budget).

**Table 2:** The 9 HSC tier-2 grade-A/B candidate detections (8 distinct systems). A literature crosscheck (Huang+2020 + NED/SIMBAD; App. A) finds **5 systems are previously-published** (*known*: H20 = Huang+2020; one also SOGRAS) and **3** genuinely new. $p_1, p_2$ = LensJudge tier-1 (DESI) / tier-2 (HSC) lens probability.

| name | RA | Dec | grade | $p_1 \to p_2$ | `v3blend8` | known |
|---|---|---|---|---|---|---|
| s_340513_3688 | 16.332 | +1.749 | A | 0.22 → 0.97 | 0.77 | Huang+2020 |
| s_327526_9059 | 9.722 | −0.409 | A | 0.04 → 0.92 | 0.74 | **new** |
| s_351303_461 | 194.534 | +3.536 | A | 0.62 → 0.92 | 0.89 | Huang+2020 |
| s_345385_1515† | 154.312 | +2.489 | A | 0.10 → 0.92 | 0.81 | Huang+2020 |
| s_340971_2860 | 130.857 | +1.784 | A | 0.12 → 0.91 | 0.76 | **new** |
| s_345385_1514† | 154.312 | +2.488 | A | 0.05 → 0.87 | 0.86 | Huang+2020 |
| s_326089_1785 | 10.288 | −0.730 | A | 0.03 → 0.82 | 0.83 | H20+SOGRAS |
| s_325163_2981 | 138.848 | −0.923 | A | 0.18 → 0.82 | 0.74 | **new** |
| s_318043_1037 | 158.789 | −2.304 | B | 0.04 → 0.45 | 0.86 | Huang+2020 |

† s_345385_1514/1515 are adjacent (∼1″) detections of the same system, so the 9 detections are **8 distinct candidate systems** (8 A-detections / 7 distinct grade-A + 1 grade-B).

**The resolution lever converts.** Every tier-2 grade-A/B candidate shows a large DESI→HSC flip: median tier-1 $p_{\rm lens} \approx 0.1$ rising to tier-2 0.82–0.97 (e.g. 0.03 → 0.95, 0.04 → 0.92). At DECaLS 1″ seeing these are ambiguous (lrg+companion ≈ lens); at HSC 0.6″ the arc/ring morphology resolves — the flip is the grader's response to resolution, not an observational confirmation.

**Independent SuGOHI validation.** Against the SuGOHI/HOLISMOKES HSC committee lens catalog — an *independent* sample with expert grades and spectroscopic redshifts (Sonnenfeld et al., 2018) — `v3blend8` recovers **79** lenses into the 95,104 survivors at a median `v3blend8` score 2.7× the survivor median (27 in the top-500), and **15 of the 24** tier-2 grade-A/B candidates are SuGOHI lenses. These 15 are a *sensitivity* check — the automated HSC grader recovers known committee lenses as grade-A/B — but they do not establish its *precision* on the genuinely-new candidates: no negative control was run through tier-2 and the certified-FDR is power-limited (§2), so the false-positive rate of the new set is unquantified.

**The candidate catalog.** Of the 24 tier-2 grade-A/B candidates, **0** are in the DESI lens-finder catalogs (by construction — the 500 were the new set after removing the 102 known), 15 are SuGOHI cross-matches, and **9** (8 distinct systems) are new to the DESI finders and SuGOHI (Table 2). A post-hoc literature crosscheck — Huang+2020 (omitted from the campaign's lineage crossmatch) plus a NED/SIMBAD cone search (App. A) — then shows **five of these eight systems are previously-published lenses** (mostly Huang+2020; one also in the SOAR Gravitational Arc Survey), leaving **3** genuinely new (`s_327526_9059`, `s_340971_2860`, `s_325163_2981`); the recoveries are an external validation of the finder. The honest caveats: LensJudge's HSC grading is automated (a single Claude grader, not a human committee, no spectroscopy), and "new" here means new to the DESI finders, SuGOHI, *and* the NED/SIMBAD literature. They are candidates warranting human-expert and spectroscopic follow-up, not confirmed lenses.

**Active-learning loop.** The cascade's agent-confirmed mimics (grade-D + named contaminant) export to **291** hard negatives in the ClaudeNet mimic-bank schema — `lrg_companion` 243-dominant, confirming that the high-`v3blend8` new frontier is lrg+companion-dominated at DECaLS

**Table 3:** Paired residual A/B on the same 500 DR11-south candidates: the prior (legacy) vetting run, a legacy-residual rerun (the LLM-nondeterminism control), and the new honest-$\chi$-residual run. The honest residual runs $\sim$2$\times$ hotter at the DESI tier-1 grade (raw A/B 90$\to$135) but the HSC tier-2 gate absorbs it, so the confirmed catalogue (tier-2 A/B) is stable (24/24/27).

| run | reached HSC tier-2 | tier-2 A/B | raw A/B (all 500) | grade A/B/C/D | cost |
|---|---|---|---|---|---|
| Prior (legacy, Jun-17) | 26 | 24 | 103 | 27/76/125/272 | \$40.38 |
| Legacy-now (control) | 28 | 24 | 90 | 28/62/135/275 | \$40.69 |
| **New ($\chi$ residual)** | 31 | 27 | 135 | 38/97/128/237 | \$48.55 |

resolution (exactly the population the HSC tier-2 rejects). Folding these into the bank and a model re-iteration are Perlmutter GPU work, deferred (below).

**Residual robustness (honest signed-$\chi$ residual).** After this vetting was complete, LensJudge's lens-light *residual* view — one of the three images the grader sees per candidate — was replaced: the legacy Gaussian high-pass (which discards the sign and imprints a four-lobe "butterfly" artifact on bright ellipticals that can mimic an arc) gave way to a signed, noise-normalised elliptical-model residual $\chi = (\text{data} - \text{model})/\sigma$, displayed per band on a fixed $\pm 5\sigma$ red–blue scale so a real arc must appear as red excess coherent across $g$ and $r$ (Fig. 2; full anatomy in the companion LensJudge residual report). To test whether the DR11 catalogue *depends* on the old residual, we re-ran the *entire* 500-candidate cascade twice in a paired design — a legacy-residual rerun (an LLM-nondeterminism control) and the new $\chi$ residual — against the stored vetting run (Table 3). The new residual runs $\sim$2$\times$ hotter at the DESI tier-1 grade (raw A/B 90$\to$135), but the HSC tier-2 gate absorbs the inflation, so the confirmed catalogue is stable (tier-2 A/B 24$\to$27). Crucially, **all 9 new-class candidate systems survive as grade A/B under the honest residual** (9/9; 13/15 SuGOHI cross-matches likewise; App. C, Table 5). The paired control is decisive: of five apparent new tier-2 promotions relative to the stored run, *four also appear in the legacy-now rerun* (run-to-run nondeterminism, which moves $\sim$18% of raw grades), leaving only one genuinely new borderline-B candidate; conversely the honest residual *recovers* two known catalogue systems that nondeterminism had dropped. The two SuGOHI lenses that fall below A/B under the new residual are independently-confirmed committee lenses scored conservatively (one already lost in the control), not lost discoveries. The rewrite is therefore a scientific-integrity improvement that leaves the DR11 catalogue intact — consistent with the discrimination-neutral result the residual report measures on its own benchmark.

# 4 DR11-north: staged, maintenance-deferred

v3 is DECam-specific — it systematically under-scores BASS/MzLS north imaging — so the north arm requires instrument-native retraining. We built the full north parent ($1.16 \times 10^7$ galaxies, *grz* only, no $i$), an 8-part manifest, and extracted **523** north-native training positives (297 BASS/MzLS curated + 226 non-leaked catalog lenses; 86 clean held-out; 33 v1 leaks tracked); the retrainer is patched to fine-tune the three swappable members from their `_b50` checkpoints on a north-native mimic blend. A full-system Perlmutter maintenance (2026-06-17 to 06-24) began mid-campaign before the north extraction could be scheduled, so the north sweep + retrain + validation gates are **deferred to post-maintenance**; all inputs are staged for a clean resume.

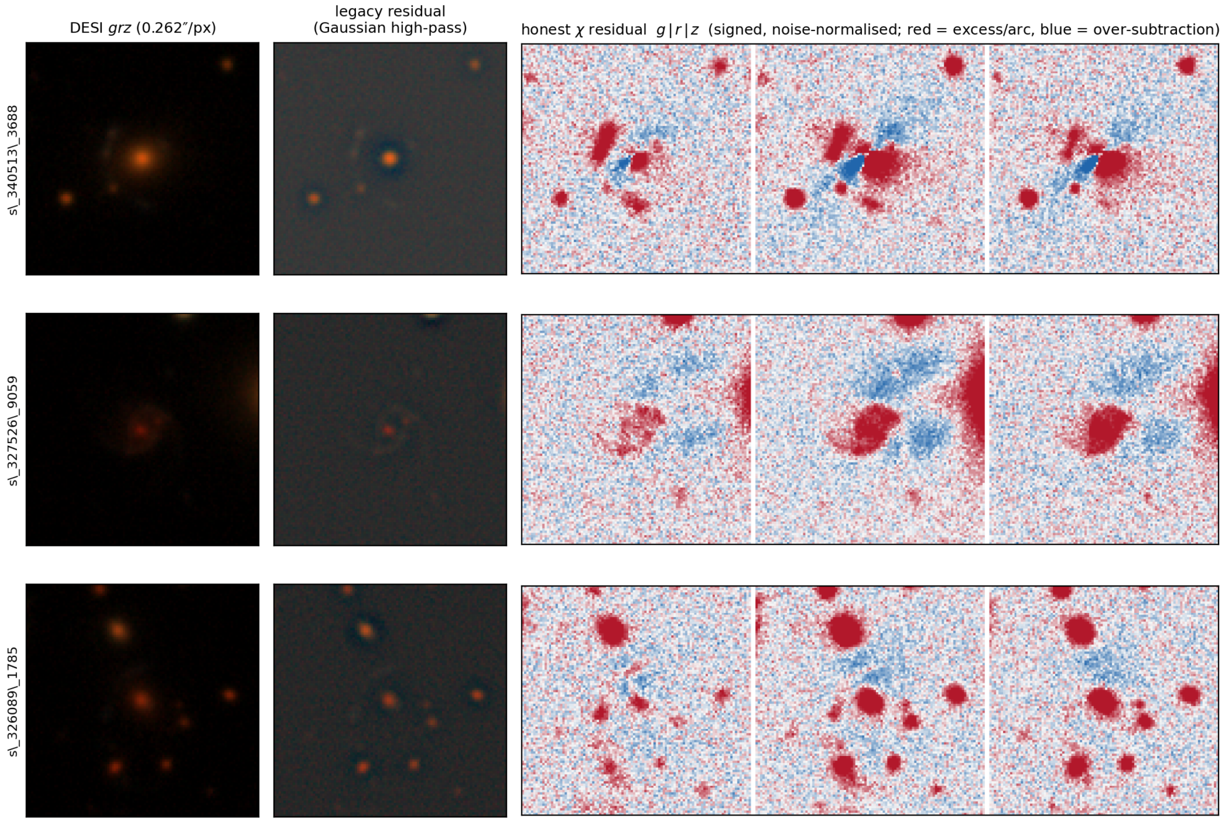



**Figure 2:** Legacy vs. honest residual for three confirmed NEW-class candidates. *Left:* the DESI *grz* cutout the finder scored (ambiguous at DECaLS 1″). *Centre:* the legacy Gaussian-high-pass residual — sign discarded, with a central dipole/butterfly on the bright lens that can read as structure. *Right:* the new signed $\chi = (\text{data} - \text{model})/\sigma$ residual as a $g \mid r \mid z$ montage on a fixed $\pm 5\sigma$ red–blue scale; the candidate arc shows as red excess *coherent across all three bands*, the real-vs-artifact discriminant. All three survive as grade A/B under the honest residual (Table 5).

# 5 Conclusion

On the final DESI Legacy Surveys release, the contaminant-aware ClaudeNet finder plus the LensJudge v3 HSC cascade is a working DESI×HSC cross-validation engine on the DECam-south footprint: it scores $5.4 \times 10^7$ galaxies, recalibrates honestly to the DR11 domain shift (with a real ∼34% relative grade-A recall loss vs DR10 — the survivor list is incomplete), and — through the new HSC tier-2 over a 20×-wider footprint than Euclid Q1 — moves 24 DESI-ambiguous candidates to HSC tier-2 grade-A/B under LensJudge's automated grader, of which 15 are SuGOHI committee lenses (a sensitivity check) and 9 (8 systems) are new to the DESI finders and SuGOHI — though a literature crosscheck (Huang+2020 + NED/SIMBAD; App. A) then shows 5 of those 8 systems are previously-published lenses, leaving **3** genuinely new. The certified-FDR is power-limited (zero) at full-population scale, the tier-2 grader is automated (no human committee or spectroscopy, precision on the new set unquantified), and *no net-new lens is established from DECaLS pixels alone* — the new candidate detections depend on HSC resolution and remain unconfirmed pending follow-up. This is the same honest ceiling the v3 program established — the resolution lever *converts* overlap candidates rather than manufacturing discoveries — now measured on DR11 with a much wider confirmable overlap. The north footprint, requiring instrument-native retraining, is staged for completion after the Perlmutter maintenance window.

## References


Hiroaki Aihara et al. Third data release of the hyper suprime-cam subaru strategic program. *Publications of the Astronomical Society of Japan*, 74(2):247–272, 2022.

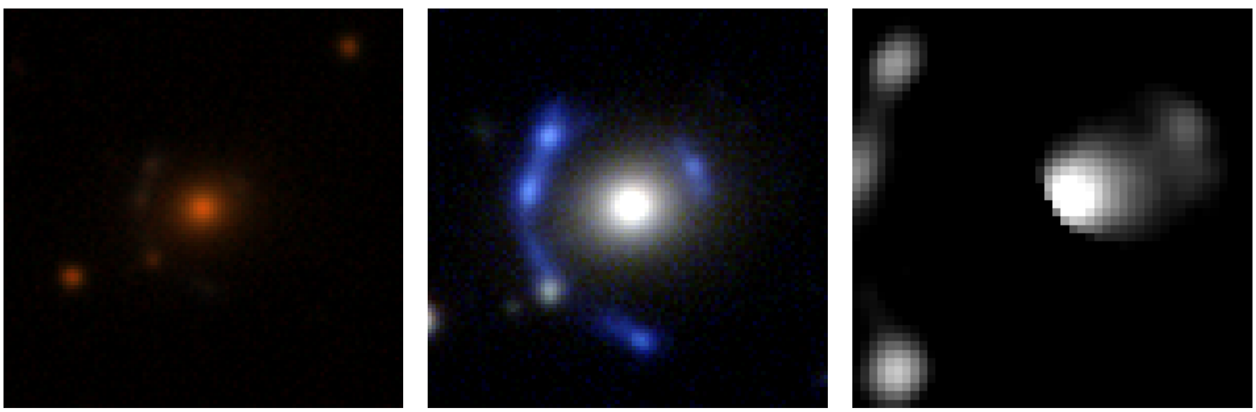
J. Inchausti Reyes et al. Strong lens discovery in DR10 with two deep-learning architectures. *arXiv:2508.20087*, 2025.

Ying Jin and Emmanuel J. Candès. Selection by prediction with conformal p-values. *JMLR*, 24, 2023.

Alessandro Sonnenfeld et al. Survey of gravitationally-lensed objects in hsc imaging (sugohi). i. *Publications of the Astronomical Society of Japan*, 70:S29, 2018.

## A The new candidate systems at DESI and HSC resolution

The figures below show the **8 distinct candidate systems** (the 9 detections of Table 2; the `s_345385` pair is one system). A post-hoc literature crosscheck (below) finds **5 of the 8 are previously-published lenses** (Huang+2020; one also SOGRAS) and **3** are genuinely new. Each is rendered as: the ClaudeNet-seen DESI *grz* cutout (0.262″/px, Lupton RGB — what the finder scored, ambiguous at DECaLS resolution); the HSC PDR3 *grizy* cutout (0.168″/px — the same view LensJudge's tier-2 grader inspected); and the HSC lens-light-subtracted view. The DESI→HSC contrast is the campaign's mechanism made visual: the finder ranks these high from ambiguous DECaLS pixels, and the resolution lever reveals the arc/ring morphology (clearest in `s_340513_3688`; fainter and correspondingly less certain in others). These are *automated-grader candidates, not confirmed lenses* — they warrant human-expert and spectroscopic follow-up. *Literature crosscheck (added post-vetting).* The campaign's lineage crossmatch omitted the Huang+2020 DECaLS catalogue and did not query the wider literature. A full check — Huang+2020 plus a NED/SIMBAD cone search — finds that **five of these eight systems are previously-published lenses** (flagged *Known* below; mostly Huang+2020, one also in the SOAR Gravitational Arc Survey). Only `s_327526_9059`, `s_340971_2860`, and `s_325163_2981` are genuinely new (absent from the lineage *and* the literature). The recoveries are an external validation of the v3+cascade finder; the net-new count here is **3**, not 8.

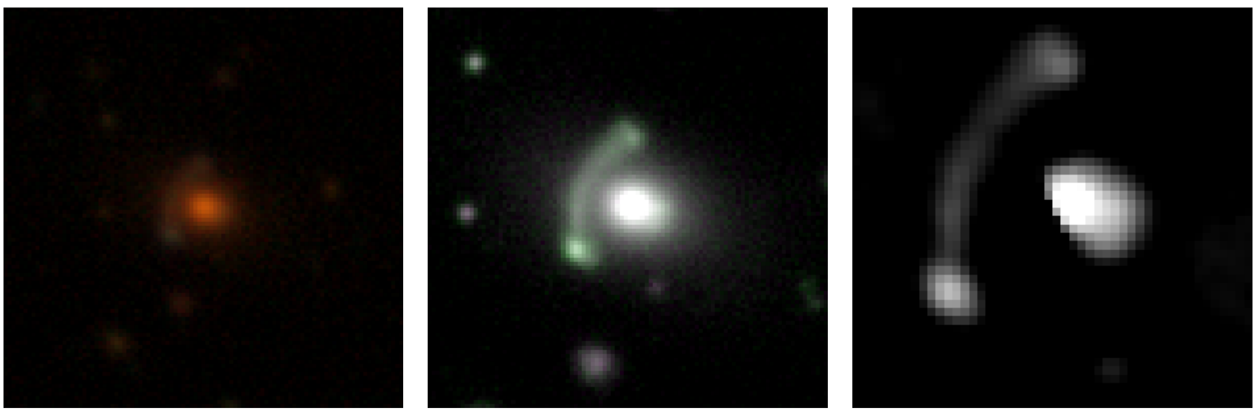


DESI *grz* (0.262″/px) HSC *grizy* (0.168″/px) HSC lens-light subtracted

**Figure 3:** `s_340513_3688` (RA 16.3319, Dec +1.7490) — HSC tier-2 grade **A**, $p_{\rm lens}$ 0.22→0.97, `v3blend8` 0.77. *Known*: Huang+2020 `DESI-016.3319+01.7490` (grade A), 0.1″.

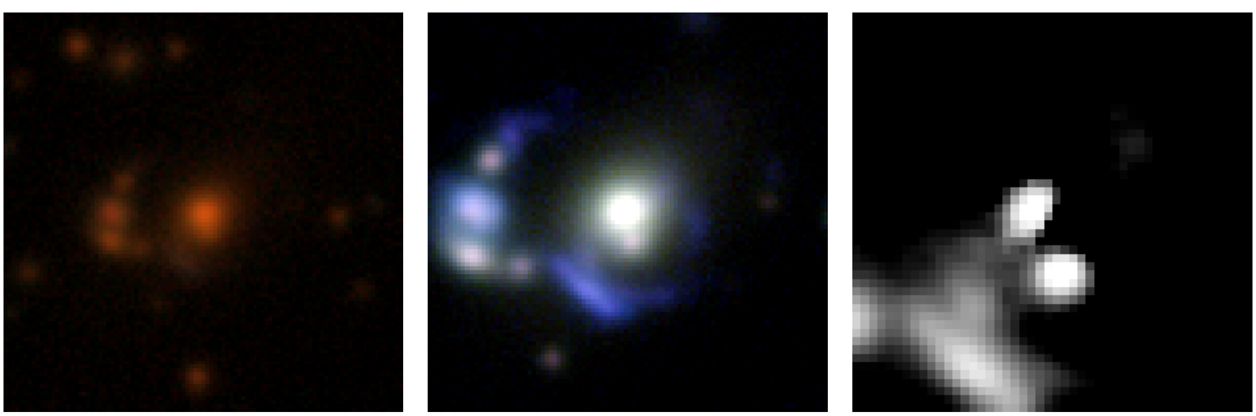
DESI *grz* (0.262″/px) HSC *grizy* (0.168″/px) HSC lens-light subtracted

**Figure 4:** s_327526_9059 (RA 9.7222, Dec -0.4093) — HSC tier-2 grade **A**, $p_{\mathrm{lens}}$ 0.04→0.92, v3blend8 0.74. *New*: absent from the lineage and from NED/SIMBAD.



DESI *grz* (0.262″/px) HSC *grizy* (0.168″/px) HSC lens-light subtracted

**Figure 5:** s_351303_461 (RA 194.5345, Dec +3.5358) — HSC tier-2 grade **A**, $p_{\mathrm{lens}}$ 0.62→0.92, v3blend8 0.89. *Known*: Huang+2020 DESI-194.5344+03.5358 (grade A), 0.2″.

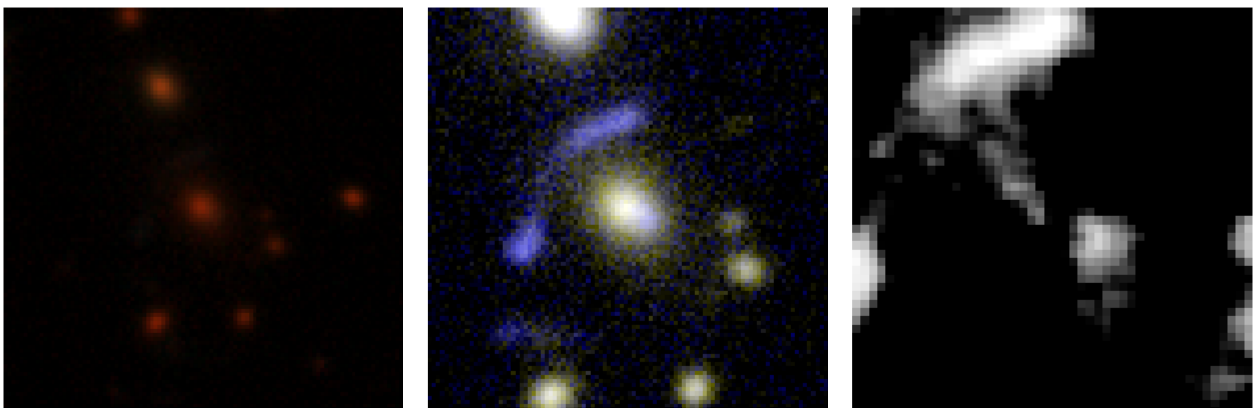


DESI *grz* (0.262″/px) HSC *grizy* (0.168″/px) HSC lens-light subtracted

**Figure 6:** s_345385_1515 (RA 154.3116, Dec +2.4885) — HSC tier-2 grade **A**, $p_{\mathrm{lens}}$ 0.10→0.92, v3blend8 0.81. *Known*: Huang+2020 DESI-154.3116+02.4885 (grade C), 0.1″. (= 8th detection s_345385_1514, the adjacent ∼1″ pair, omitted)

DESI *grz* (0.262″/px) HSC *grizy* (0.168″/px) HSC lens-light subtracted

**Figure 7:** `s_340971_2860` (RA 130.8571, Dec +1.7840) — HSC tier-2 grade **A**, $p_{\rm lens}$ 0.12→0.91, `v3blend8` 0.76. *New*: absent from the lineage and from NED/SIMBAD.



DESI *grz* (0.262″/px) HSC *grizy* (0.168″/px) HSC lens-light subtracted

**Figure 8:** `s_326089_1785` (RA 10.2876, Dec -0.7303) — HSC tier-2 grade **A**, $p_{\rm lens}$ 0.03→0.82, `v3blend8` 0.83. *Known*: Huang+2020 `DESI-010.2876-00.7303` (grade B), also SOGRAS J0041−0043, 0.1″.

DESI *grz* (0.262″/px) HSC *grizy* (0.168″/px) HSC lens-light subtracted

**Figure 9:** `s_325163_2981` (RA 138.8482, Dec -0.9226) — HSC tier-2 grade **A**, $p_{\rm lens}$ 0.18→0.82, `v3blend8` 0.74. *New*: absent from the lineage and from NED/SIMBAD.

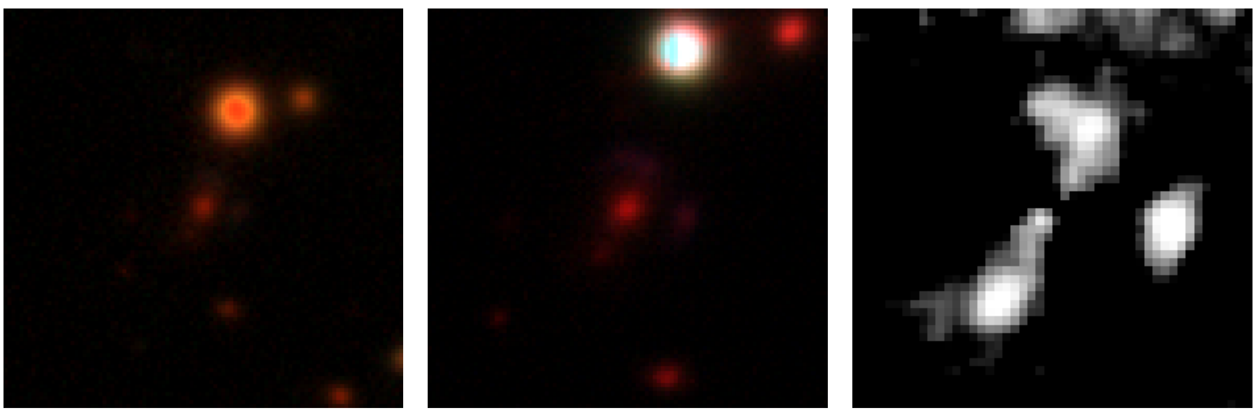



**Figure 10:** `s_318043_1037` (RA 158.7893, Dec -2.3037) — HSC tier-2 grade **B**, $p_{\rm lens}$ 0.04→0.45, `v3blend8` 0.86. *Known*: Huang+2020 `DESI-158.7893-02.3037` (grade B), 0.1″.

## B DESI-resolution candidates without high-resolution coverage

Only ∼7% of the DR11-south candidates fall in HSC-SSP (§3); the rest cannot be validated beyond DECaLS resolution with current high-res data. **This appendix is regenerated from the honest-$\chi$-residual cascade run** (§3, "Residual robustness"): Table 4 lists all **108** candidates graded A/B at DECaLS 1″ by the LensJudge cascade that are *not* in HSC-SSP *or* Euclid Q1 (17 A, 91 B), ranked by tier-1 $p_{\rm lens}$ — versus 78 under the legacy residual. This DESI-resolution-only set is the *least stable* part of the pipeline: with no high-resolution anchor its membership is sensitive both to the residual (the honest one runs ∼2× hotter at tier-1) and to LLM nondeterminism (a legacy-residual rerun alone gives 66), so the larger list is hotter, not more complete, and its exact membership is not load-bearing. Each is then shown below as an expanded DESI *grz* triptych — *full* (the whole 101 px, 0.262″/px cutout the finder scored), *zoom* (2.5× center crop), and *residual* (the signed $\chi = (\text{data} - \text{model})/\sigma$ of the $r{+}z$ luminance after subtracting a robust elliptical-galaxy model, on a red–blue diverging scale at fixed $\pm 5\sigma$: red marks unmodelled excess such as an arc/ring, blue marks over-subtraction; the noise $\sigma$ is estimated per band, so the residual is honest about significance rather than re-floored to hide its sign) — all at DECaLS resolution, carrying its full metadata and a KNOWN/NEW tag (KNOWN naming the previously-published lens it recovers). The panels are grouped into a KNOWN sub-section (previously-published lenses recovered) followed by a NEW sub-section (candidates new to the literature), ranked by tier-1 $p_{\rm lens}$ within each. A literature crosscheck (`crosscheck_candidates.py`; Huang/Storfer/Inchausti + NED/SIMBAD) finds **69 of the 108 are previously-published lenses** — 48 from the Dark Energy Survey lens searches, 11 Huang+2020, 6 AGEL, 1 CASSOWARY, and 3 via SIMBAD (the *known* column of Table 4) — because this DECaLS-south footprint heavily overlaps DES, which the campaign crossmatch omitted; the **39 genuinely new** are mostly grade B, with **3 grade A** that the honest residual newly surfaces (`s_230539_4159`, `s_194300_2796`, `s_320069_835`) — still DESI-resolution-only and unvalidatable. A NED cone search corroborates the new set (no NED lens match; reached 105/108, with 3 timeouts). At 1″ a genuine lens and an lrg+companion mimic are not separable (the resolution ceiling of §3), so these are *candidates*, an unknown and likely large fraction of which are mimics — the per-candidate panels show both clear arc/ring morphologies and ambiguous lrg+companion configurations. They are listed in full for completeness and as a follow-up target set, awaiting Euclid DR1 / Rubin / Roman or targeted high-resolution and spectroscopic follow-up.

**Table 4:** The 108 DESI-resolution grade-A/B candidates (LensJudge `direct/escalate` grade at DECaLS 0.262″) with *no* HSC-SSP or Euclid Q1 coverage, ranked by tier-1 $p_{\rm lens}$. A literature crosscheck (`crosscheck_candidates.py`: the Huang/Storfer/Inchausti catalogues + a NED/SIMBAD cone search) finds **69 are previously-published lenses** — 48 DES, 11 Huang+2020, 6 AGEL, 3 SIMBAD, 1 CASSOWARY (*known* column: survey code; per-object matches in `data/dr11s_resolution_xmatch.csv`) — and **39 are genuinely new** (—; 3 grade A and 36 grade B). The high known fraction is because this DECaLS-south footprint heavily overlaps the Dark Energy Survey lens searches, which the campaign crossmatch omitted. The new ones cannot be validated beyond DECaLS resolution; at 1″ a genuine lens and an lrg+companion mimic are not separable, so an unknown fraction are mimics. Per-candidate full/zoom/residual panels follow (App. B).

| # | name | RA | Dec | grade | $p_{\rm lens}$ | `v3blend8` | known |
|---|---|---|---|---|---|---|---|
| 1 | `s_441355_1188` | 177.1381 | +19.5009 | A | 0.95 | 0.83 | CSWA |
| 2 | `s_98012_247` | 56.1462 | -44.7956 | A | 0.92 | 0.86 | DES |
| 3 | `s_310364_6388` | 38.2078 | -3.3906 | A | 0.88 | 0.82 | H20 |

continued. . .

Table 4 continued

| # | name | RA | Dec | grade | $p_{\mathrm{lens}}$ | `v3blend8` | known |
|---|---|---|---|---|---|---|---|
| 4 | s_301694_7426 | 26.2679 | -4.9310 | A | 0.88 | 0.85 | H20 |
| 5 | s_181942_4122 | 61.6016 | -26.7733 | A | 0.87 | 0.86 | DES |
| 6 | s_137502_1047 | 26.4450 | -35.6910 | A | 0.87 | 0.80 | DES |
| 7 | s_230539_4159 | 222.0715 | -17.8489 | A | 0.85 | 0.84 | — |
| 8 | s_71042_1653 | 20.1760 | -51.7314 | A | 0.85 | 0.82 | DES |
| 9 | s_180593_5604 | 44.0974 | -27.1218 | A | 0.85 | 0.88 | DES |
| 10 | s_194300_2796 | 242.6052 | -24.4155 | A | 0.83 | 0.74 | — |
| 11 | s_304994_4662 | 133.5531 | -4.4026 | A | 0.82 | 0.84 | H20 |
| 12 | s_61472_9399 | 359.5200 | -54.6765 | A | 0.80 | 0.88 | DES |
| 13 | s_91761_4679 | 350.3683 | -46.5138 | A | 0.80 | 0.86 | DES |
| 14 | s_91796_2958 | 2.9714 | -46.2395 | A | 0.78 | 0.79 | DES |
| 15 | s_320069_835 | 305.2638 | -1.9376 | A | 0.78 | 0.89 | — |
| 16 | s_292418_3907 | 216.8280 | -6.7541 | A | 0.78 | 0.89 | H20 |
| 17 | s_81254_12458 | 57.3313 | -48.9592 | A | 0.76 | 0.84 | DES |
| 18 | s_124958_7349 | 58.1768 | -38.4292 | B | 0.68 | 0.89 | DES |
| 19 | s_59958_9730 | 64.5412 | -54.9597 | B | 0.65 | 0.83 | DES |
| 20 | s_197474_9756 | 30.7668 | -23.6341 | B | 0.65 | 0.81 | DES |
| 21 | s_313198_1471 | 27.5380 | -3.0773 | B | 0.65 | 0.81 | AGEL |
| 22 | s_232538_8716 | 25.2755 | -17.2232 | B | 0.65 | 0.85 | H20 |
| 23 | s_280236_3353 | 25.8623 | -8.8393 | B | 0.65 | 0.84 | AGEL |
| 24 | s_234689_542 | 227.0427 | -16.8771 | B | 0.65 | 0.80 | — |
| 25 | s_502450_1680 | 161.1145 | +31.2344 | B | 0.63 | 0.79 | — |
| 26 | s_94959_7228 | 56.8054 | -45.5850 | B | 0.62 | 0.84 | DES |
| 27 | s_119342_4136 | 50.7164 | -39.7694 | B | 0.62 | 0.90 | DES |
| 28 | s_243378_3831 | 326.5170 | -15.4964 | B | 0.62 | 0.84 | — |
| 29 | s_240846_8850 | 30.2832 | -15.8547 | B | 0.62 | 0.80 | H20 |
| 30 | s_188493_9685 | 83.4556 | -25.6151 | B | 0.62 | 0.88 | DES |
| 31 | s_94933_7964 | 47.5548 | -45.5743 | B | 0.60 | 0.78 | DES |
| 32 | s_95070_5466 | 96.2505 | -45.4358 | B | 0.60 | 0.78 | DES |
| 33 | s_369309_653 | 30.4465 | +6.6551 | B | 0.60 | 0.74 | — |
| 34 | s_34114_11363 | 353.7468 | -64.0687 | B | 0.58 | 0.90 | DES |
| 35 | s_102952_10164 | 334.8017 | -43.8098 | B | 0.58 | 0.85 | DES |
| 36 | s_191004_9637 | 56.9356 | -24.9088 | B | 0.58 | 0.88 | DES |
| 37 | s_43736_6056 | 67.3945 | -60.1905 | B | 0.58 | 0.83 | DES |
| 38 | s_137518_1133 | 31.3588 | -35.6632 | B | 0.58 | 0.84 | DES |
| 39 | s_129372_3768 | 16.4504 | -37.4286 | B | 0.58 | 0.82 | DES |
| 40 | s_98882_7820 | 1.6860 | -44.4974 | B | 0.58 | 0.80 | DES |
| 41 | s_218735_721 | 350.2028 | -20.0497 | B | 0.58 | 0.79 | — |
| 42 | s_360639_3636 | 13.4868 | +5.3590 | B | 0.55 | 0.85 | — |
| 43 | s_381188_1199 | 144.1511 | +8.8633 | B | 0.55 | 0.90 | H20 |
| 44 | s_114736_1046 | 343.5127 | -40.9303 | B | 0.55 | 0.86 | DES |
| 45 | s_87091_1800 | 84.5193 | -47.5872 | B | 0.55 | 0.89 | DES |
| 46 | s_327710_4950 | 55.6649 | -0.3820 | B | 0.55 | 0.74 | — |
| 47 | s_289350_3716 | 164.8051 | -7.1449 | B | 0.55 | 0.81 | — |

continued. . .

Table 4 continued

| # | name | RA | Dec | grade | $p_{\mathrm{lens}}$ | `v3blend8` | known |
|---|---|---|---|---|---|---|---|
| 48 | s_384032_2232 | 143.3887 | +9.3219 | B | 0.55 | 0.84 | H20 |
| 49 | s_73683_6069 | 357.3753 | -51.2276 | B | 0.55 | 0.83 | DES |
| 50 | s_80200_6872 | 15.4919 | -49.2940 | B | 0.55 | 0.88 | DES |
| 51 | s_58263_3751 | 42.7179 | -55.4032 | B | 0.55 | 0.88 | DES |
| 52 | s_491286_1438 | 180.3616 | +29.0116 | B | 0.52 | 0.74 | — |
| 53 | s_164140_7181 | 41.2060 | -30.3214 | B | 0.50 | 0.79 | DES |
| 54 | s_71871_246 | 353.9664 | -51.8716 | B | 0.50 | 0.76 | AGEL |
| 55 | s_61626_1035 | 65.1782 | -54.3770 | B | 0.50 | 0.87 | DES |
| 56 | s_262449_2840 | 188.7861 | -11.9798 | B | 0.50 | 0.87 | — |
| 57 | s_107312_5223 | 23.8451 | -42.5398 | B | 0.50 | 0.85 | DES |
| 58 | s_90920_9419 | 45.9517 | -46.4407 | B | 0.50 | 0.85 | DES |
| 59 | s_217461_3974 | 12.0451 | -19.9206 | B | 0.50 | 0.78 | — |
| 60 | s_260341_325 | 9.9798 | -12.2097 | B | 0.50 | 0.84 | — |
| 61 | s_207576_2272 | 248.1195 | -22.0355 | B | 0.48 | 0.76 | — |
| 62 | s_162970_1777 | 62.5554 | -30.6238 | B | 0.48 | 0.82 | DES |
| 63 | s_234043_6467 | 58.4427 | -17.1109 | B | 0.47 | 0.88 | AGEL |
| 64 | s_283917_2408 | 235.6200 | -8.3490 | B | 0.45 | 0.89 | — |
| 65 | s_50907_11900 | 303.5808 | -57.9504 | B | 0.45 | 0.74 | DES |
| 66 | s_61411_4481 | 332.9866 | -54.6445 | B | 0.45 | 0.76 | DES |
| 67 | s_165468_9648 | 64.7771 | -29.9937 | B | 0.45 | 0.78 | DES |
| 68 | s_148704_6732 | 183.8176 | -33.6006 | B | 0.45 | 0.80 | — |
| 69 | s_280264_9147 | 33.1051 | -8.8697 | B | 0.45 | 0.84 | literature (SIMBAD) |
| 70 | s_266046_1185 | 25.9989 | -11.2770 | B | 0.45 | 0.76 | — |
| 71 | s_251013_4202 | 137.4230 | -13.9601 | B | 0.45 | 0.79 | — |
| 72 | s_52008_8091 | 96.5963 | -57.5083 | B | 0.45 | 0.89 | DES |
| 73 | s_209853_1103 | 139.8053 | -21.4604 | B | 0.45 | 0.85 | — |
| 74 | s_504880_4027 | 152.0763 | +31.7005 | B | 0.45 | 0.84 | H20 |
| 75 | s_181762_9592 | 11.4347 | -26.6294 | B | 0.45 | 0.86 | — |
| 76 | s_69087_268 | 307.2325 | -52.5218 | B | 0.43 | 0.81 | DES |
| 77 | s_256199_9097 | 29.6819 | -13.0258 | B | 0.42 | 0.78 | DES |
| 78 | s_36700_530 | 347.9904 | -63.1176 | B | 0.42 | 0.78 | literature (SIMBAD) |
| 79 | s_23764_4161 | 132.6603 | -68.1876 | B | 0.42 | 0.79 | — |
| 80 | s_29019_24056 | 248.9430 | -66.0747 | B | 0.42 | 0.74 | — |
| 81 | s_204169_8722 | 49.1618 | -22.6093 | B | 0.42 | 0.77 | DES |
| 82 | s_112158_2867 | 209.6741 | -41.4932 | B | 0.42 | 0.83 | — |
| 83 | s_139825_11098 | 18.9226 | -35.3386 | B | 0.42 | 0.76 | literature (SIMBAD) |
| 84 | s_60524_3113 | 309.7973 | -54.9952 | B | 0.42 | 0.81 | DES |
| 85 | s_38835_314 | 56.6543 | -61.9792 | B | 0.42 | 0.84 | DES |
| 86 | s_405220_1292 | 136.4655 | +13.0307 | B | 0.42 | 0.89 | — |
| 87 | s_68464_11443 | 52.7373 | -52.4703 | B | 0.42 | 0.88 | DES |
| 88 | s_206743_10995 | 24.2120 | -22.0076 | B | 0.42 | 0.85 | DES |
| 89 | s_41562_4180 | 42.2402 | -60.9010 | B | 0.40 | 0.87 | DES |
| 90 | s_188312_3322 | 33.2537 | -25.3945 | B | 0.40 | 0.76 | — |
| 91 | s_120781_3317 | 156.8525 | -39.6093 | B | 0.40 | 0.83 | — |

continued...

Table 4 continued

| # | name | RA | Dec | grade | $p_{\text{lens}}$ | `v3blend8` | known |
|---|---|---|---|---|---|---|---|
| 92 | s_266620_918 | 172.1320 | -11.3108 | B | 0.40 | 0.80 | — |
| 93 | s_410935_2097 | 164.1204 | +14.0606 | B | 0.38 | 0.83 | — |
| 94 | s_32666_26056 | 248.6090 | -64.5569 | B | 0.38 | 0.73 | — |
| 95 | s_244220_4253 | 184.3042 | -15.2610 | B | 0.38 | 0.80 | — |
| 96 | s_119499_8020 | 101.8093 | -39.8359 | B | 0.38 | 0.75 | — |
| 97 | s_156771_3446 | 54.3219 | -31.8704 | B | 0.38 | 0.89 | AGEL |
| 98 | s_205319_6061 | 359.9772 | -22.4596 | B | 0.38 | 0.88 | — |
| 99 | s_416616_6090 | 189.5370 | +15.0309 | B | 0.38 | 0.93 | H20 |
| 100 | s_364281_4525 | 207.3133 | +5.6308 | B | 0.35 | 0.90 | — |
| 101 | s_250611_2161 | 33.7898 | -14.1185 | B | 0.35 | 0.77 | — |
| 102 | s_212207_5081 | 50.6597 | -20.9567 | B | 0.35 | 0.84 | — |
| 103 | s_128091_3374 | 332.6355 | -37.9263 | B | 0.35 | 0.85 | — |
| 104 | s_194944_4494 | 59.2045 | -24.1447 | B | 0.35 | 0.85 | DES |
| 105 | s_94861_9433 | 21.9716 | -45.5428 | B | 0.32 | 0.87 | DES |
| 106 | s_235253_4230 | 14.4452 | -16.7457 | B | 0.32 | 0.76 | H20 |
| 107 | s_394431_6727 | 261.7637 | +11.0021 | B | 0.30 | 0.77 | AGEL |
| 108 | s_369977_4434 | 198.5174 | +6.6887 | B | 0.28 | 0.75 | — |

### Known — previously-published lenses recovered (69 of 108)

These coincide (≤few″) with a catalogued strong lens — 48 DES, 11 Huang+2020, 6 AGEL, 3 SIMBAD, 1 CASSOWARY — an external validation of the finder (the campaign crossmatch had omitted these catalogues). Ranked by tier-1 $p_{\text{lens}}$.

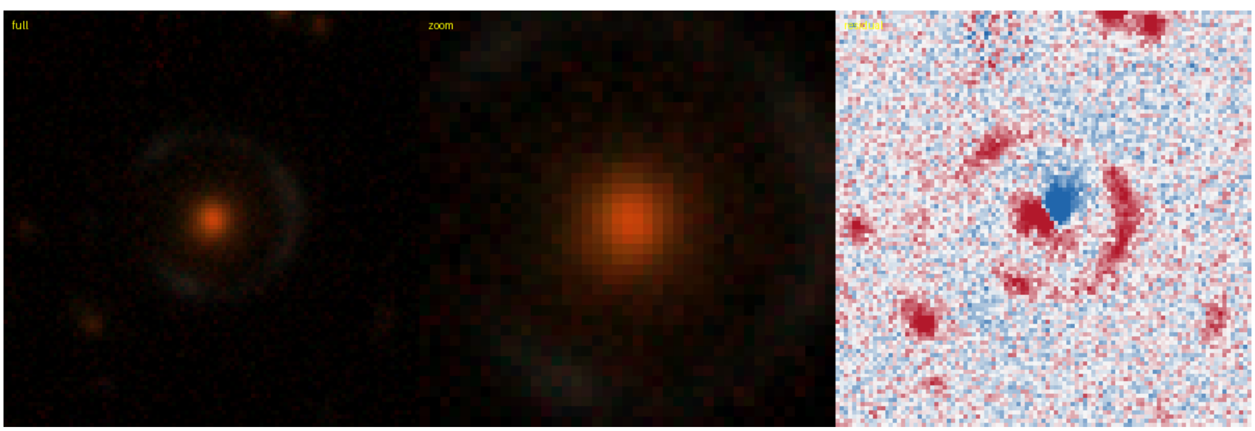


**Figure 11:** **#1** `s_441355_1188` — RA 177.1381, Dec +19.5009; grade **A**, tier-1 $p_{\text{lens}}$ 0.95, `v3blend8` 0.83. KNOWN — CASSOWARY: `CSWA 1`, 0.01″ (lineage).

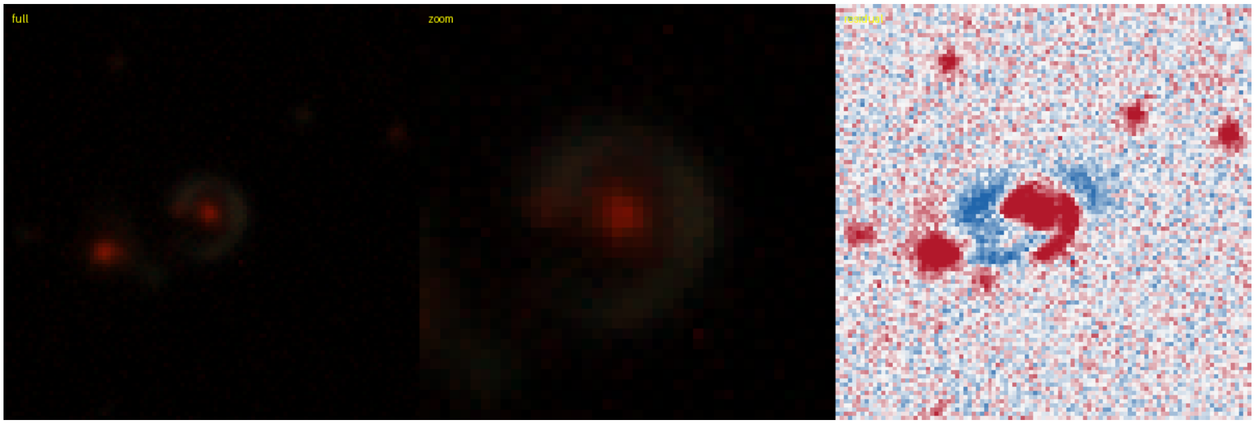



**Figure 12:** **#2** `s_98012_247` — RA 56.1462, Dec -44.7956; grade **A**, tier-1 $p_{\mathrm{lens}}$ 0.92, `v3blend8` 0.86. KNOWN — DES: `DES J034435.09-444743.9`, 0.18″ (SIMBAD).

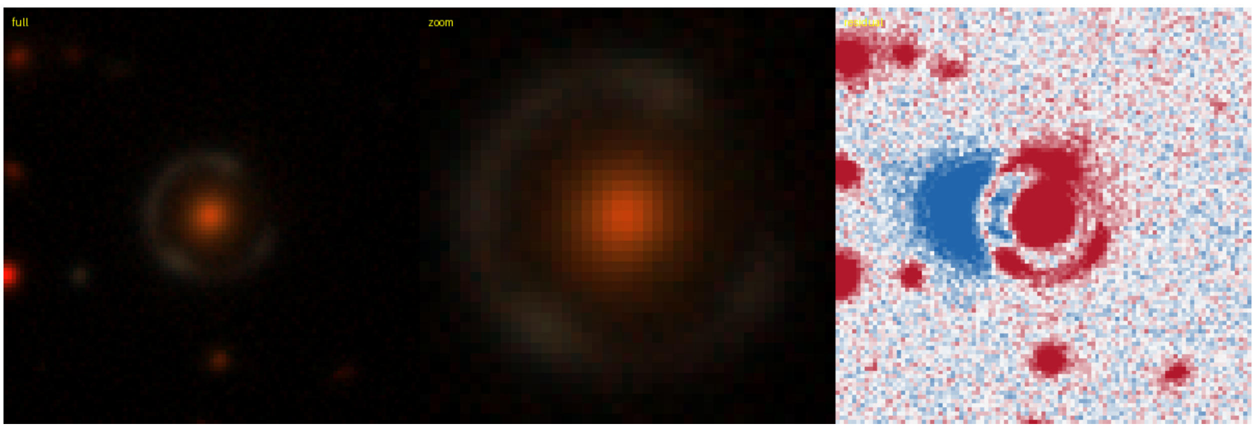



**Figure 13:** **#3** `s_310364_6388` — RA 38.2078, Dec -3.3906; grade **A**, tier-1 $p_{\mathrm{lens}}$ 0.88, `v3blend8` 0.82. KNOWN — Huang+2020: `DESI-038.2078-03.3906`, 0.16″ (lineage).

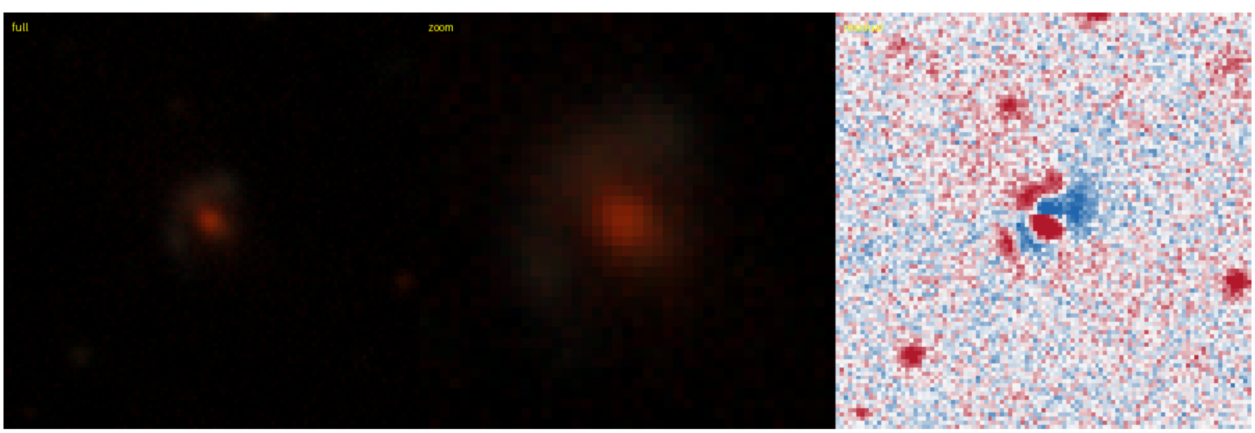



**Figure 14:** **#4** `s_301694_7426` — RA 26.2679, Dec -4.9310; grade **A**, tier-1 $p_{\mathrm{lens}}$ 0.88, `v3blend8` 0.85. KNOWN — Huang+2020: `DESI-026.2679-04.9310`, 0.13″ (lineage).

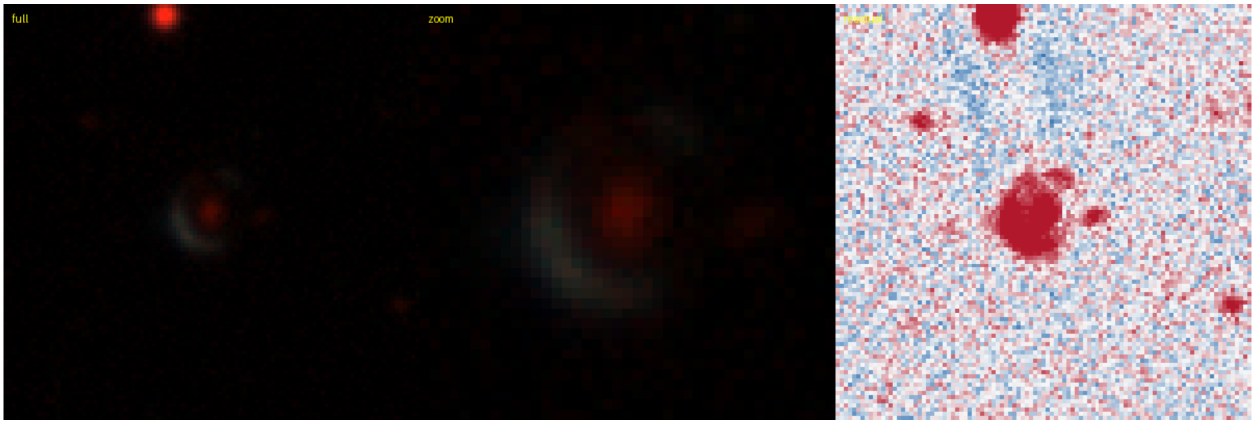



**Figure 15:** **#5** `s_181942_4122` — RA 61.6016, Dec -26.7733; grade **A**, tier-1 $p_{\rm lens}$ 0.87, `v3blend8` 0.86. KNOWN — DES: `DES J040624.49-264625.0`, 1.74″ (SIMBAD).

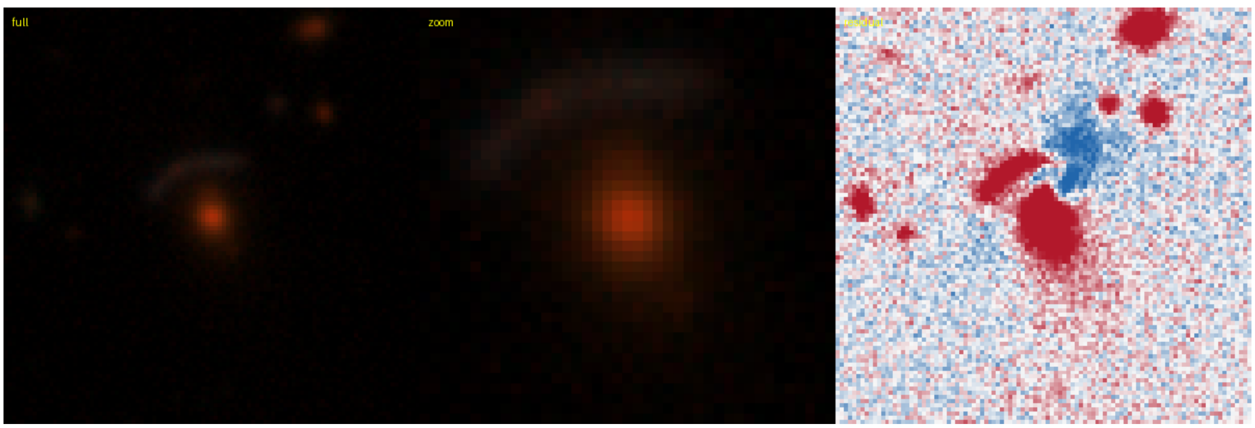



**Figure 16:** **#6** `s_137502_1047` — RA 26.4450, Dec -35.6910; grade **A**, tier-1 $p_{\rm lens}$ 0.87, `v3blend8` 0.80. KNOWN — DES: `DES J014546.78-354127.3`, 0.20″ (SIMBAD).


**Figure 17:** **#8** `s_71042_1653` — RA 20.1760, Dec -51.7314; grade **A**, tier-1 $p_{\rm lens}$ 0.85, `v3blend8` 0.82. KNOWN — DES: `[DBL2017] DES J0120-5143 (A)`, 0.19″ (SIMBAD).

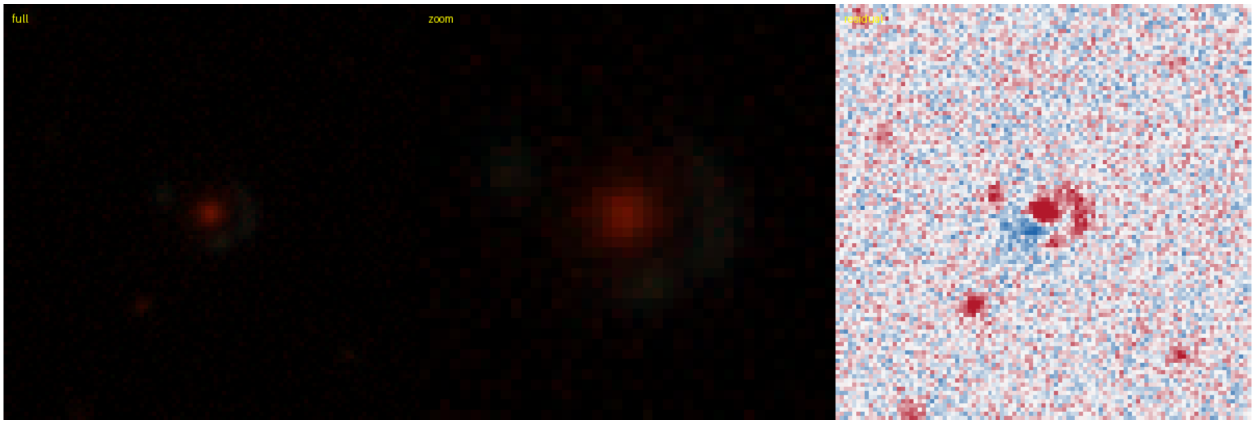



**Figure 18:** **#9** `s_180593_5604` — RA 44.0974, Dec -27.1218; grade **A**, tier-1 $p_{\rm lens}$ 0.85, `v3blend8` 0.88. KNOWN — DES: `DES J025623.36-270718.5`, 0.28″ (SIMBAD).


**Figure 19:** **#11** `s_304994_4662` — RA 133.5531, Dec -4.4026; grade **A**, tier-1 $p_{\rm lens}$ 0.82, `v3blend8` 0.84. KNOWN — Huang+2020: `DESI-133.5531-04.4026`, 0.14″ (lineage).

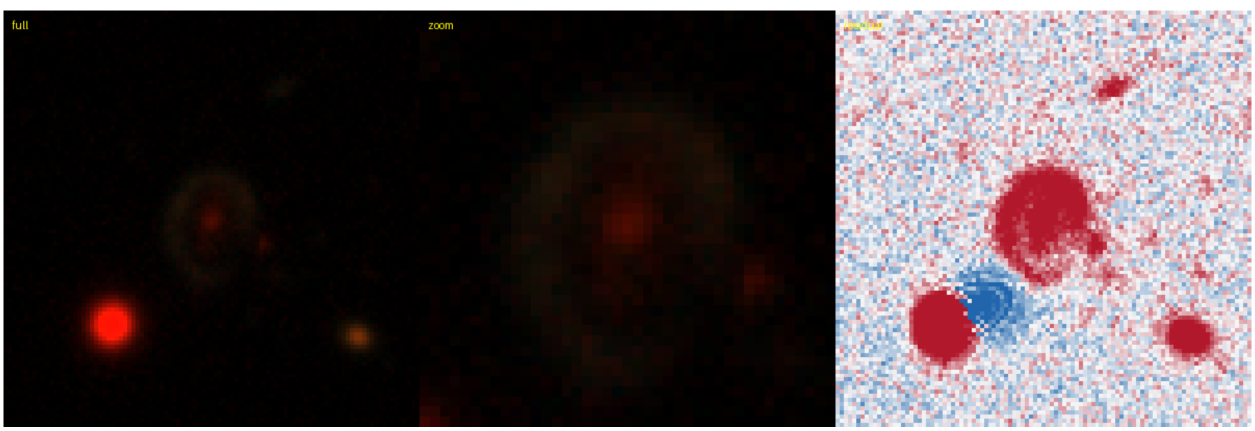



**Figure 20:** **#12** `s_61472_9399` — RA 359.5200, Dec -54.6765; grade **A**, tier-1 $p_{\rm lens}$ 0.80, `v3blend8` 0.88. KNOWN — DES: `DES J235804.95-544034.4`, 1.73″ (SIMBAD).

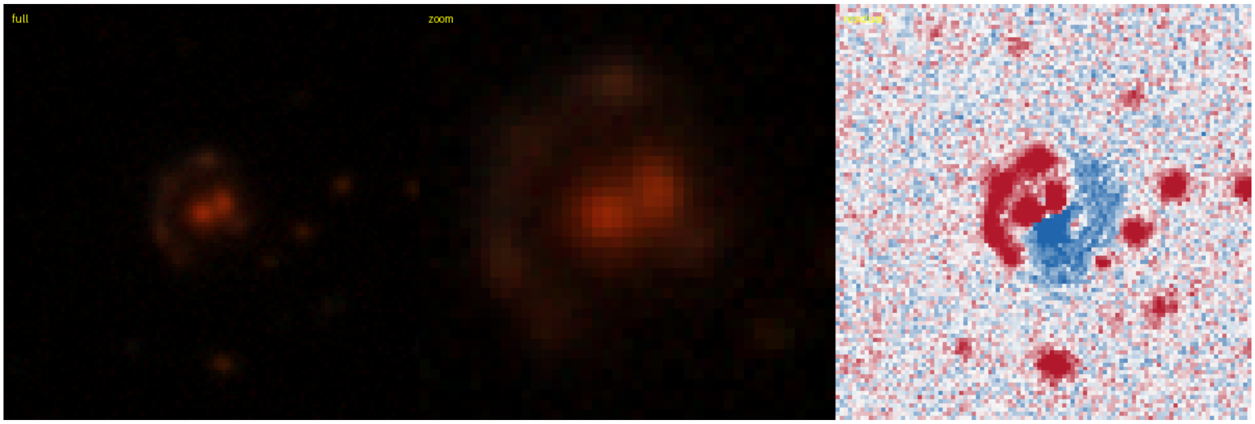



**Figure 21:** **#13** `s_91761_4679` — RA 350.3683, Dec -46.5138; grade **A**, tier-1 $p_{\text{lens}}$ 0.80, `v3blend8` 0.86. KNOWN — DES: `DES J232128.37-463049.3`, 0.25″ (SIMBAD).

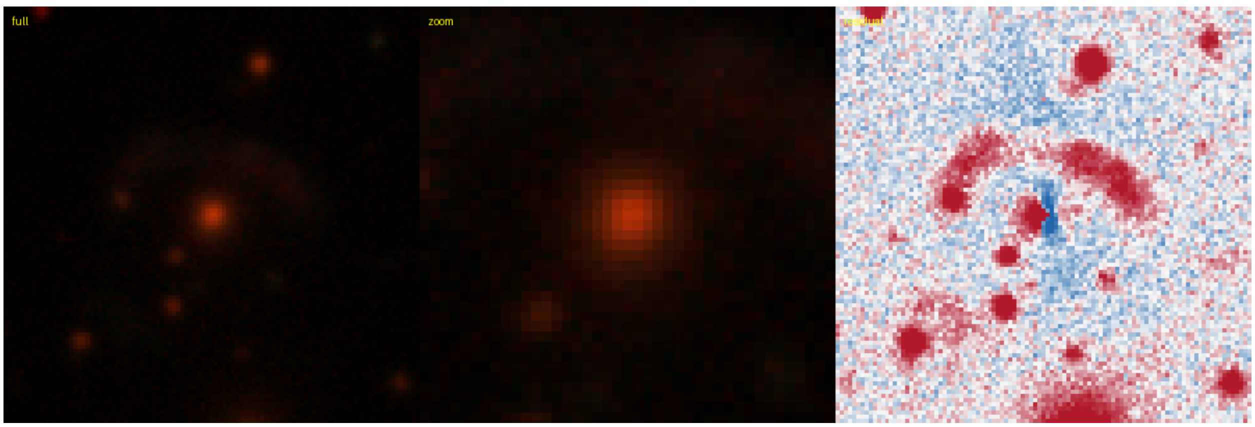



**Figure 22:** **#14** `s_91796_2958` — RA 2.9714, Dec -46.2395; grade **A**, tier-1 $p_{\text{lens}}$ 0.78, `v3blend8` 0.79. KNOWN — DES: `[DBL2017] DES J0011-4614 (A)`, 0.15″ (SIMBAD).

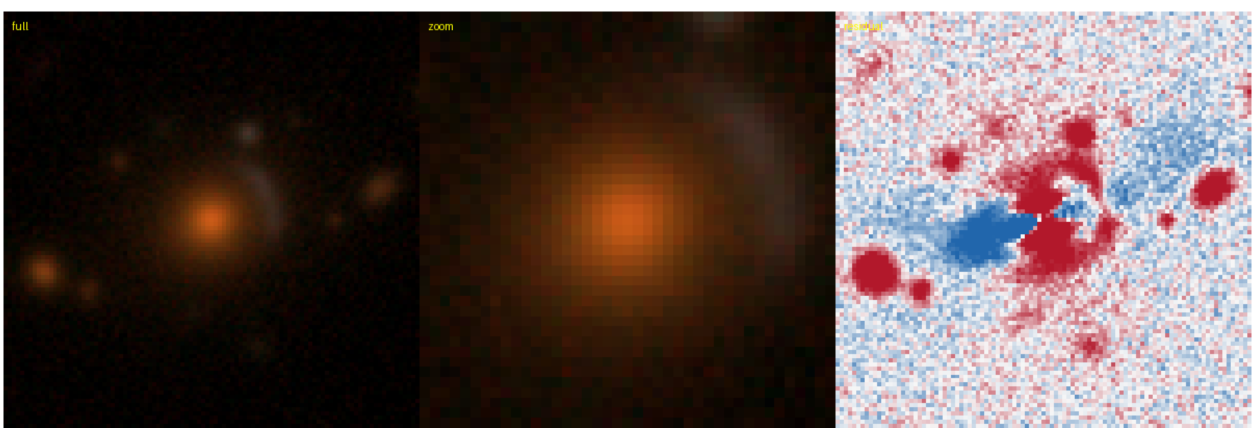



**Figure 23:** **#16** `s_292418_3907` — RA 216.8280, Dec -6.7541; grade **A**, tier-1 $p_{\text{lens}}$ 0.78, `v3blend8` 0.89. KNOWN — Huang+2020: `DESI-216.8280-06.7541`, 0.15″ (lineage).

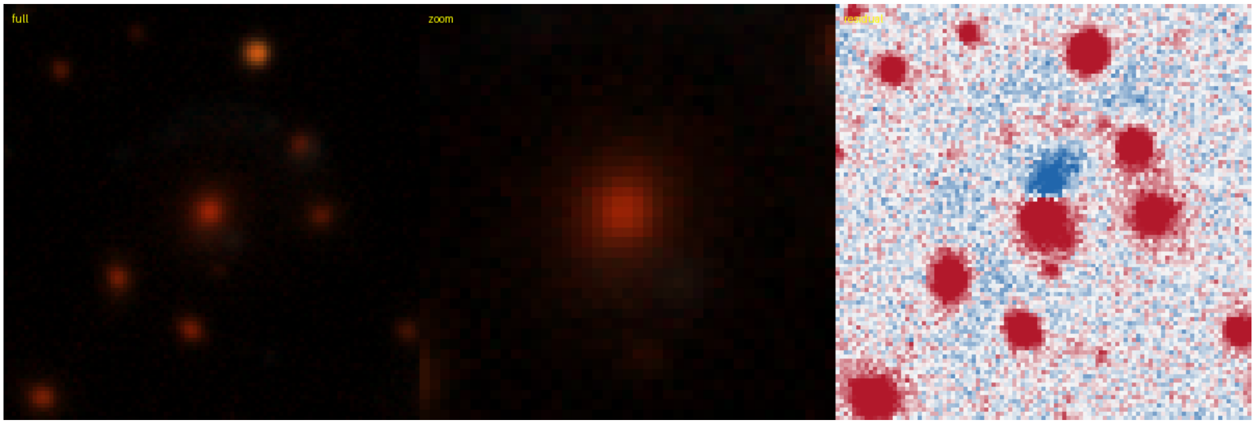



**Figure 24:** **#17** `s_81254_12458` — RA 57.3313, Dec -48.9592; grade **A**, tier-1 $p_{\rm lens}$ 0.76, `v3blend8` 0.84. KNOWN — DES: `[DBL2017] DES J0349-4857 (A)`, 0.16″ (SIMBAD).

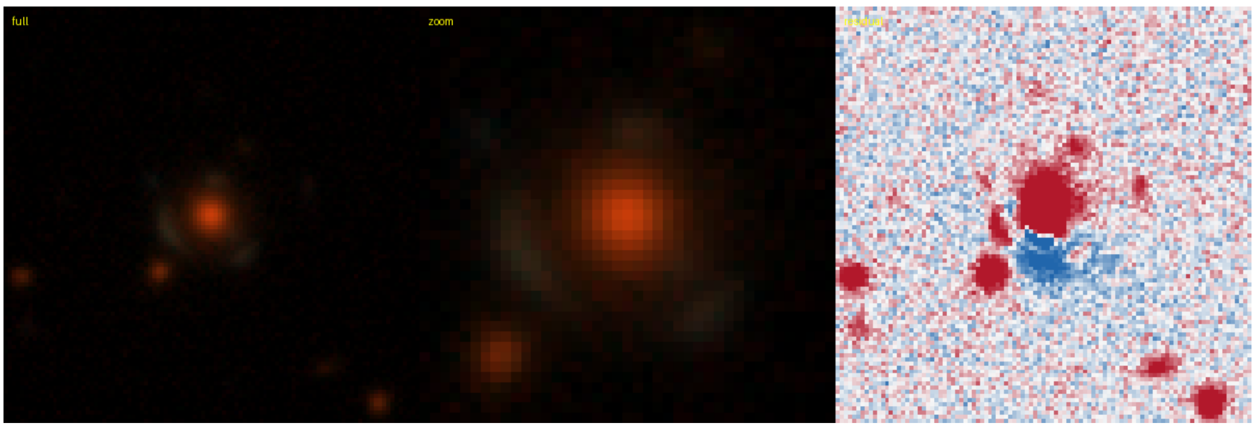



**Figure 25:** **#18** `s_124958_7349` — RA 58.1768, Dec -38.4292; grade **B**, tier-1 $p_{\rm lens}$ 0.68, `v3blend8` 0.89. KNOWN — DES: `DES J035242.40-382544.9`, 0.01″ (lineage).

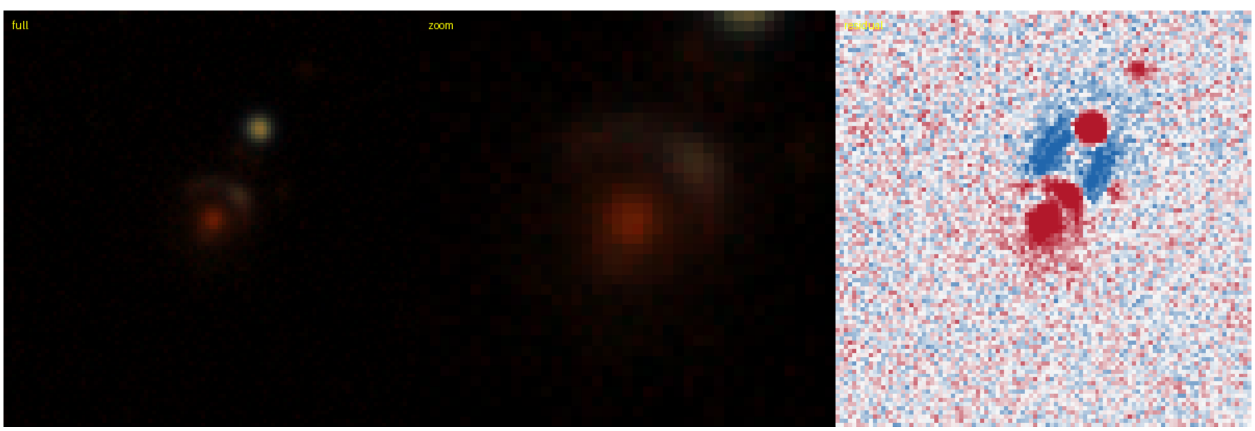



**Figure 26:** **#19** `s_59958_9730` — RA 64.5412, Dec -54.9597; grade **B**, tier-1 $p_{\rm lens}$ 0.65, `v3blend8` 0.83. KNOWN — DES: `[DBL2017] DES J0418-5457 (A)`, 0.14″ (SIMBAD).

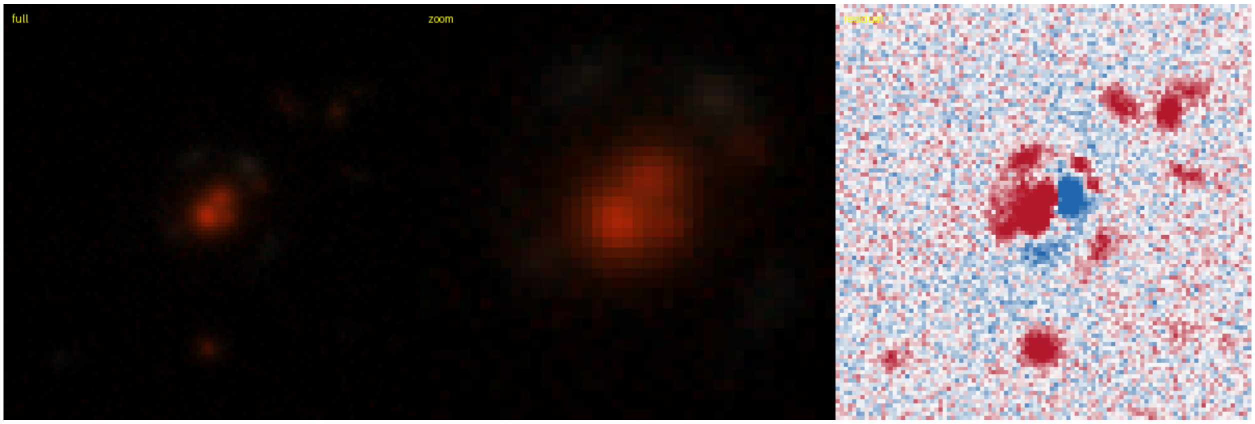



**Figure 27:** **#20** `s_197474_9756` — RA 30.7668, Dec -23.6341; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.65, `v3blend8` 0.81. KNOWN — DES: `DES J020304.01-233802.5`, 0.22″ (SIMBAD).


**Figure 28:** **#21** `s_313198_1471` — RA 27.5380, Dec -3.0773; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.65, `v3blend8` 0.81. KNOWN — AGEL: `AGEL J015009-030438 system`, 0.34″ (SIMBAD).

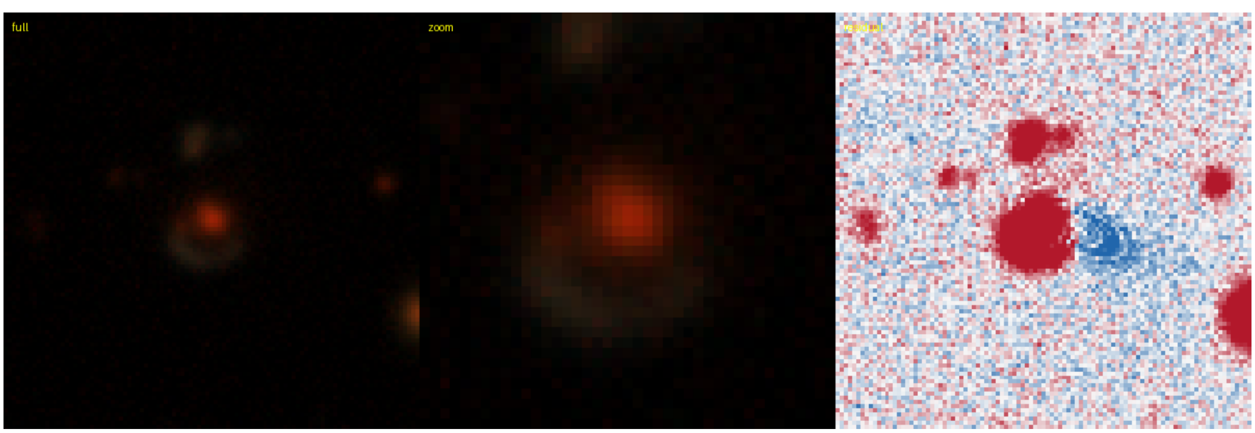



**Figure 29:** **#22** `s_232538_8716` — RA 25.2755, Dec -17.2232; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.65, `v3blend8` 0.85. KNOWN — Huang+2020: `DESI-025.2755-17.2232`, 0.19″ (lineage).

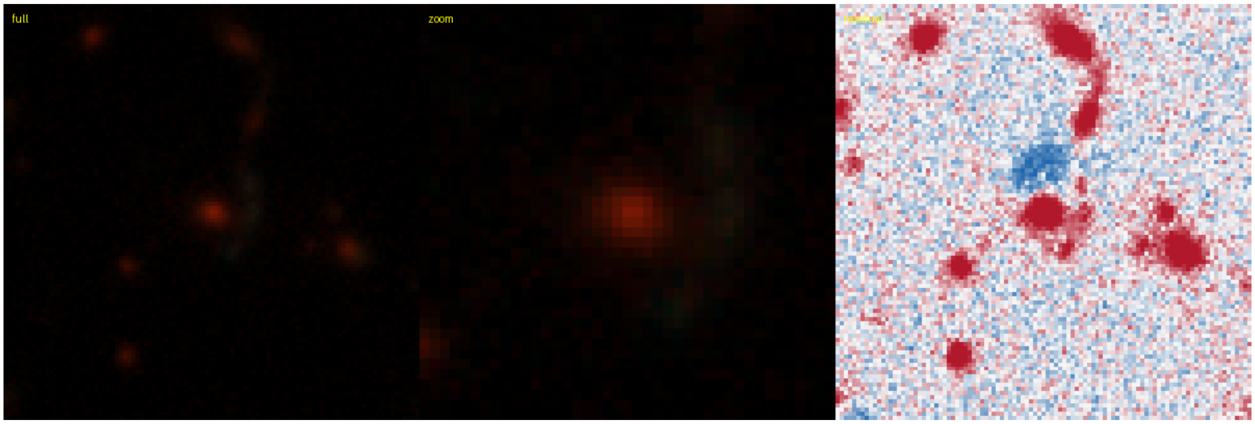



**Figure 30:** **#23** `s_280236_3353` — RA 25.8623, Dec -8.8393; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.65, `v3blend8` 0.84. KNOWN — AGEL: `AGEL J014327-085021 system`, 0.93″ (SIMBAD).

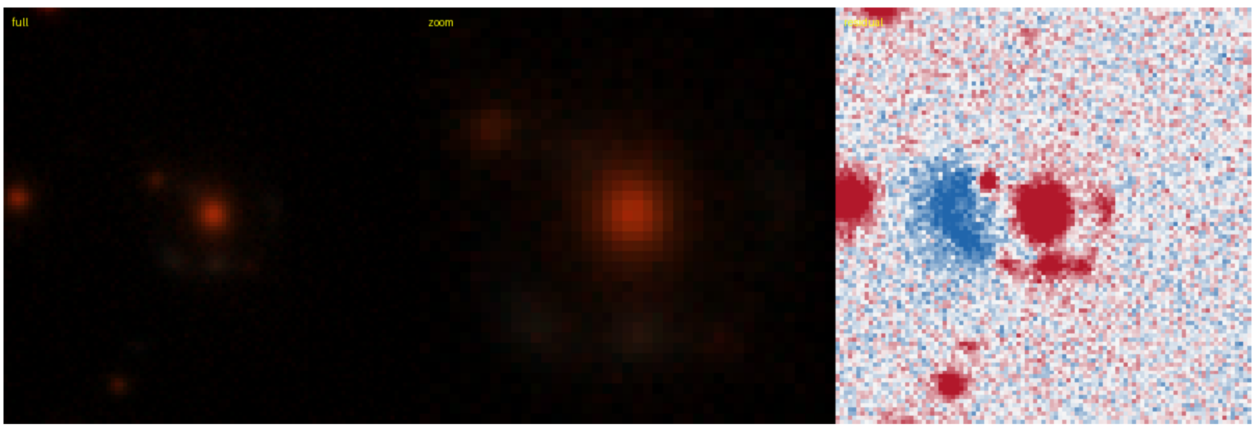



**Figure 31:** **#26** `s_94959_7228` — RA 56.8054, Dec -45.5850; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.62, `v3blend8` 0.84. KNOWN — DES: `[DBL2017] DES J0347-4535 (A)`, 0.09″ (SIMBAD).


**Figure 32:** **#27** `s_119342_4136` — RA 50.7164, Dec -39.7694; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.62, `v3blend8` 0.90. KNOWN — DES: `DES J032251.91-394609.6`, 0.17″ (SIMBAD).

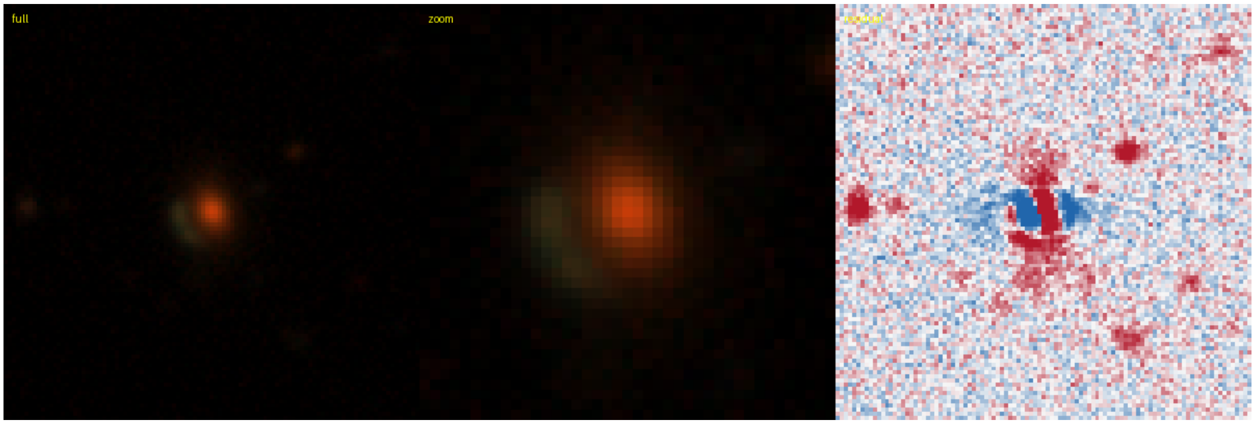



**Figure 33:** **#29** `s_240846_8850` — RA 30.2832, Dec -15.8547; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.62, `v3blend8` 0.80. KNOWN — Huang+2020: `DESI-030.2832-15.8547`, 0.15″ (lineage).

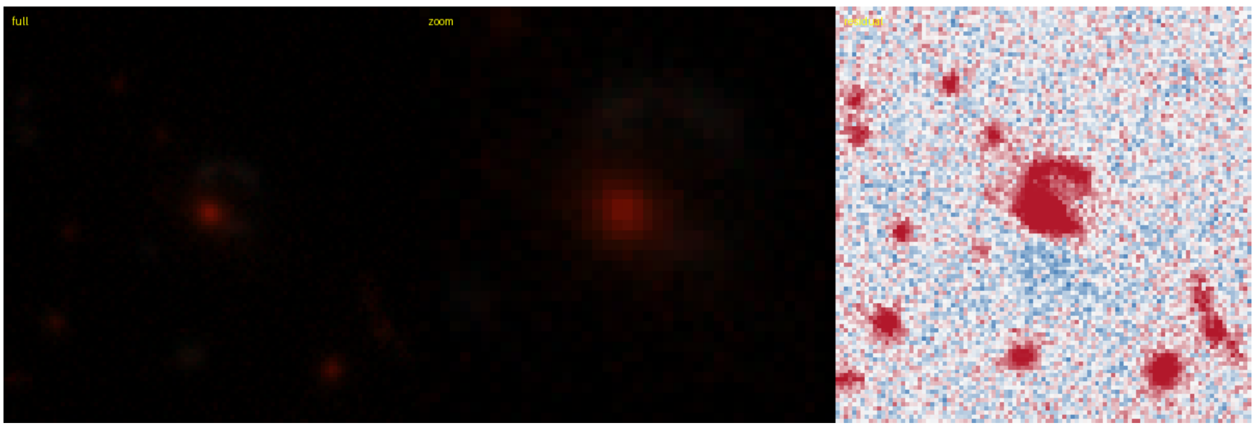



**Figure 34:** **#30** `s_188493_9685` — RA 83.4556, Dec -25.6151; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.62, `v3blend8` 0.88. KNOWN — DES: `DES J053349.32-253654.3`, 0.15″ (SIMBAD).

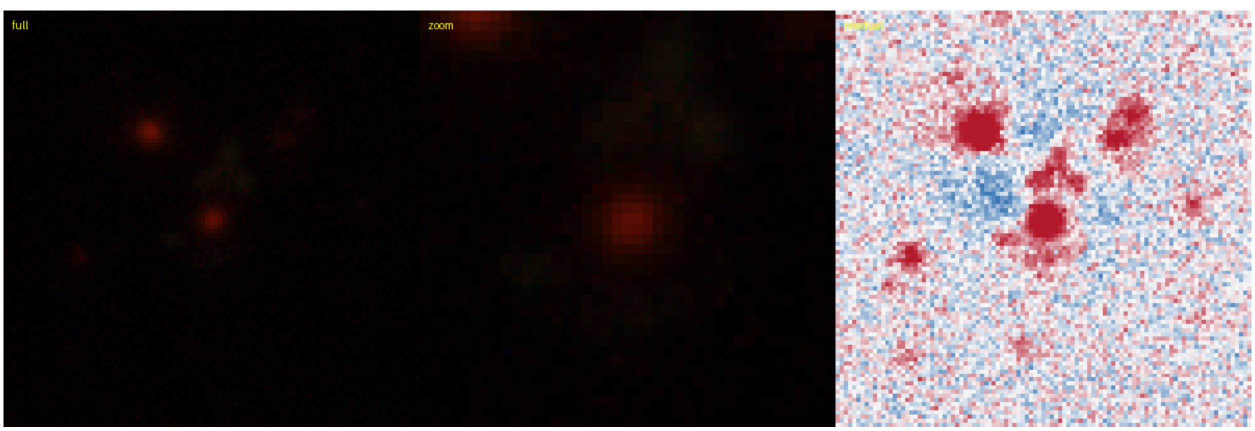



**Figure 35:** **#31** `s_94933_7964` — RA 47.5548, Dec -45.5743; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.60, `v3blend8` 0.78. KNOWN — DES: `[DBL2017] DES J0310-4534 (A)`, 0.14″ (SIMBAD).

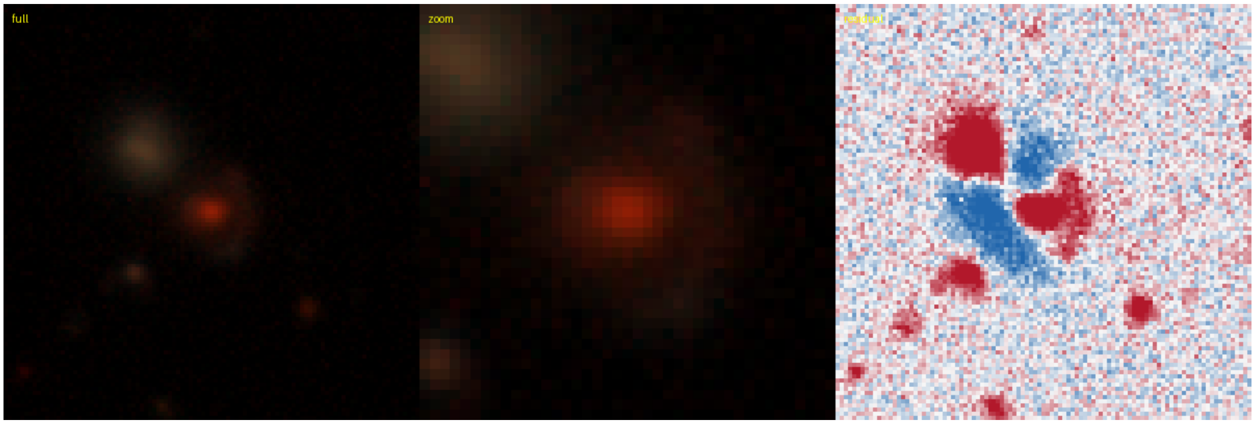



**Figure 36:** **#32** `s_95070_5466` — RA 96.2505, Dec -45.4358; grade **B**, tier-1 $p_{\rm lens}$ 0.60, `v3blend8` 0.78. KNOWN — DES: `[DBL2017] DES J0625-4526 (A)`, 0.08″ (SIMBAD).

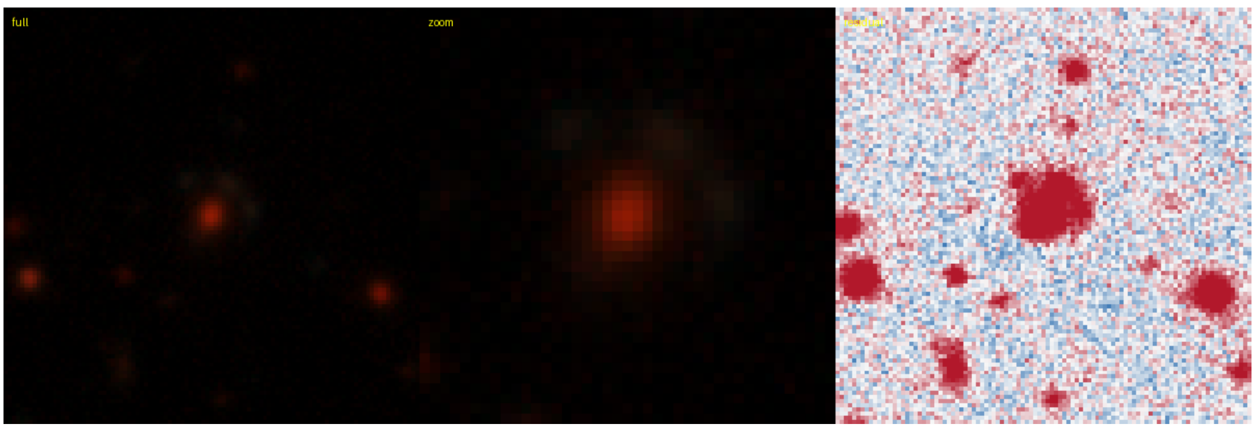



**Figure 37:** **#34** `s_34114_11363` — RA 353.7468, Dec -64.0687; grade **B**, tier-1 $p_{\rm lens}$ 0.58, `v3blend8` 0.90. KNOWN — DES: `DES J233459.19-640406.9`, 0.32″ (SIMBAD).

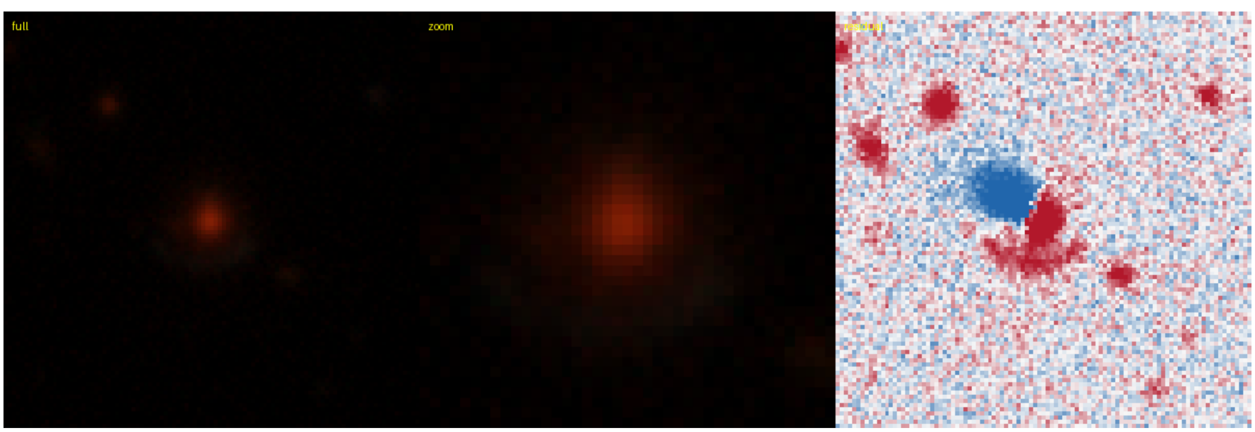



**Figure 38:** **#35** `s_102952_10164` — RA 334.8017, Dec -43.8098; grade **B**, tier-1 $p_{\rm lens}$ 0.58, `v3blend8` 0.85. KNOWN — DES: `DES J221912.39-434835.1`, 0.01″ (lineage).

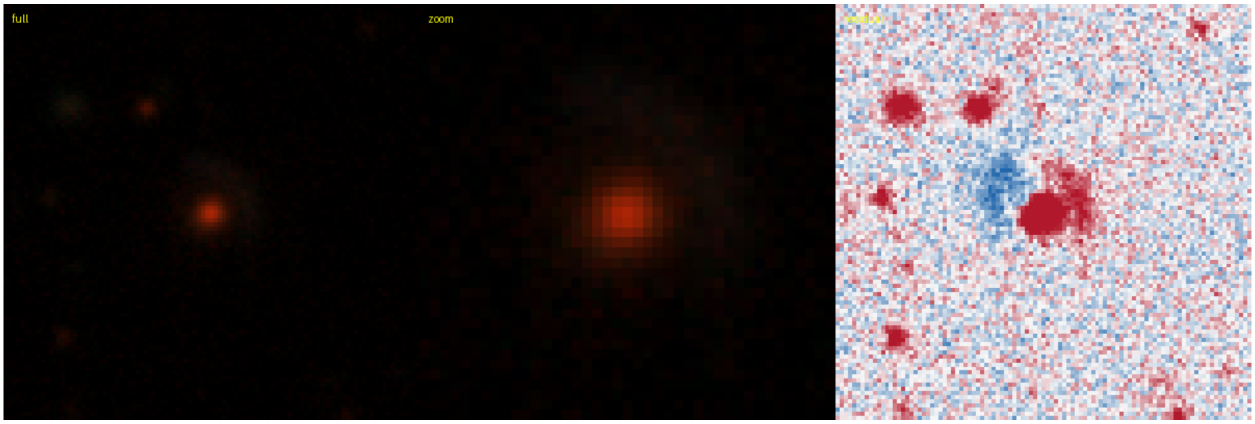



**Figure 39:** **#36** `s_191004_9637` — RA 56.9356, Dec -24.9088; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.58, `v3blend8` 0.88. KNOWN — DES: `DES J034744.53-245431.4`, 0.20″ (SIMBAD).


**Figure 40:** **#37** `s_43736_6056` — RA 67.3945, Dec -60.1905; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.58, `v3blend8` 0.83. KNOWN — DES: `DES J042934.66-601126.3`, 0.35″ (SIMBAD).

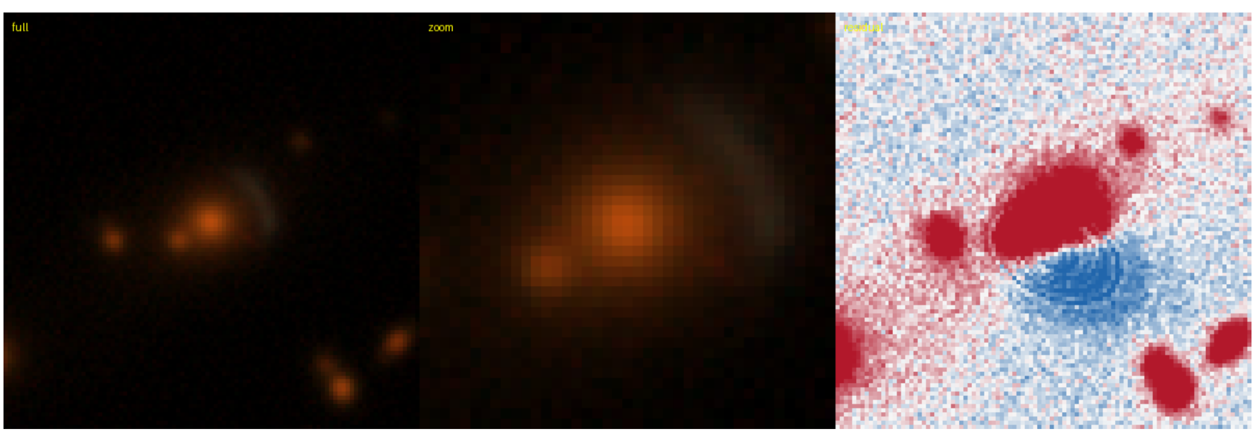



**Figure 41:** **#38** `s_137518_1133` — RA 31.3588, Dec -35.6632; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.58, `v3blend8` 0.84. KNOWN — DES: `DES J020526.09-353947.4`, 0.14″ (SIMBAD).

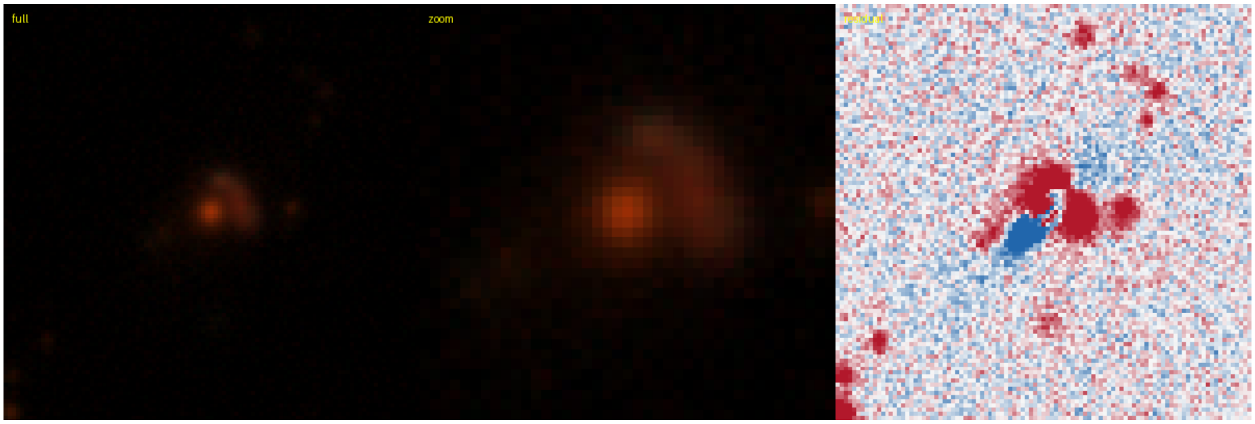



**Figure 42: #39** `s_129372_3768` — RA 16.4504, Dec -37.4286; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.58, `v3blend8` 0.82. KNOWN — DES: `DES J010548.04-372542.4`, 0.71″ (SIMBAD).

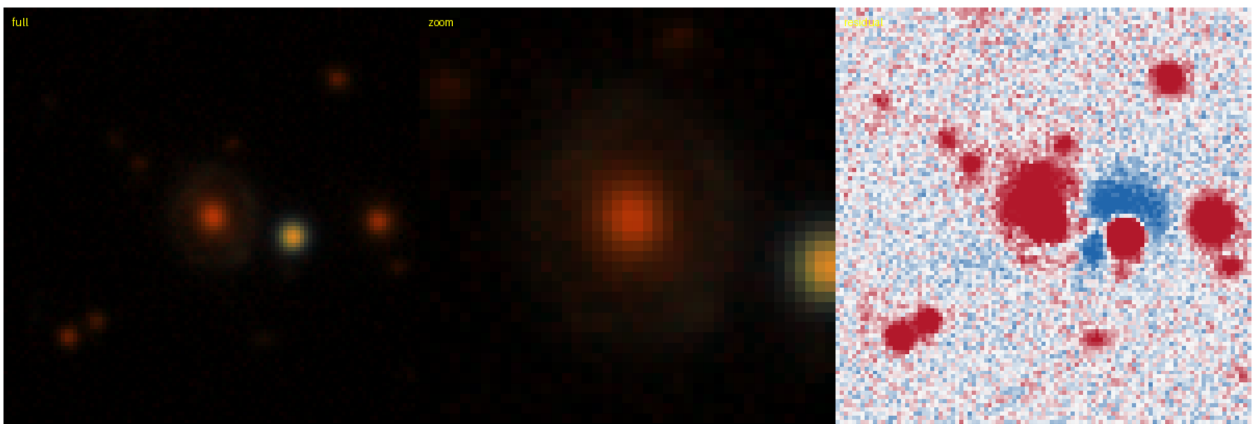



**Figure 43: #40** `s_98882_7820` — RA 1.6860, Dec -44.4974; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.58, `v3blend8` 0.80. KNOWN — DES: `[DBL2017] DES J0006-4429 (A)`, 0.18″ (SIMBAD).

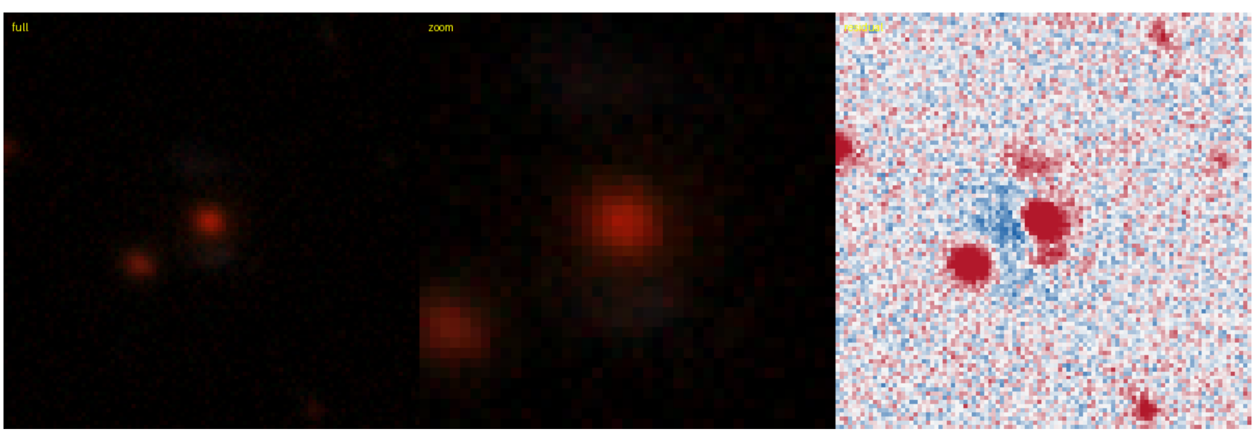



**Figure 44: #43** `s_381188_1199` — RA 144.1511, Dec +8.8633; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.55, `v3blend8` 0.90. KNOWN — Huang+2020: `DESI-144.1511+08.8633`, 0.20″ (lineage).

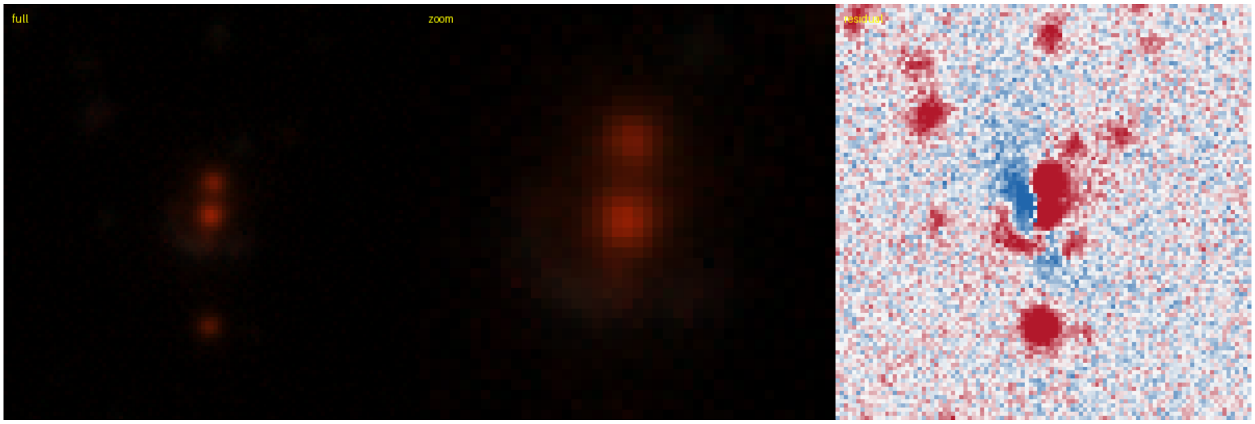



**Figure 45:** **#44** `s_114736_1046` — RA 343.5127, Dec -40.9303; grade **B**, tier-1 $p_{\rm lens}$ 0.55, `v3blend8` 0.86. KNOWN — DES: `[DBL2017] DES J2254-4055 (A)`, 0.11″ (SIMBAD).

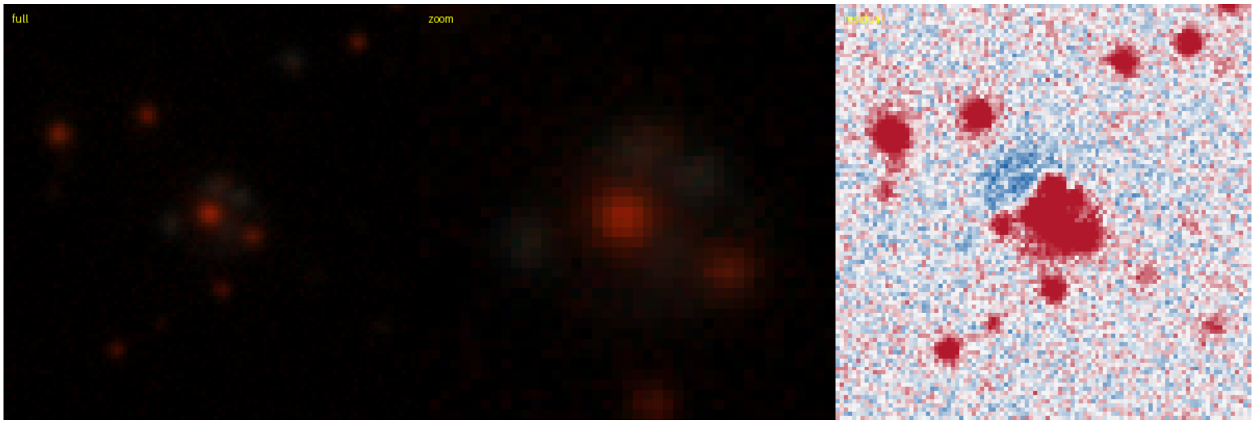



**Figure 46:** **#45** `s_87091_1800` — RA 84.5193, Dec -47.5872; grade **B**, tier-1 $p_{\rm lens}$ 0.55, `v3blend8` 0.89. KNOWN — DES: `[DBL2017] DES J0538-4735 (A)`, 0.11″ (SIMBAD).

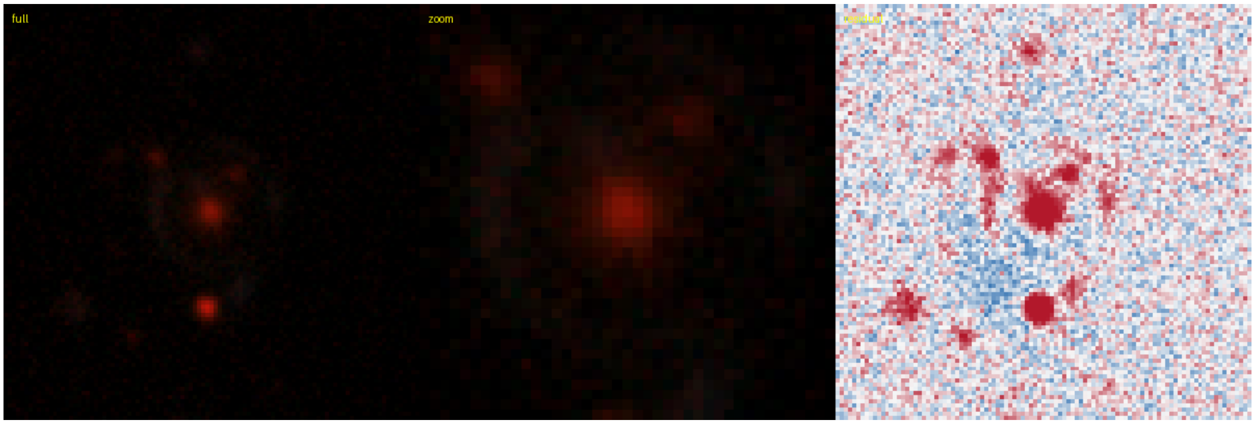



**Figure 47:** **#48** `s_384032_2232` — RA 143.3887, Dec +9.3219; grade **B**, tier-1 $p_{\rm lens}$ 0.55, `v3blend8` 0.84. KNOWN — Huang+2020: `DESI-143.3887+09.3219`, 0.17″ (lineage).

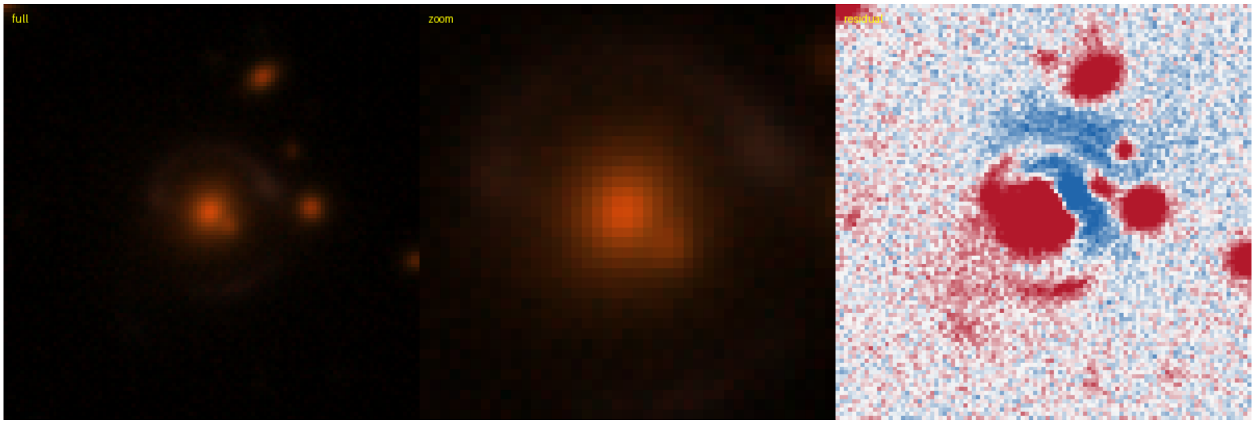



**Figure 48:** **#49** `s_73683_6069` — RA 357.3753, Dec -51.2276; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.55, `v3blend8` 0.83. KNOWN — DES: `[DBL2017] DES J2349-5113 (A)`, $0.00''$ (SIMBAD).

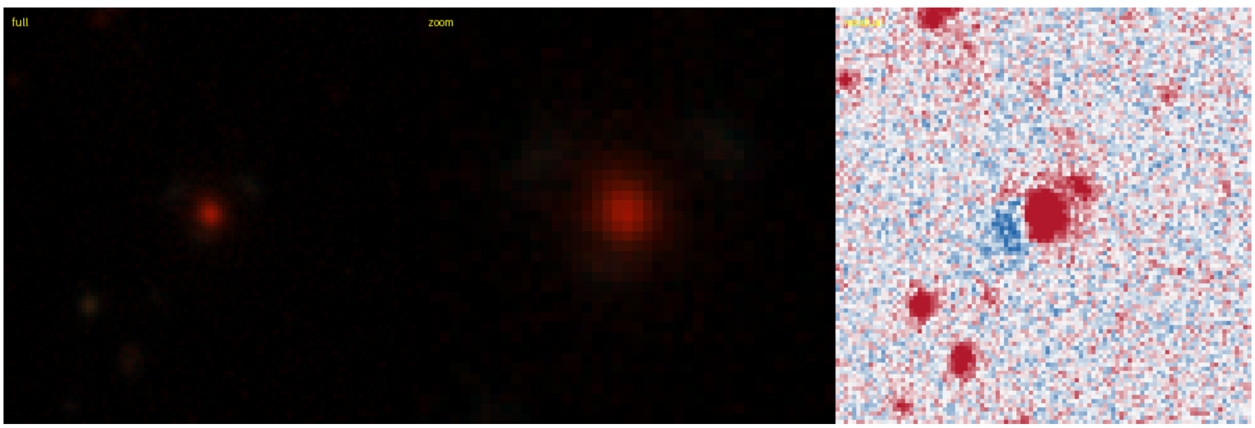



**Figure 49:** **#50** `s_80200_6872` — RA 15.4919, Dec -49.2940; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.55, `v3blend8` 0.88. KNOWN — DES: `DES J010158.03-491738.1`, $0.12''$ (SIMBAD).

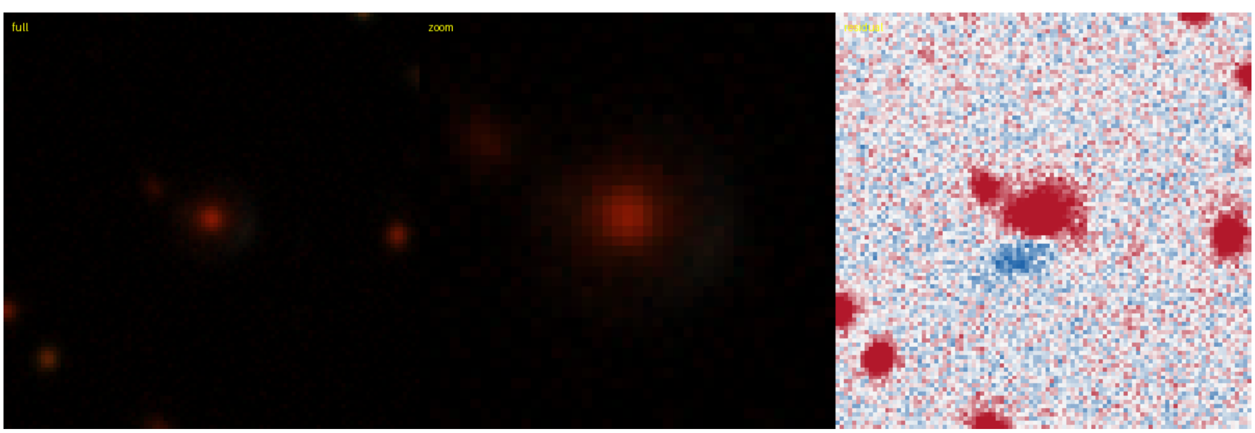



**Figure 50:** **#51** `s_58263_3751` — RA 42.7179, Dec -55.4032; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.55, `v3blend8` 0.88. KNOWN — DES: `DES J025052.27-552411.7`, $0.22''$ (SIMBAD).

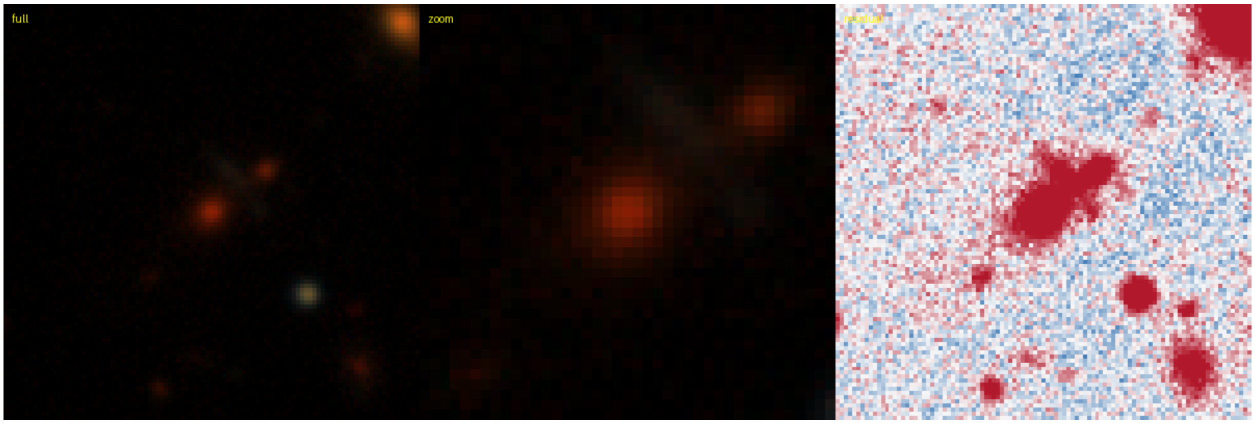



**Figure 51:** **#53** `s_164140_7181` — RA 41.2060, Dec -30.3214; grade **B**, tier-1 $p_{\rm lens}$ 0.50, `v3blend8` 0.79. KNOWN — DES: `DES J024449.42-301916.7`, 0.24″ (SIMBAD).

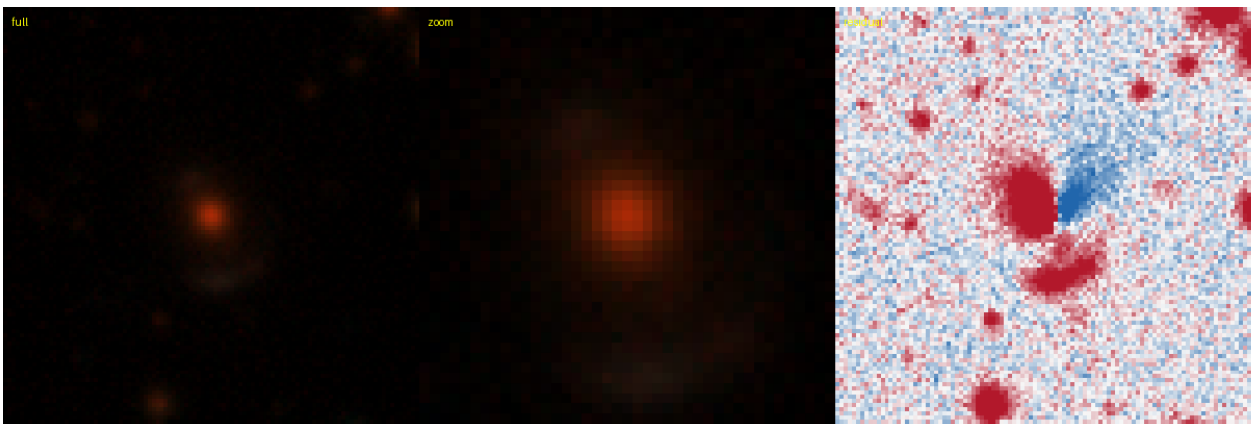



**Figure 52:** **#54** `s_71871_246` — RA 353.9664, Dec -51.8716; grade **B**, tier-1 $p_{\rm lens}$ 0.50, `v3blend8` 0.76. KNOWN — AGEL: `AGEL J233552-515218 system`, 0.95″ (SIMBAD).

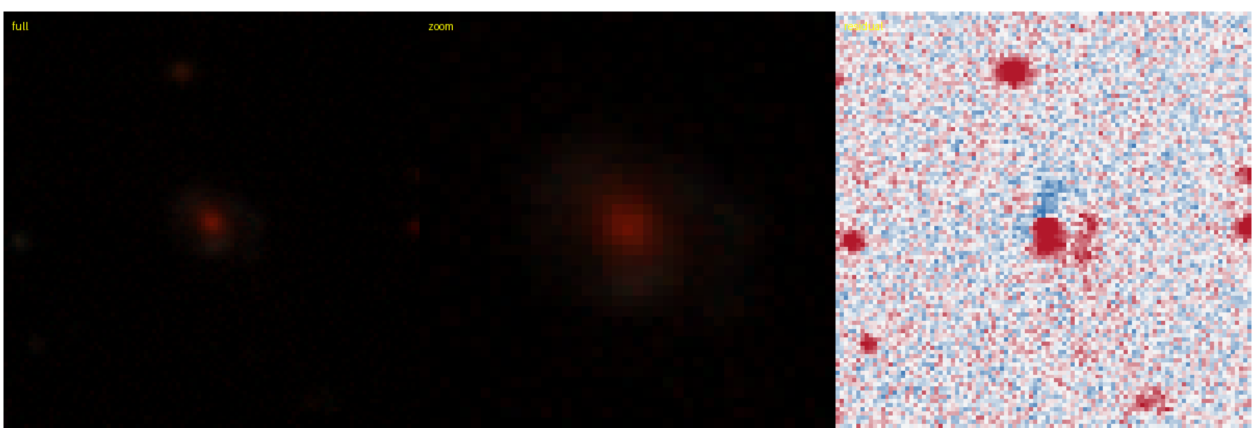



**Figure 53:** **#55** `s_61626_1035` — RA 65.1782, Dec -54.3770; grade **B**, tier-1 $p_{\rm lens}$ 0.50, `v3blend8` 0.87. KNOWN — DES: `DES J042042.76-542237.4`, 0.18″ (SIMBAD).

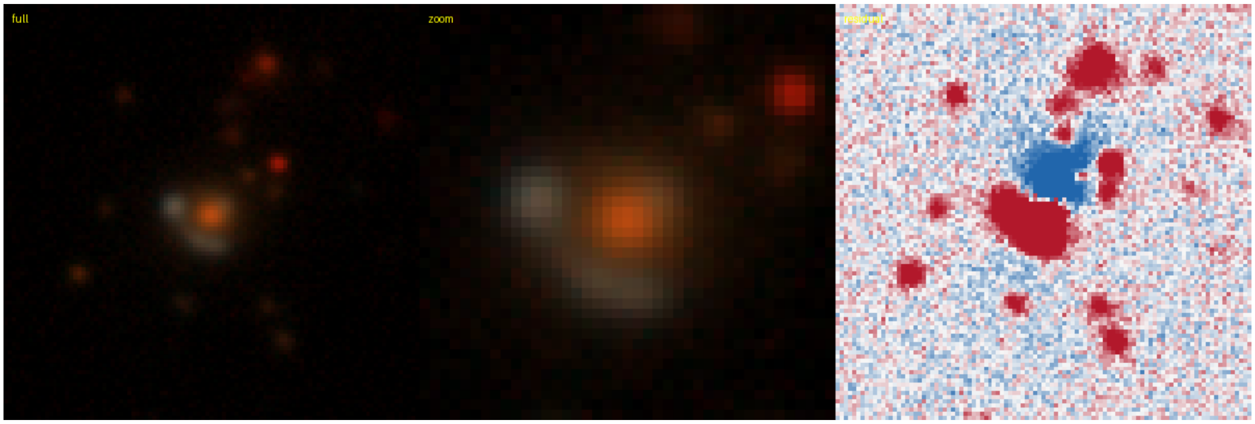



**Figure 54:** **#57** `s_107312_5223` — RA 23.8451, Dec -42.5398; grade **B**, tier-1 $p_{\rm lens}$ 0.50, `v3blend8` 0.85. KNOWN — DES: `[DBL2017] DES J0135-4232 (A)`, 0.18″ (SIMBAD).

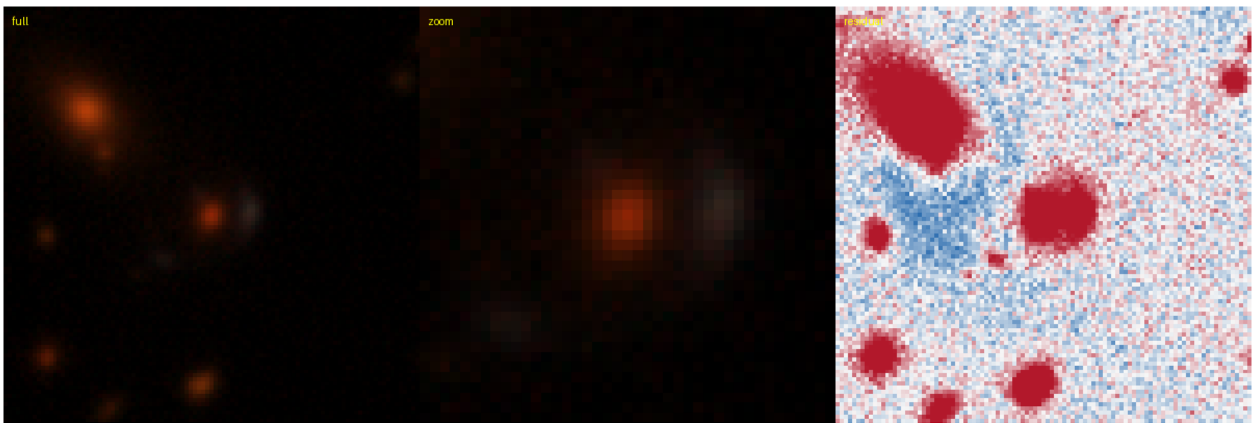



**Figure 55:** **#58** `s_90920_9419` — RA 45.9517, Dec -46.4407; grade **B**, tier-1 $p_{\rm lens}$ 0.50, `v3blend8` 0.85. KNOWN — DES: `[DBL2017] DES J0303-4626 (A)`, 0.15″ (SIMBAD).

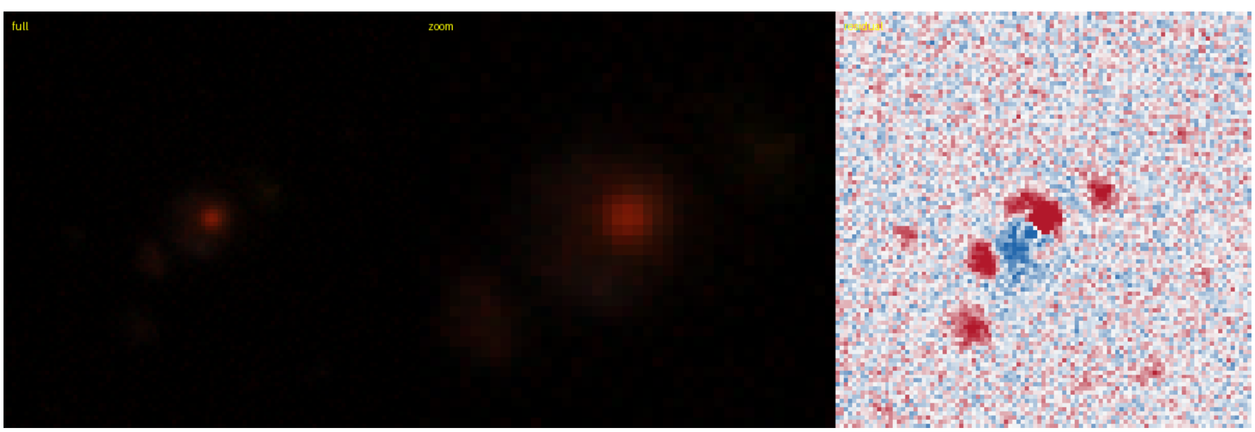



**Figure 56:** **#62** `s_162970_1777` — RA 62.5554, Dec -30.6238; grade **B**, tier-1 $p_{\rm lens}$ 0.48, `v3blend8` 0.82. KNOWN — DES: `DES J041013.30-303725.8`, 0.09″ (SIMBAD).

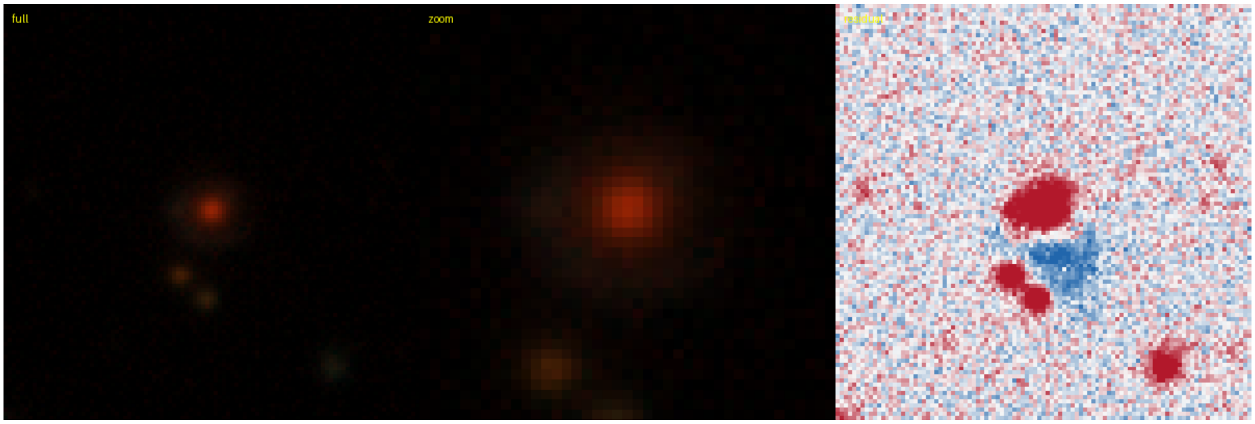



**Figure 57:** **#63** s_234043_6467 — RA 58.4427, Dec -17.1109; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.47, v3blend8 0.88. KNOWN — AGEL: AGEL J035346-170639 lens, 0.04″ (SIMBAD).

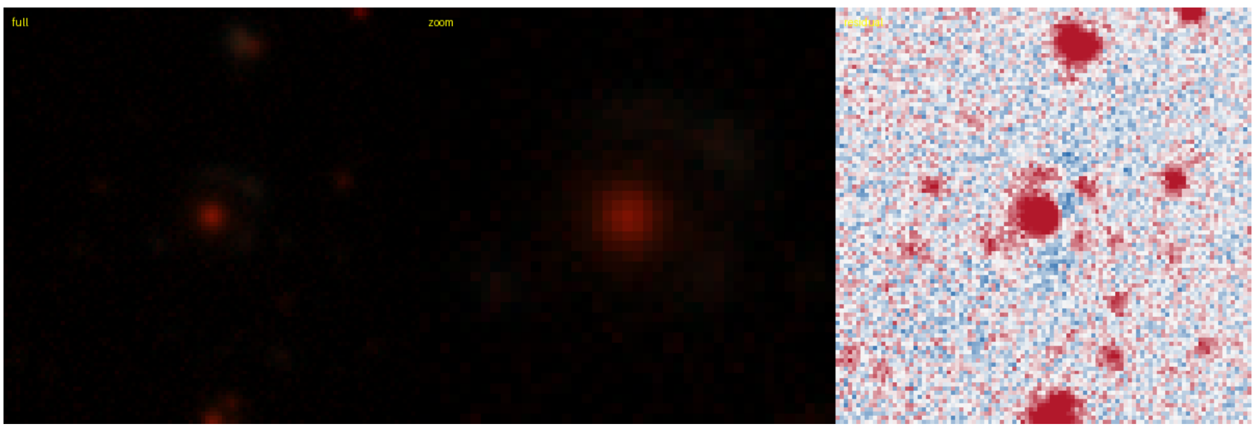



**Figure 58:** **#65** s_50907_11900 — RA 303.5808, Dec -57.9504; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.45, v3blend8 0.74. KNOWN — DES: DES J201419.38-575701.4, 0.15″ (SIMBAD).

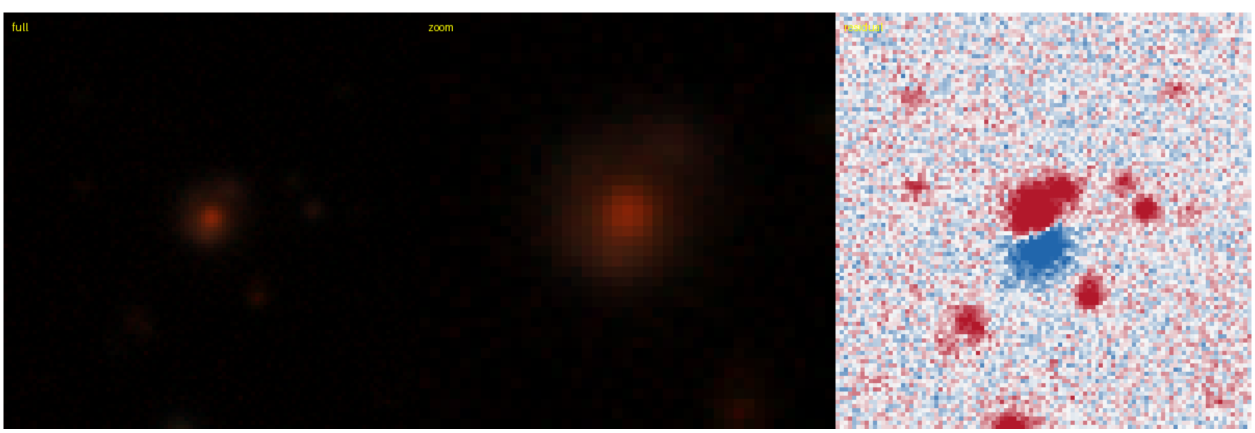



**Figure 59:** **#66** s_61411_4481 — RA 332.9866, Dec -54.6445; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.45, v3blend8 0.76. KNOWN — DES: DES J221156.77-543839.9, 0.22″ (SIMBAD).

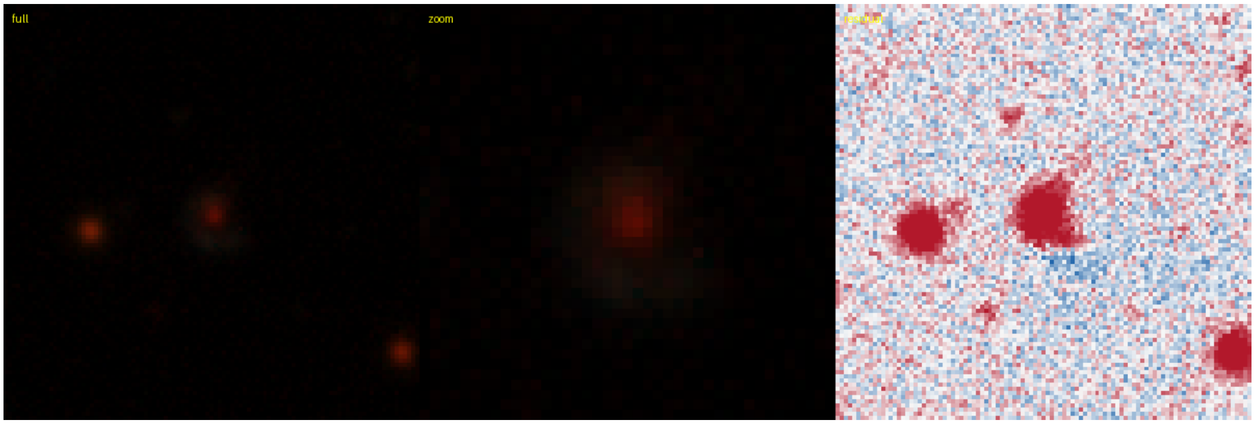



**Figure 60:** **#67** `s_165468_9648` — RA 64.7771, Dec -29.9937; grade **B**, tier-1 $p_{\rm lens}$ 0.45, `v3blend8` 0.78. KNOWN — DES: `DES J041906.50-295937.7`, 0.49″ (SIMBAD).

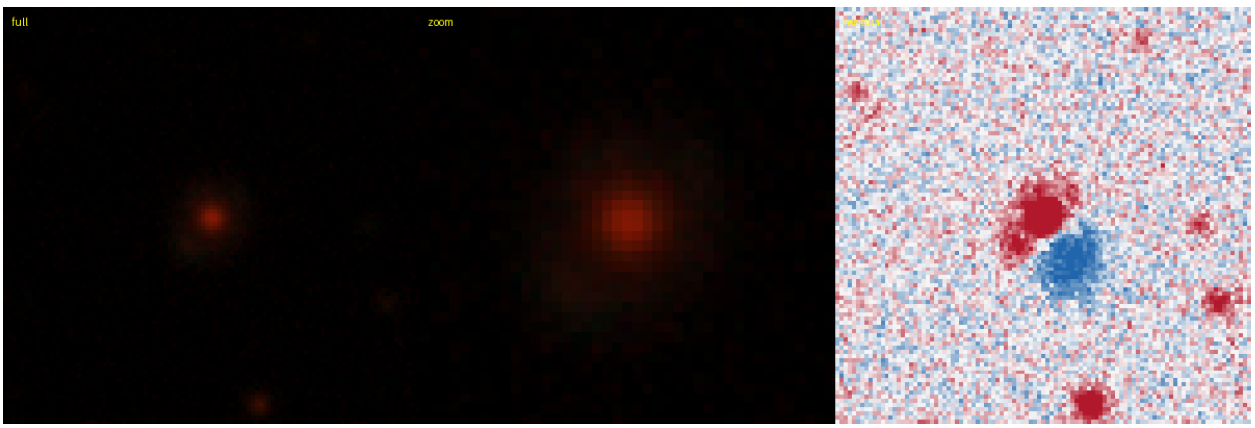



**Figure 61:** **#69** `s_280264_9147` — RA 33.1051, Dec -8.8697; grade **B**, tier-1 $p_{\rm lens}$ 0.45, `v3blend8` 0.84. KNOWN — literature (SIMBAD): `[MVM2016] SW45`, 0.12″ (SIMBAD).

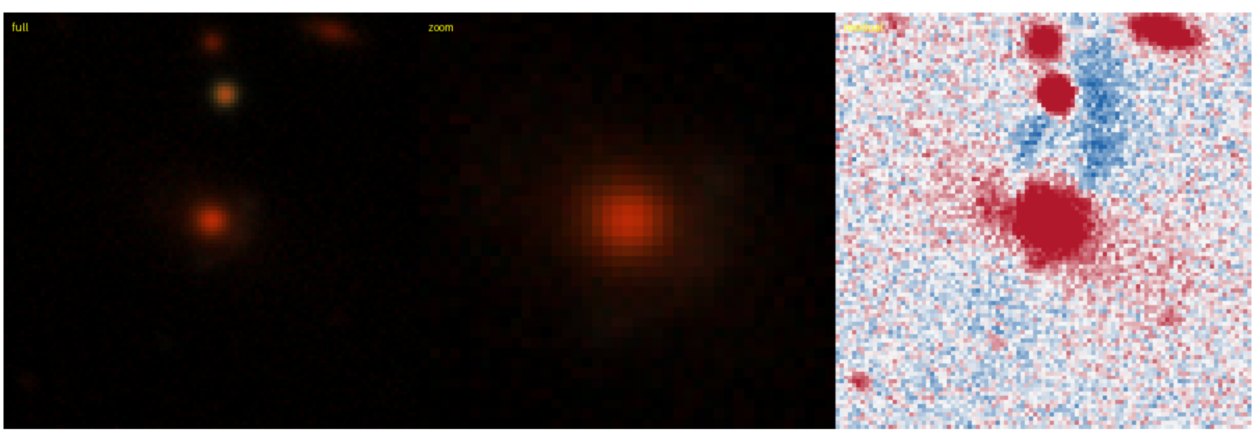



**Figure 62:** **#72** `s_52008_8091` — RA 96.5963, Dec -57.5083; grade **B**, tier-1 $p_{\rm lens}$ 0.45, `v3blend8` 0.89. KNOWN — DES: `DES J062623.11-573030.1`, 0.09″ (SIMBAD).

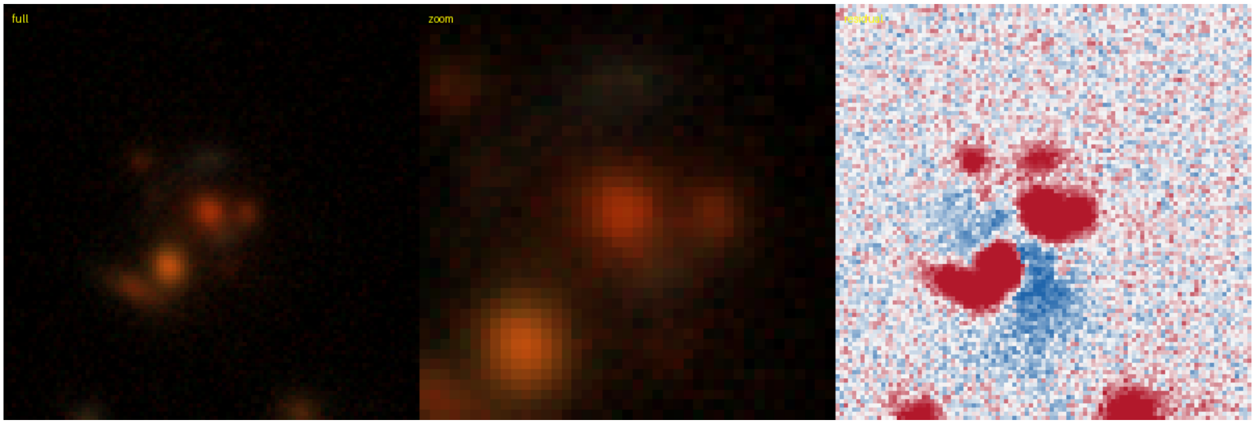


**Figure 63:** **#74** `s_504880_4027` — RA 152.0763, Dec +31.7005; grade **B**, tier-1 $p_{\rm lens}$ 0.45, `v3blend8` 0.84. KNOWN — Huang+2020: `DESI-152.0763+31.7005`, 0.12″ (lineage).

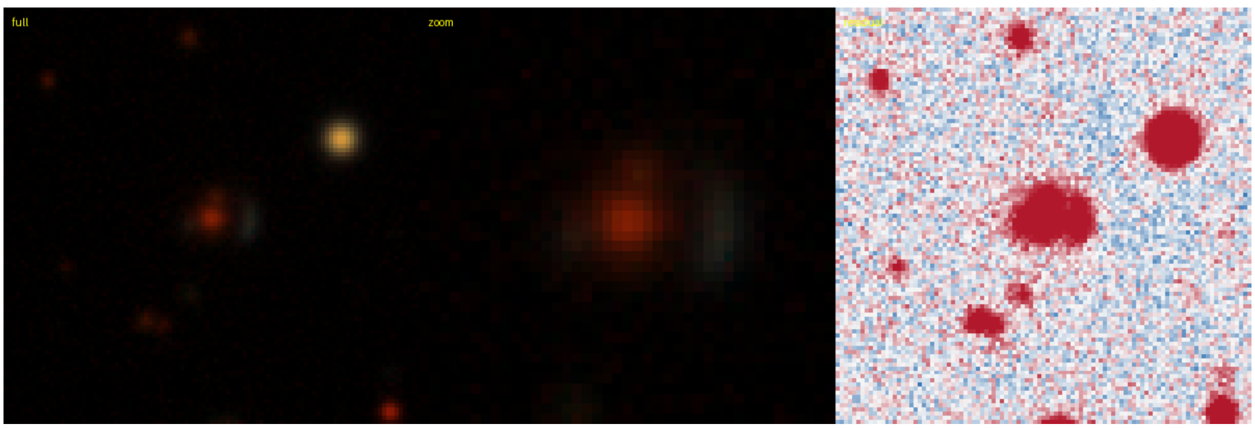


**Figure 64:** **#76** `s_69087_268` — RA 307.2325, Dec -52.5218; grade **B**, tier-1 $p_{\rm lens}$ 0.43, `v3blend8` 0.81. KNOWN — DES: `DES J202855.79-523118.3`, 0.12″ (SIMBAD).

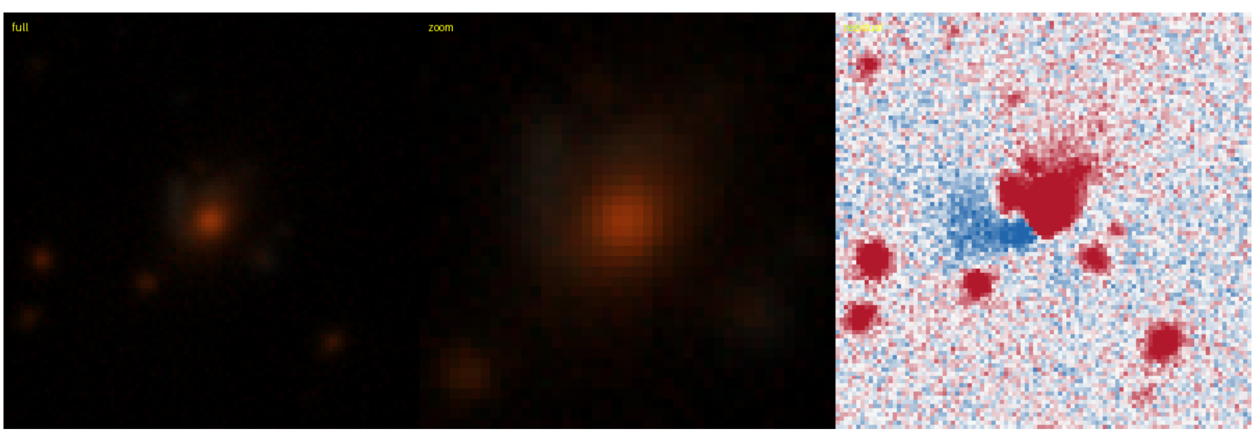


**Figure 65:** **#77** `s_256199_9097` — RA 29.6819, Dec -13.0258; grade **B**, tier-1 $p_{\rm lens}$ 0.42, `v3blend8` 0.78. KNOWN — DES: `DES J015843.66-130132.8`, 0.18″ (SIMBAD).

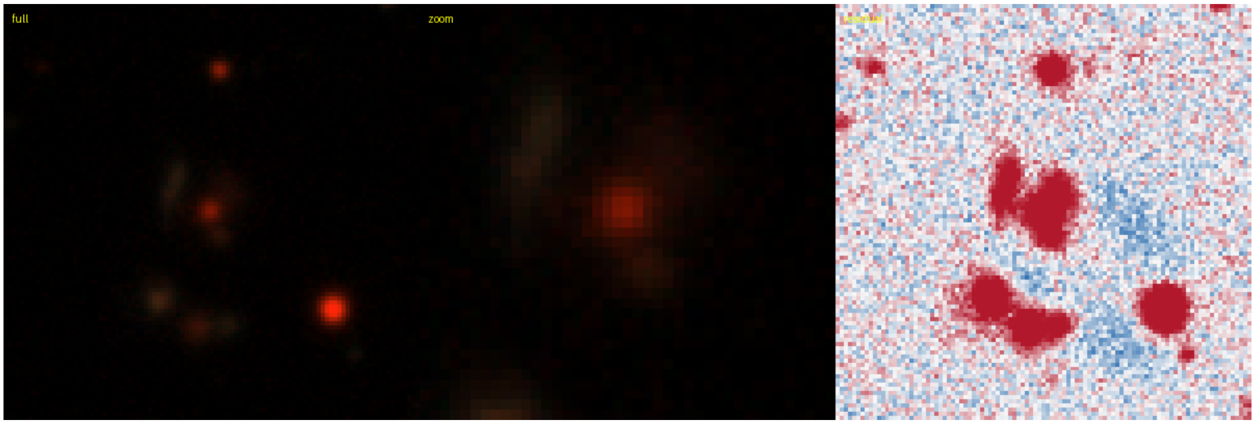



**Figure 66:** **#78** `s_36700_530` — RA 347.9904, Dec -63.1176; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.42, `v3blend8` 0.78. KNOWN — literature (SIMBAD): `DES J2311-6307:[OWD2022] 1`, 2.80″ (NED).

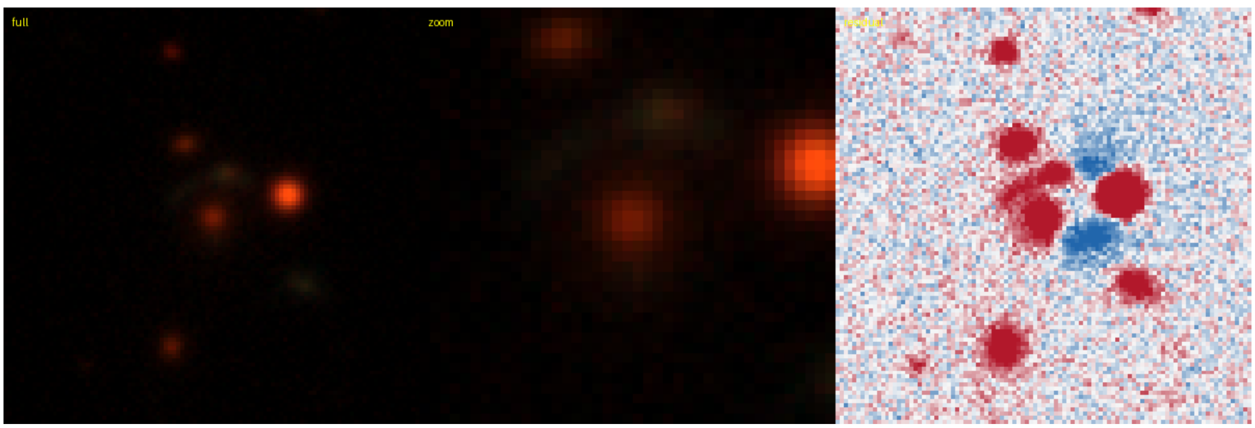



**Figure 67:** **#81** `s_204169_8722` — RA 49.1618, Dec -22.6093; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.42, `v3blend8` 0.77. KNOWN — DES: `DES J031638.82-223633.3`, 0.15″ (SIMBAD).

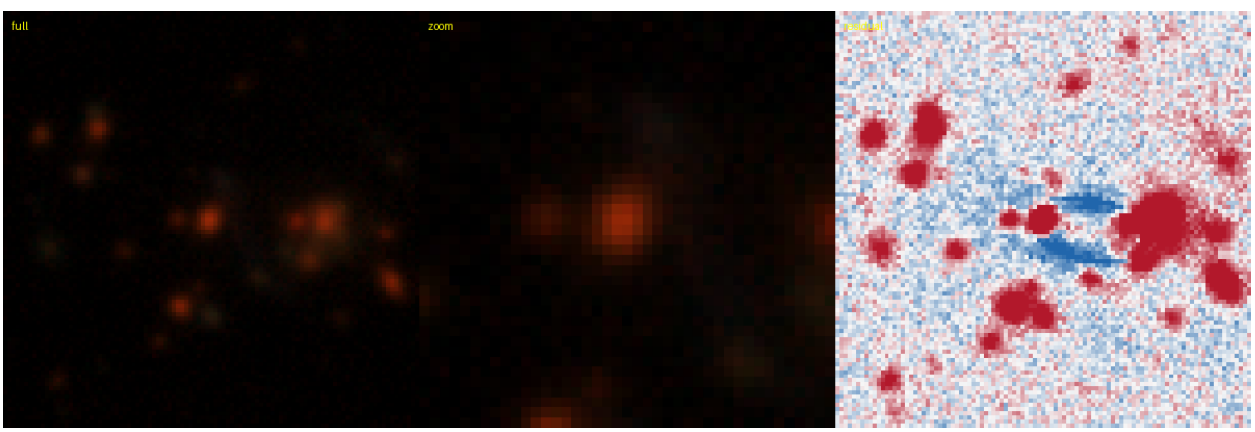



**Figure 68:** **#83** `s_139825_11098` — RA 18.9226, Dec -35.3386; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.42, `v3blend8` 0.76. KNOWN — literature (SIMBAD): `DES J0115-3520:[OWD2022] 1`, 4.88″ (NED).


**Figure 69:** **#84** `s_60524_3113` — RA 309.7973, Dec -54.9952; grade **B**, tier-1 $p_{\rm lens}$ 0.42, `v3blend8` 0.81. KNOWN — DES: `[DBL2017] DES J2039-5459 (A)`, 0.16″ (SIMBAD).

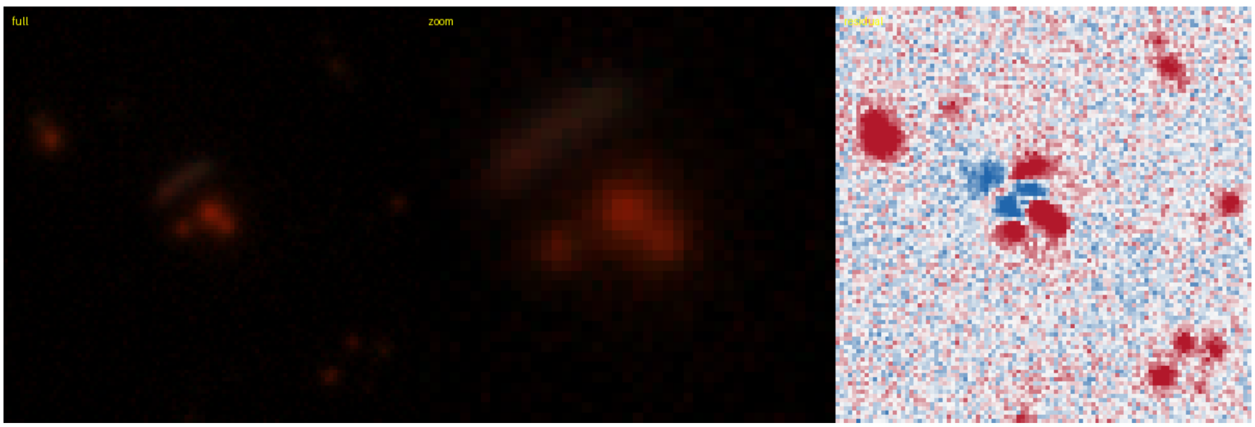



**Figure 70:** **#85** `s_38835_314` — RA 56.6543, Dec -61.9792; grade **B**, tier-1 $p_{\rm lens}$ 0.42, `v3blend8` 0.84. KNOWN — DES: `[DBL2017] DES J0346-6158 (A)`, 0.23″ (SIMBAD).

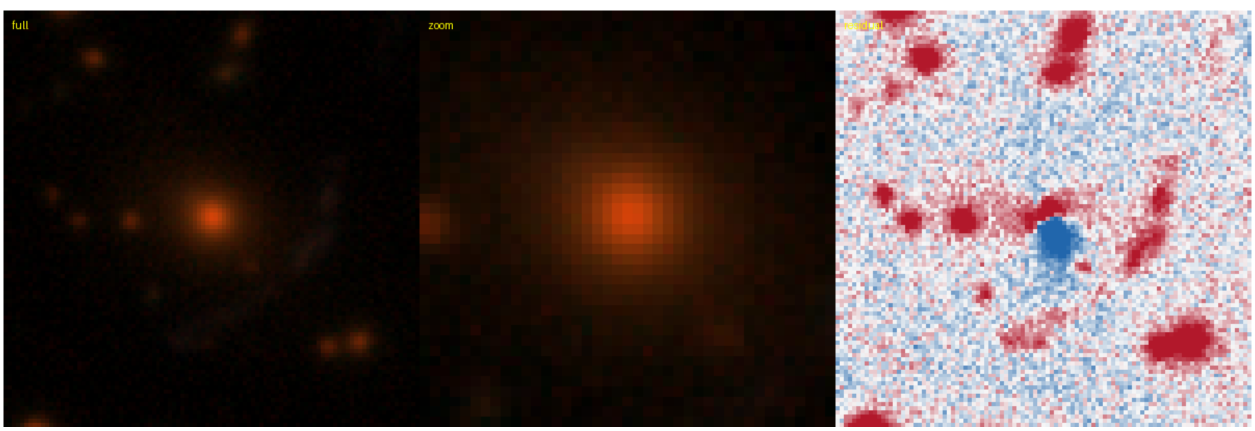



**Figure 71:** **#87** `s_68464_11443` — RA 52.7373, Dec -52.4703; grade **B**, tier-1 $p_{\rm lens}$ 0.42, `v3blend8` 0.88. KNOWN — DES: `[DBL2017] DES J0330-5228`, 0.11″ (SIMBAD).

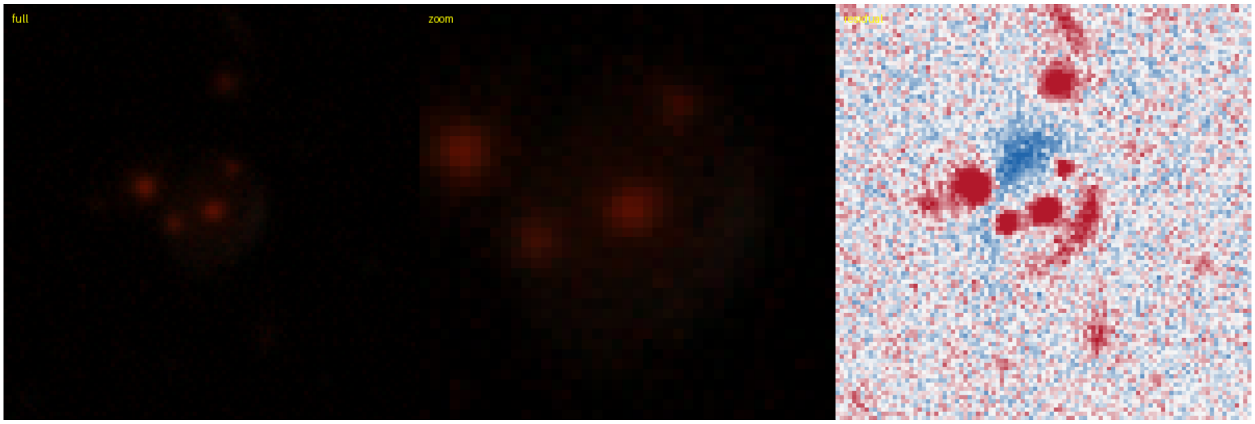



**Figure 72:** **#88** `s_206743_10995` — RA 24.2120, Dec -22.0076; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.42, `v3blend8` 0.85. KNOWN — DES: `DES J013650.85-220027.2`, 0.23″ (SIMBAD).

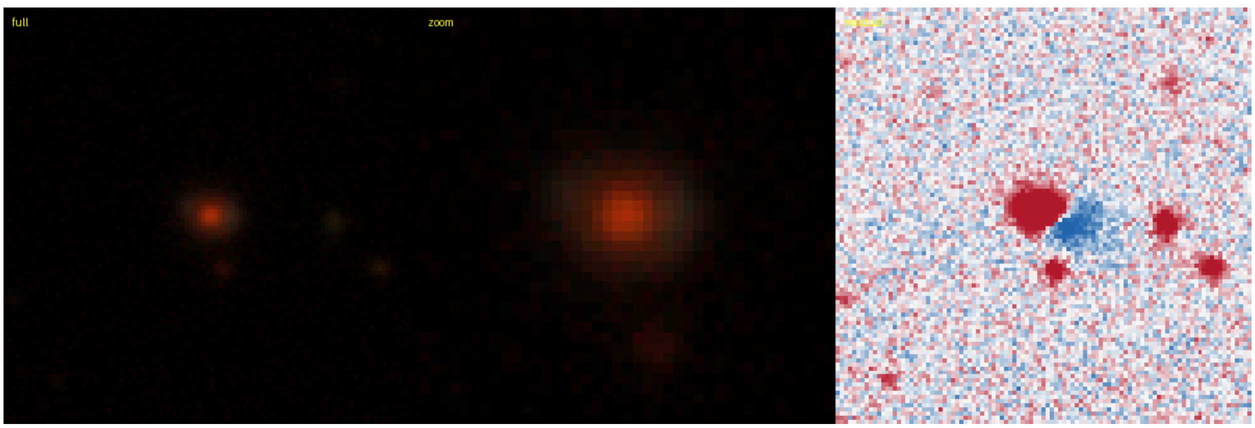



**Figure 73:** **#89** `s_41562_4180` — RA 42.2402, Dec -60.9010; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.40, `v3blend8` 0.87. KNOWN — DES: `DES J024857.61-605403.4`, 0.19″ (SIMBAD).

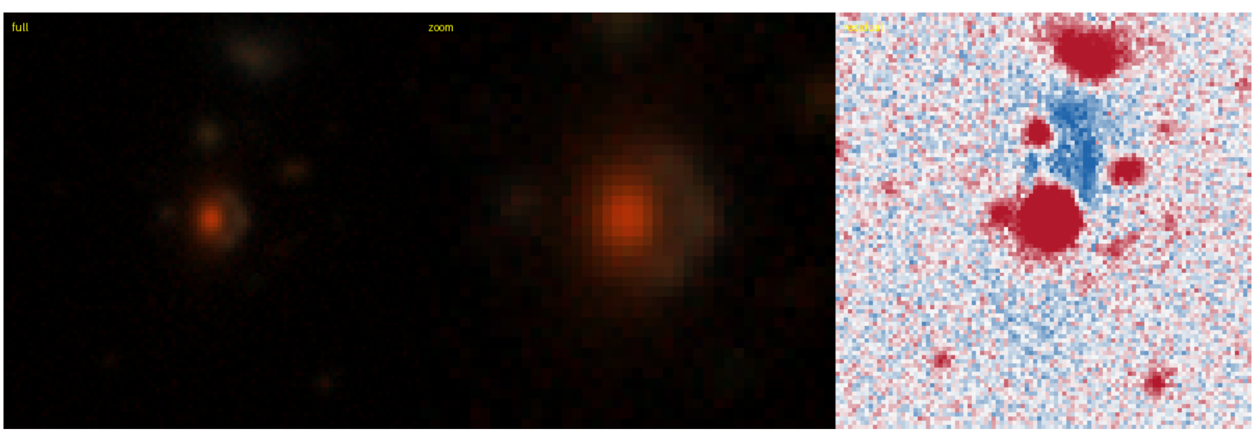



**Figure 74:** **#97** `s_156771_3446` — RA 54.3219, Dec -31.8704; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.38, `v3blend8` 0.89. KNOWN — AGEL: `AGEL J033717-315214 system`, 0.99″ (SIMBAD).

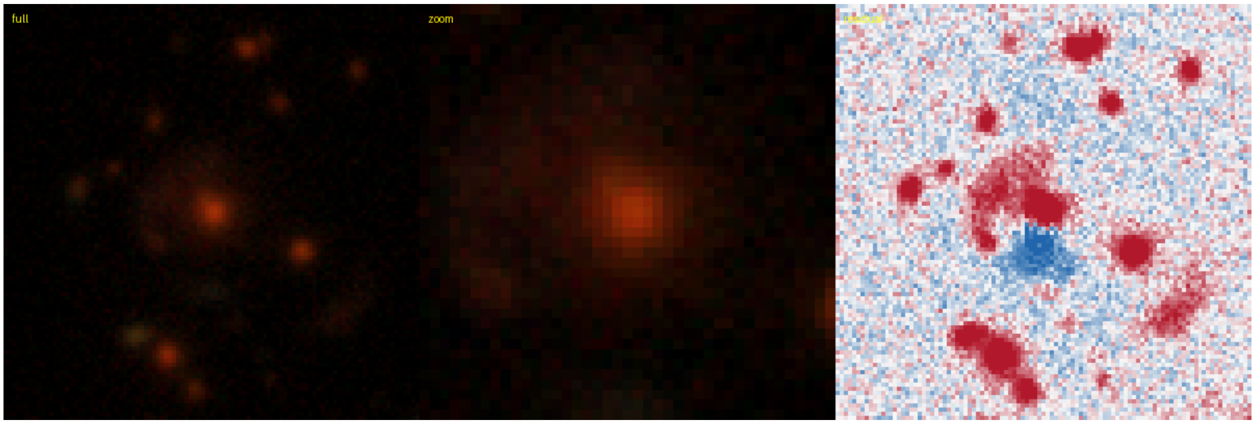



**Figure 75:** **#99** `s_416616_6090` — RA 189.5370, Dec +15.0309; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.38, `v3blend8` 0.93. KNOWN — Huang+2020: `DESI-189.5370+15.0309`, 0.06″ (lineage).

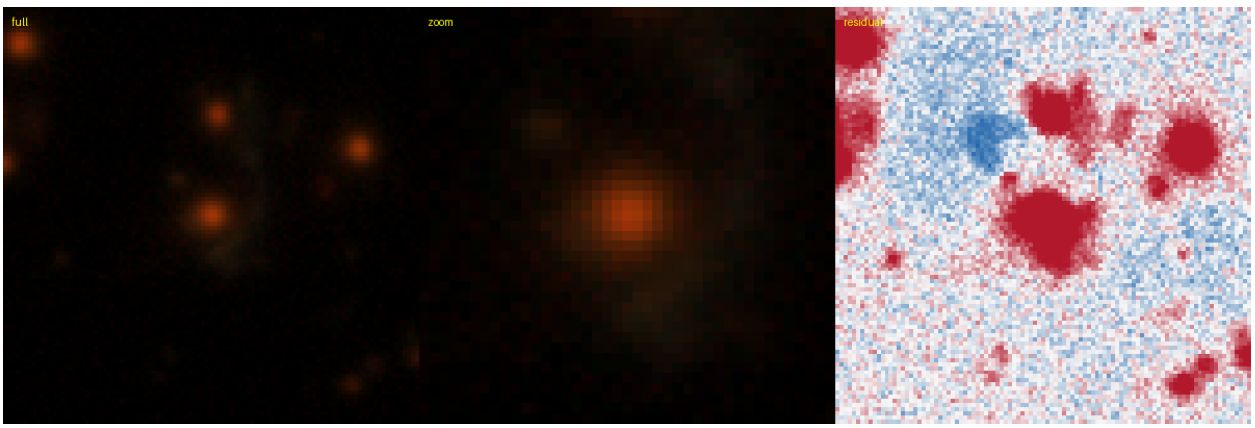



**Figure 76:** **#104** `s_194944_4494` — RA 59.2045, Dec -24.1447; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.35, `v3blend8` 0.85. KNOWN — DES: `DES J035649.05-240841.1`, 0.28″ (SIMBAD).

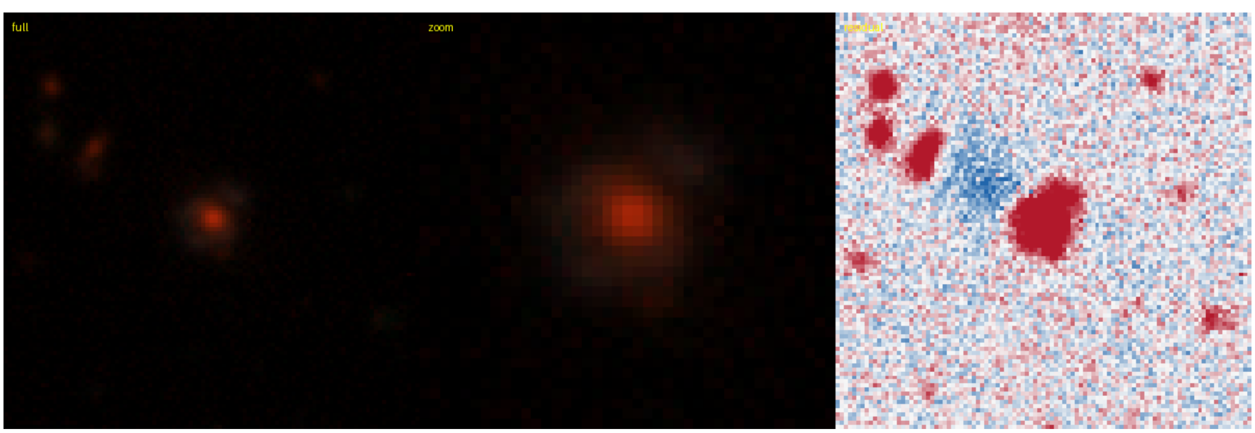



**Figure 77:** **#105** `s_94861_9433` — RA 21.9716, Dec -45.5428; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.32, `v3blend8` 0.87. KNOWN — DES: `DES J012753.17-453233.8`, 0.17″ (SIMBAD).

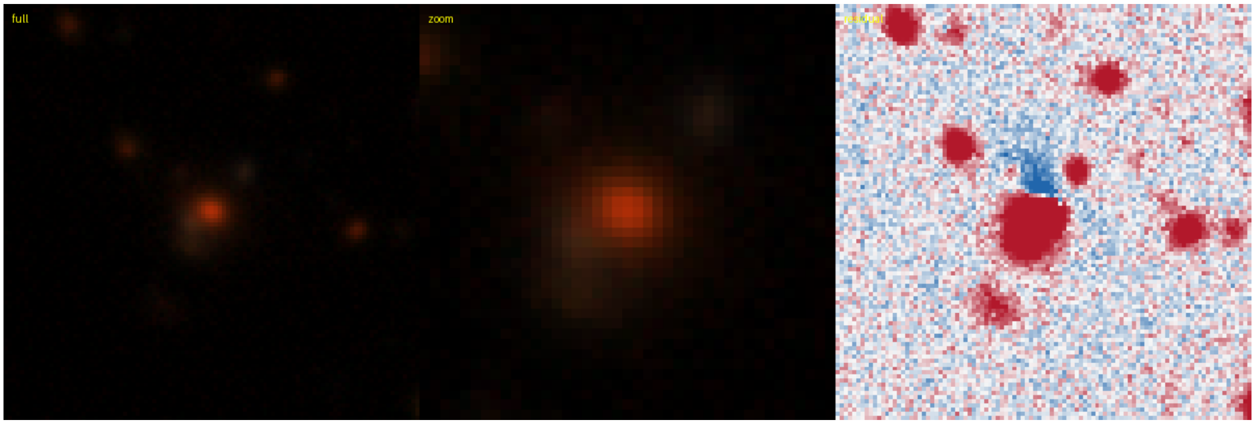



**Figure 78: #106** `s_235253_4230` — RA 14.4452, Dec -16.7457; grade **B**, tier-1 $p_{\text{lens}}$ 0.32, `v3blend8` 0.76. KNOWN — Huang+2020: `DESI-014.4452-16.7457`, $0.17''$ (lineage).

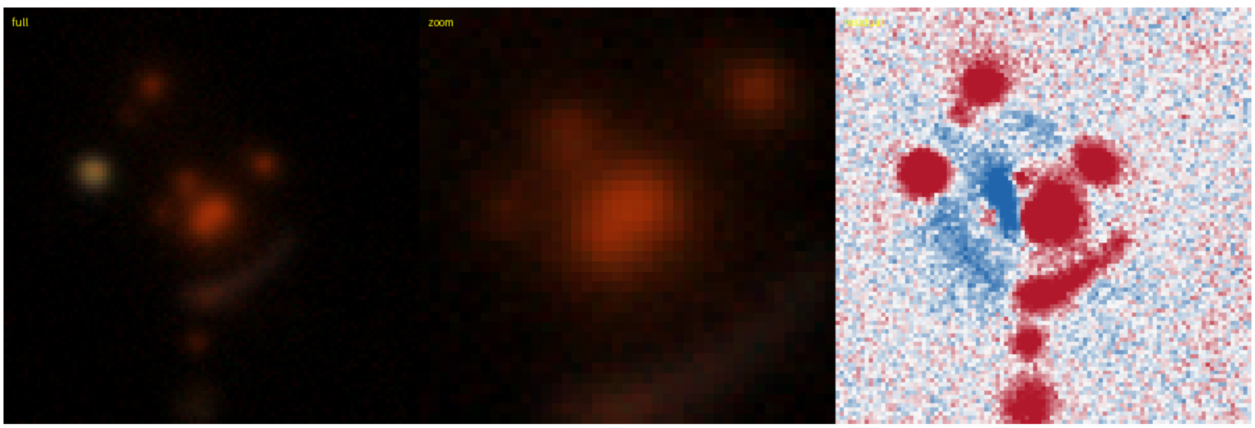



**Figure 79: #107** `s_394431_6727` — RA 261.7637, Dec +11.0021; grade **B**, tier-1 $p_{\text{lens}}$ 0.30, `v3blend8` 0.77. KNOWN — AGEL: `AGEL J172703+110008 system`, $0.61''$ (SIMBAD).

## New — candidates new to the literature (39 of 108)

These are absent from the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature (3 grade A and 36 grade B); at DECaLS 1″ an unknown (likely large) fraction are lrg+companion mimics. Ranked by tier-1 $p_{\text{lens}}$.

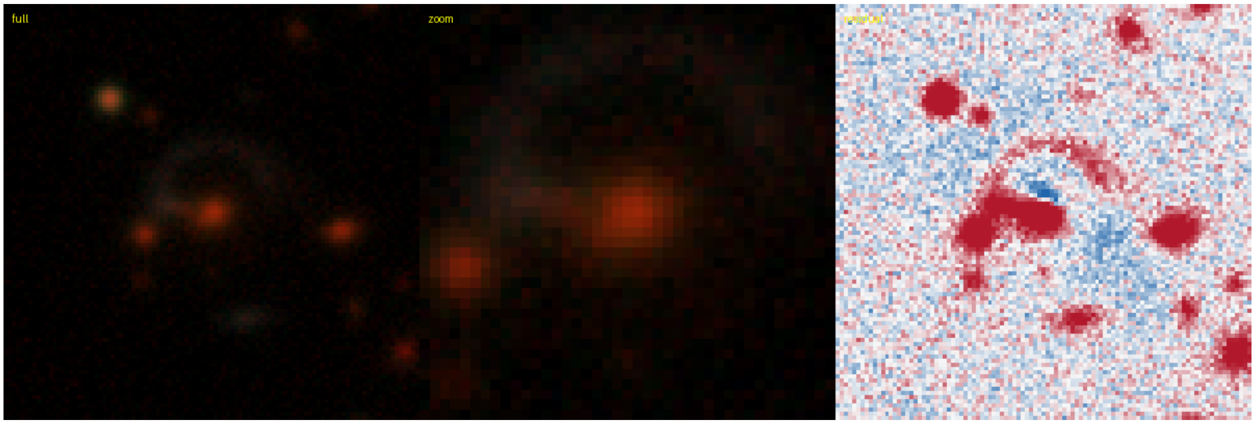



**Figure 80: #7** `s_230539_4159` — RA 222.0715, Dec -17.8489; grade **A**, tier-1 $p_{\text{lens}}$ 0.85, `v3blend8` 0.84. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

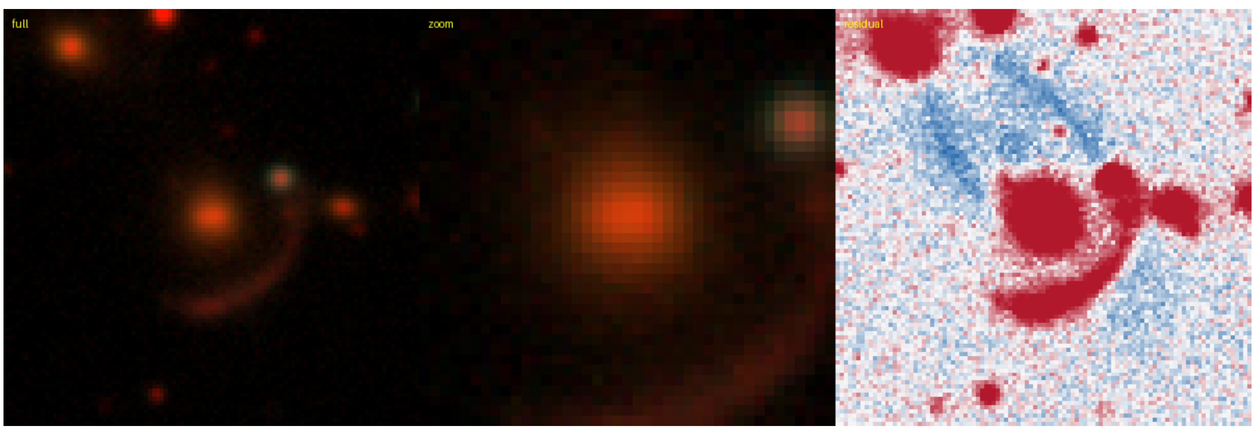



**Figure 81: #10** `s_194300_2796` — RA 242.6052, Dec -24.4155; grade **A**, tier-1 $p_{\text{lens}}$ 0.83, `v3blend8` 0.74. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

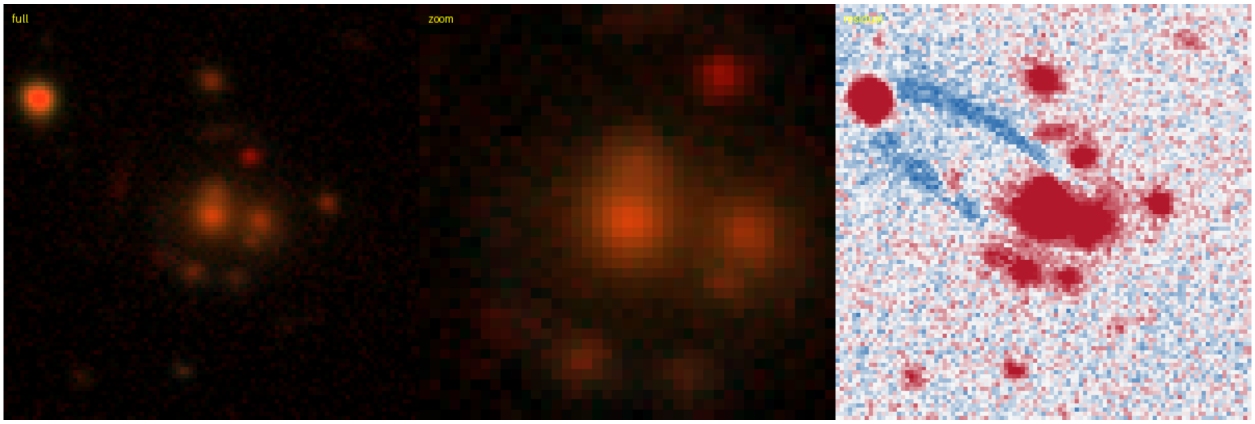



**Figure 82: #15** s_320069_835 — RA 305.2638, Dec -1.9376; grade **A**, tier-1 $p_{\mathrm{lens}}$ 0.78, v3blend8 0.89. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

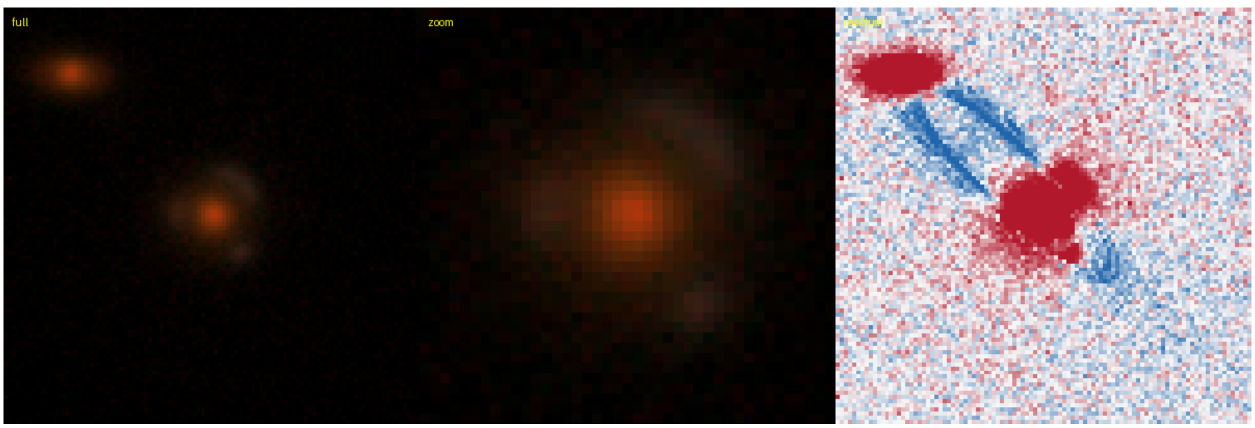



**Figure 83: #24** s_234689_542 — RA 227.0427, Dec -16.8771; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.65, v3blend8 0.80. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

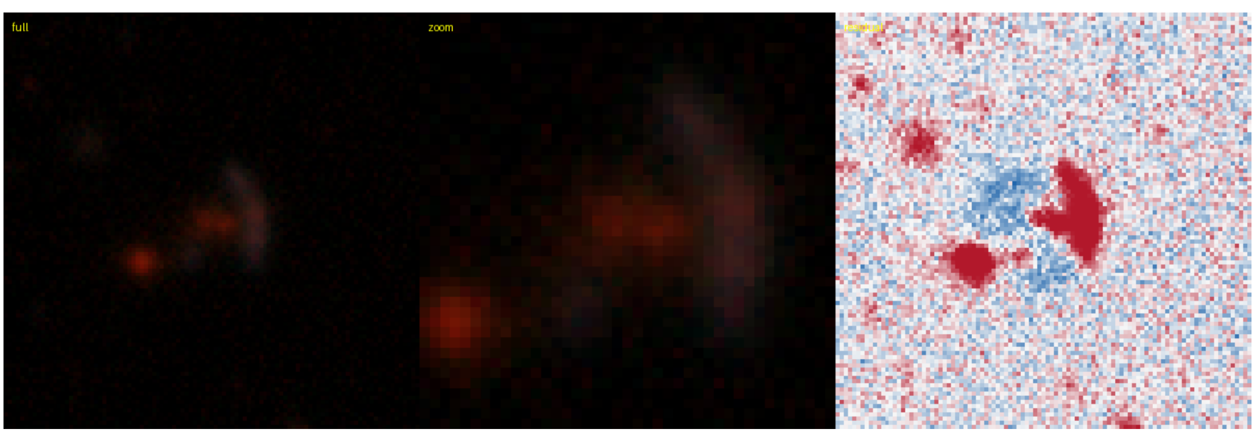



**Figure 84: #25** s_502450_1680 — RA 161.1145, Dec +31.2344; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.63, v3blend8 0.79. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

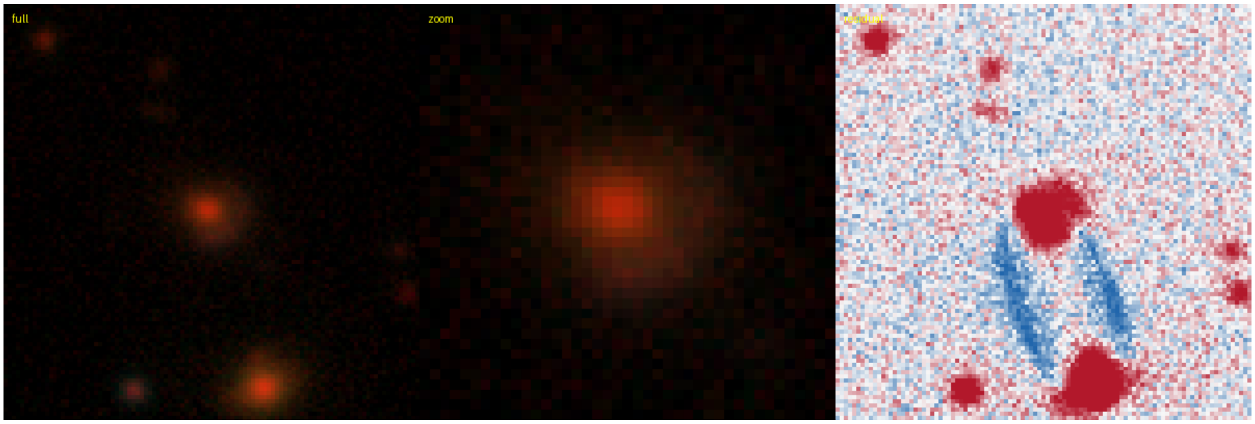



**Figure 85:** **#28** `s_243378_3831` — RA 326.5170, Dec -15.4964; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.62, `v3blend8` 0.84. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

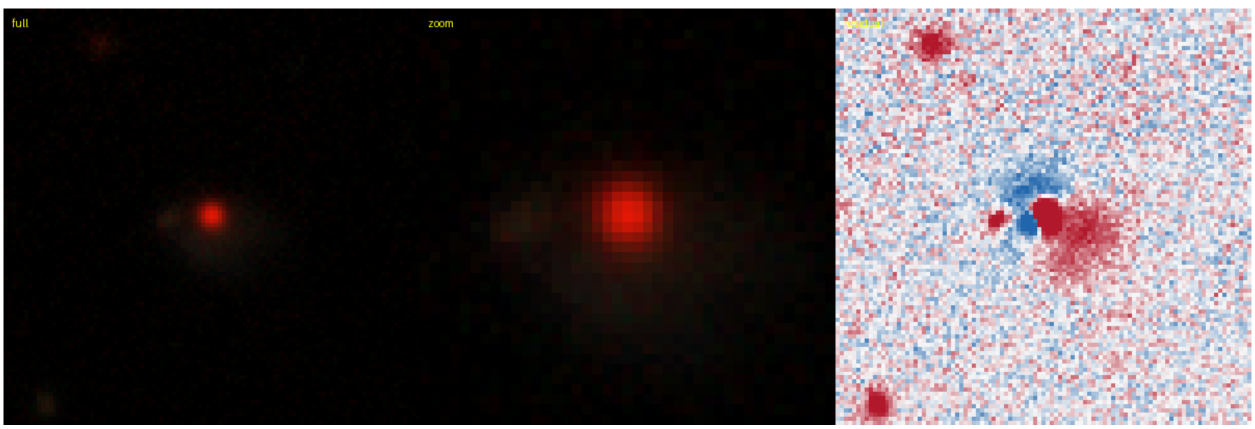



**Figure 86:** **#33** `s_369309_653` — RA 30.4465, Dec +6.6551; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.60, `v3blend8` 0.74. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

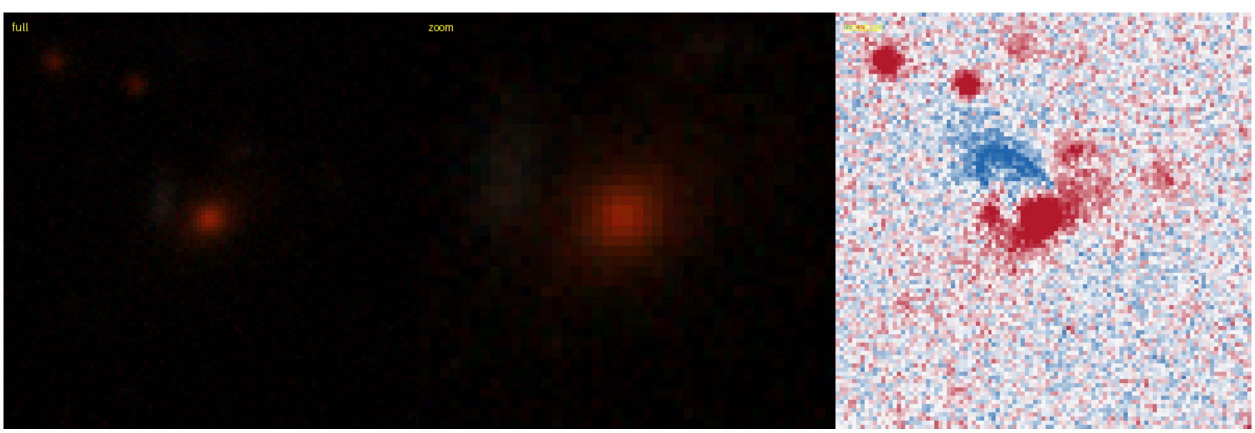



**Figure 87:** **#41** `s_218735_721` — RA 350.2028, Dec -20.0497; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.58, `v3blend8` 0.79. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

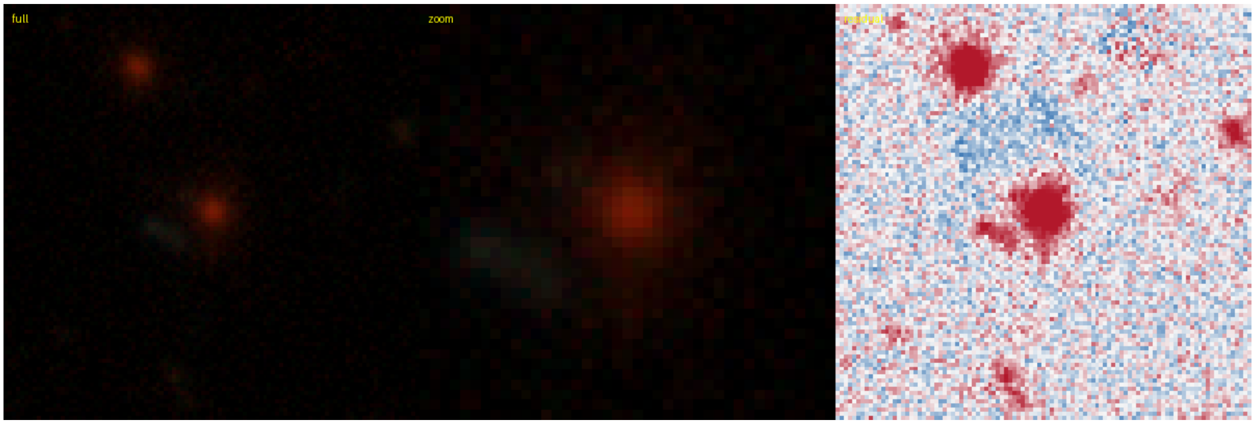



**Figure 88:** **#42** `s_360639_3636` — RA 13.4868, Dec +5.3590; grade **B**, tier-1 $p_{\text{lens}}$ 0.55, `v3blend8` 0.85. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

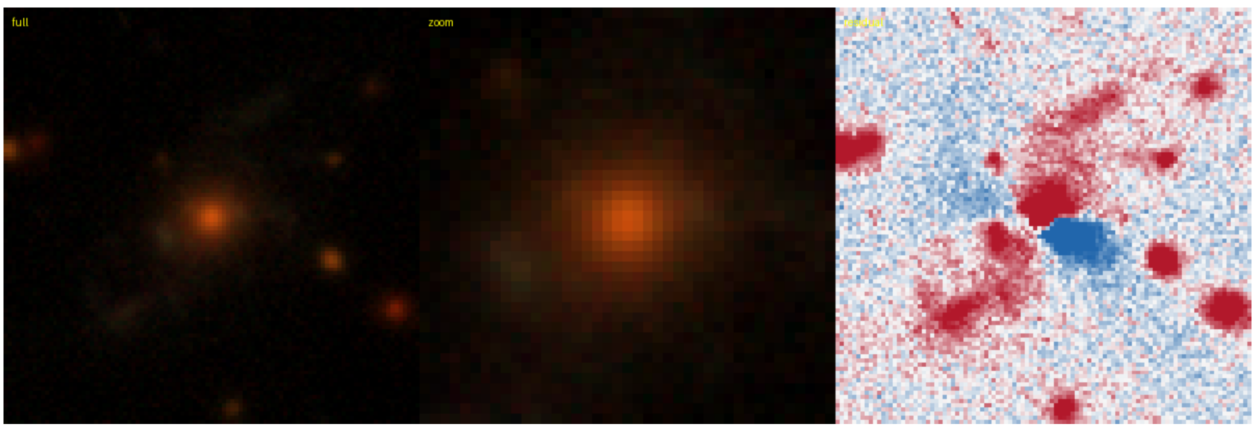



**Figure 89:** **#46** `s_327710_4950` — RA 55.6649, Dec -0.3820; grade **B**, tier-1 $p_{\text{lens}}$ 0.55, `v3blend8` 0.74. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.


**Figure 90:** **#47** `s_289350_3716` — RA 164.8051, Dec -7.1449; grade **B**, tier-1 $p_{\text{lens}}$ 0.55, `v3blend8` 0.81. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

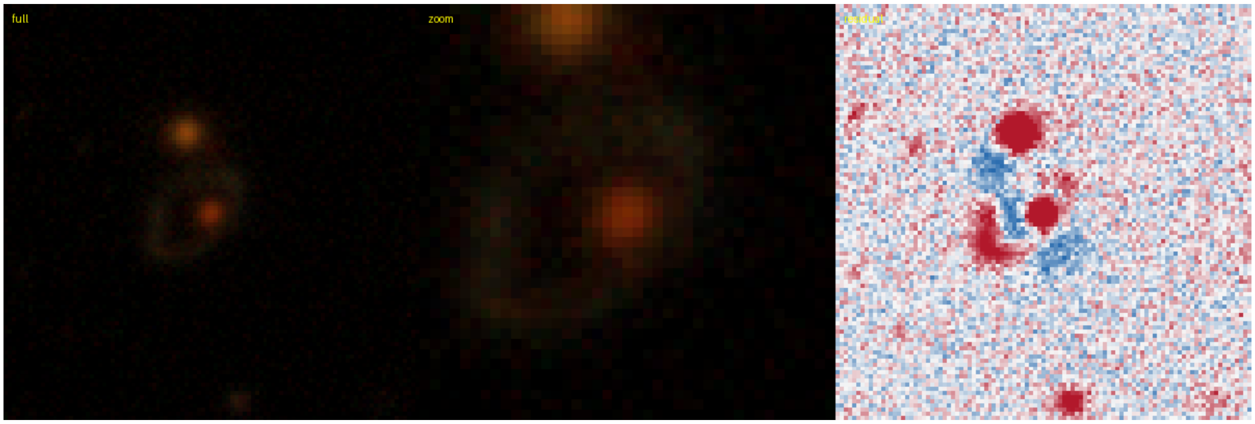



**Figure 91:** **#52** s_491286_1438 — RA 180.3616, Dec +29.0116; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.52, v3blend8 0.74. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

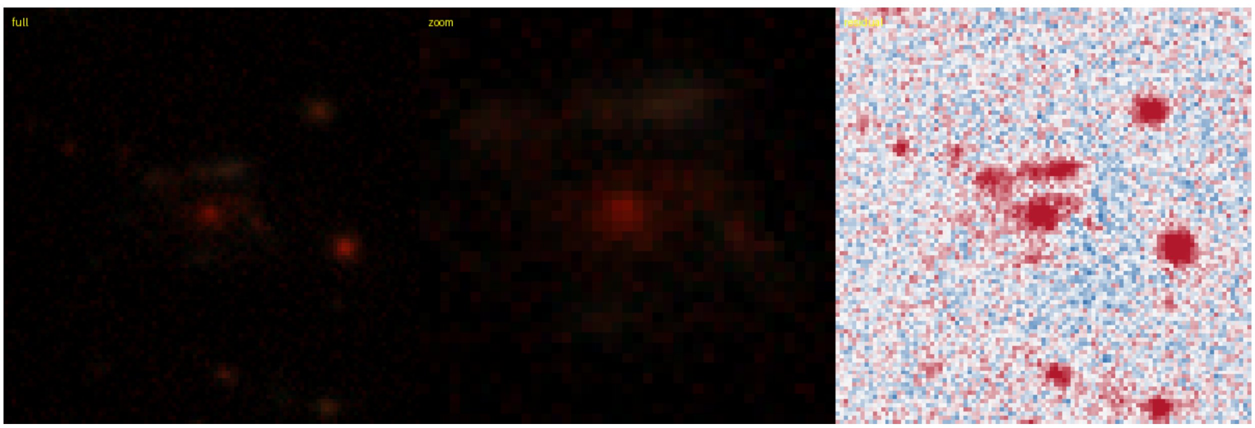



**Figure 92:** **#56** s_262449_2840 — RA 188.7861, Dec -11.9798; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.50, v3blend8 0.87. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

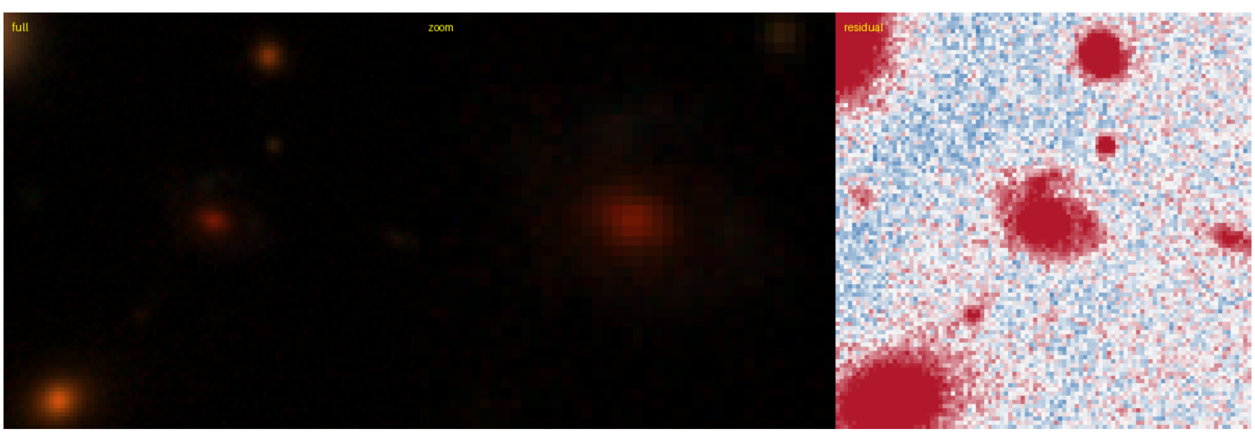



**Figure 93:** **#59** s_217461_3974 — RA 12.0451, Dec -19.9206; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.50, v3blend8 0.78. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.


**Figure 94:** **#60** `s_260341_325` — RA 9.9798, Dec -12.2097; grade **B**, tier-1 $p_{\rm lens}$ 0.50, `v3blend8` 0.84. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

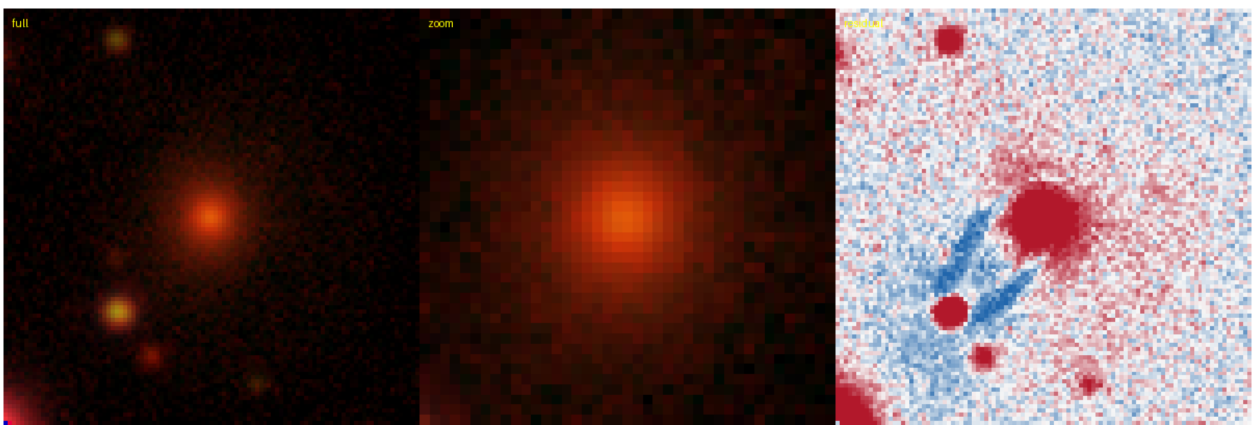



**Figure 95:** **#61** `s_207576_2272` — RA 248.1195, Dec -22.0355; grade **B**, tier-1 $p_{\rm lens}$ 0.48, `v3blend8` 0.76. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

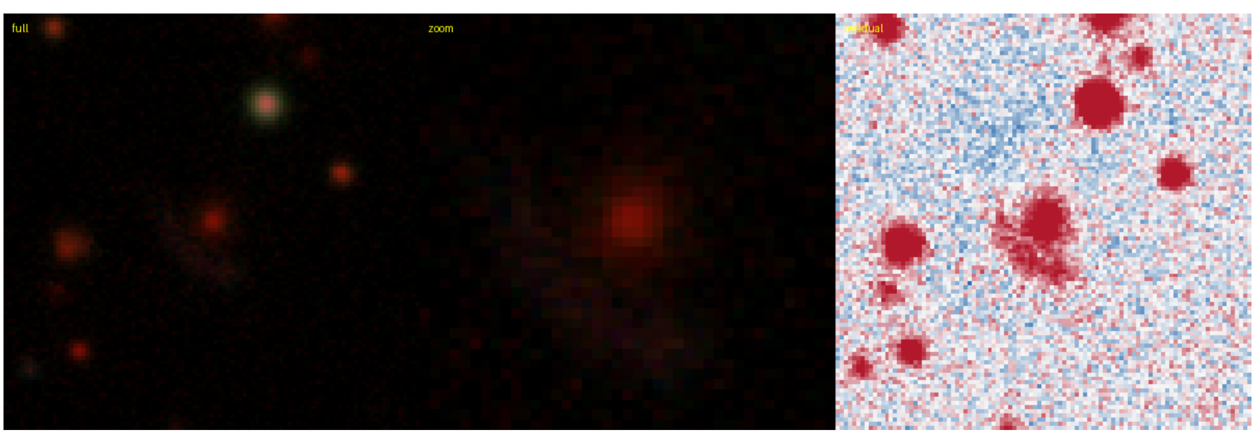



**Figure 96:** **#64** `s_283917_2408` — RA 235.6200, Dec -8.3490; grade **B**, tier-1 $p_{\rm lens}$ 0.45, `v3blend8` 0.89. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

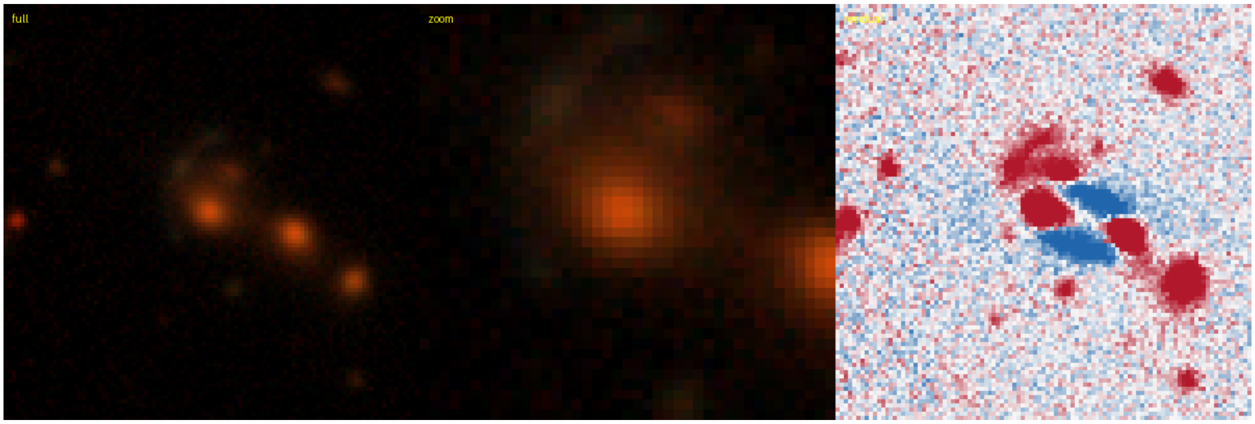



**Figure 97:** **#68** `s_148704_6732` — RA 183.8176, Dec -33.6006; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.45, `v3blend8` 0.80. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

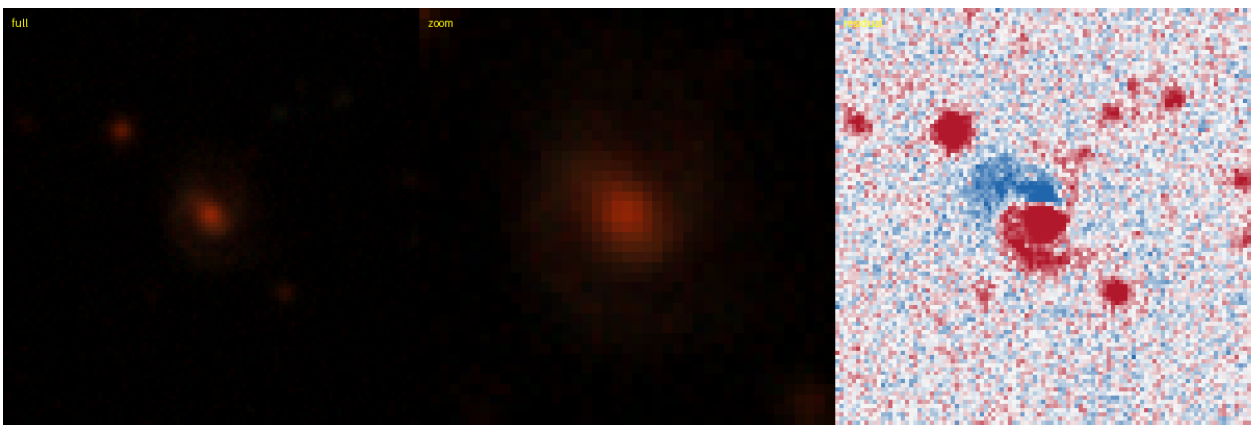



**Figure 98:** **#70** `s_266046_1185` — RA 25.9989, Dec -11.2770; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.45, `v3blend8` 0.76. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

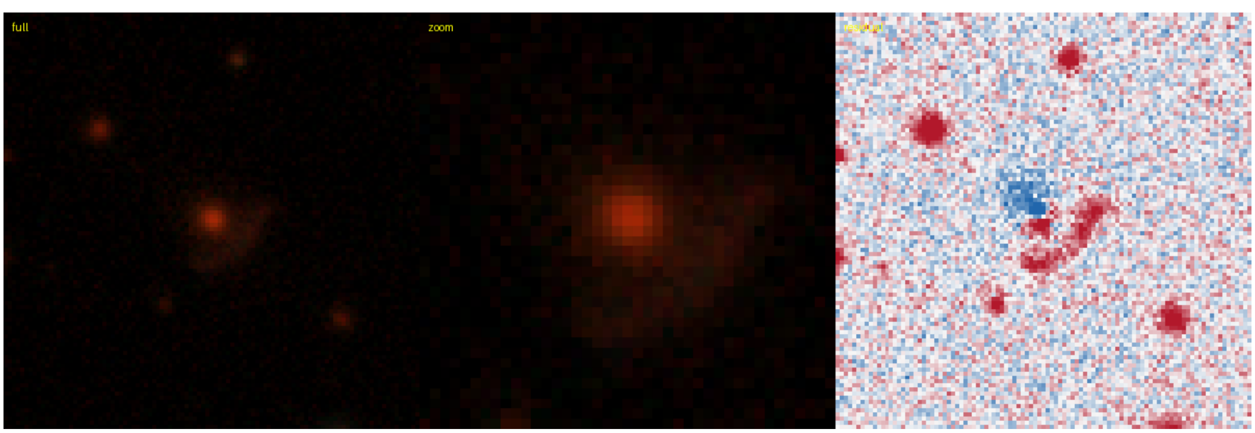



**Figure 99:** **#71** `s_251013_4202` — RA 137.4230, Dec -13.9601; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.45, `v3blend8` 0.79. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

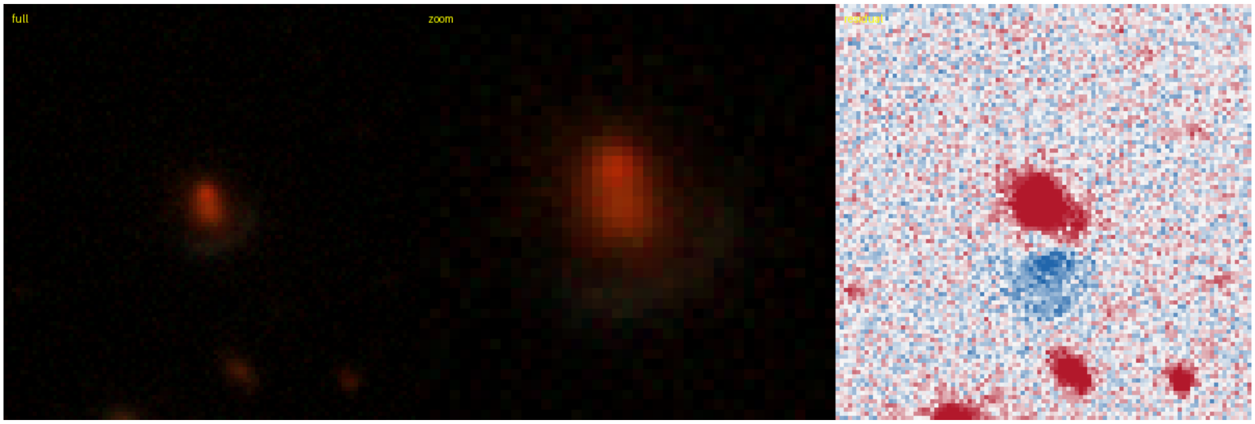



**Figure 100:** **#73** `s_209853_1103` — RA 139.8053, Dec -21.4604; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.45, `v3blend8` 0.85. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

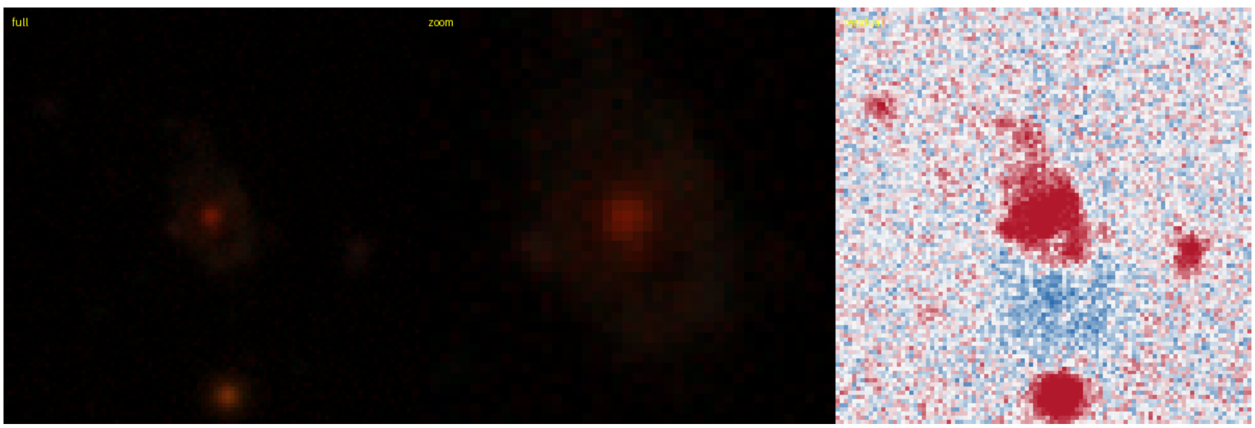



**Figure 101:** **#75** `s_181762_9592` — RA 11.4347, Dec -26.6294; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.45, `v3blend8` 0.86. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

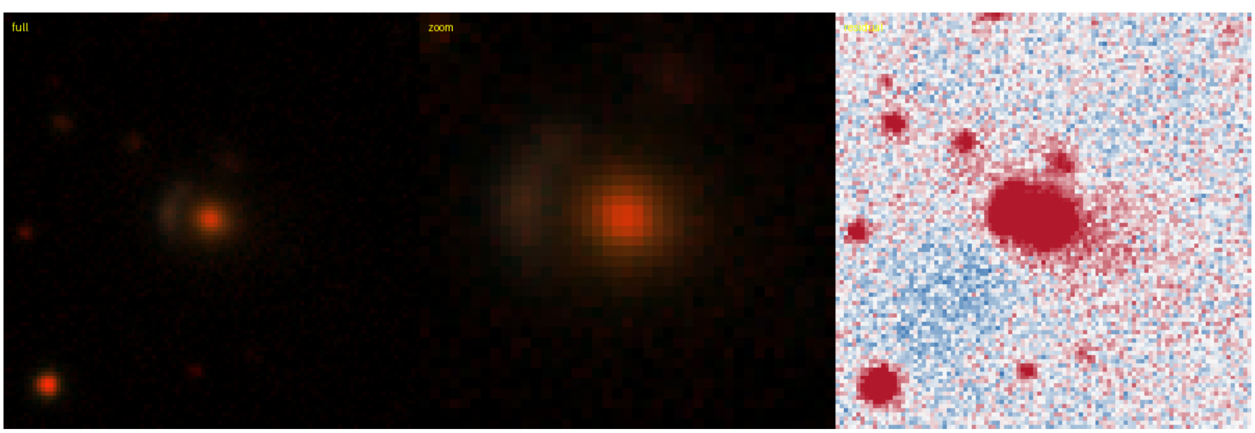



**Figure 102:** **#79** `s_23764_4161` — RA 132.6603, Dec -68.1876; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.42, `v3blend8` 0.79. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

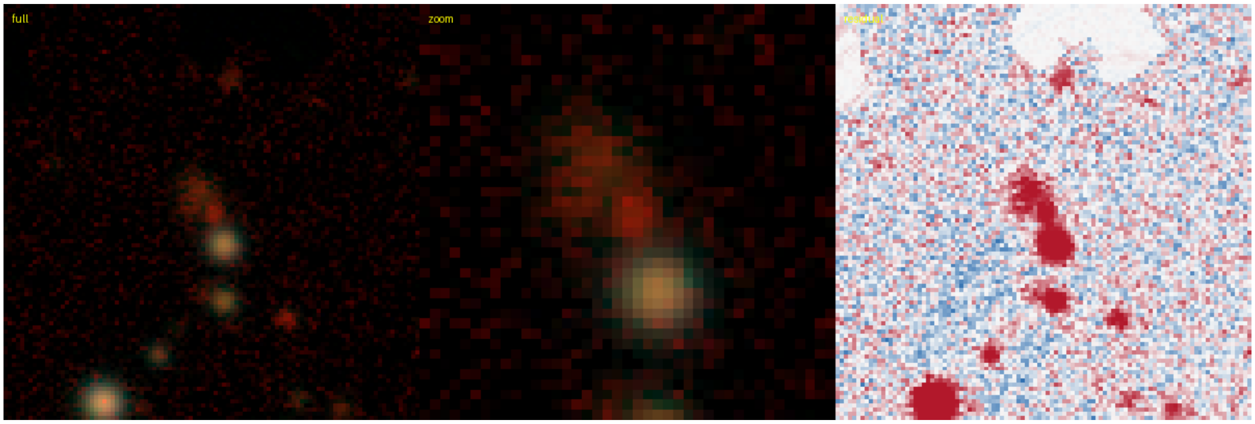



**Figure 103:** **#80** `s_29019_24056` — RA 248.9430, Dec -66.0747; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.42, `v3blend8` 0.74. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

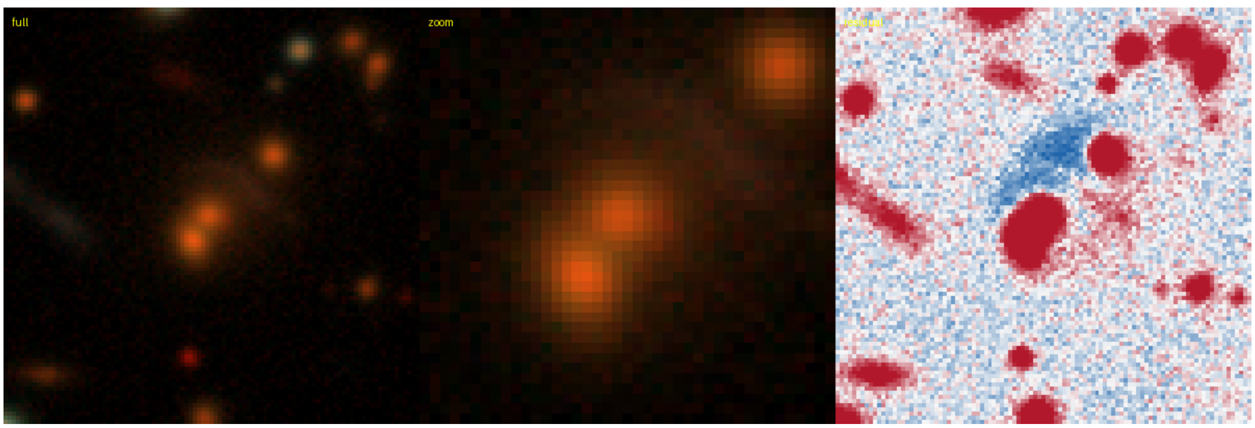



**Figure 104:** **#82** `s_112158_2867` — RA 209.6741, Dec -41.4932; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.42, `v3blend8` 0.83. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

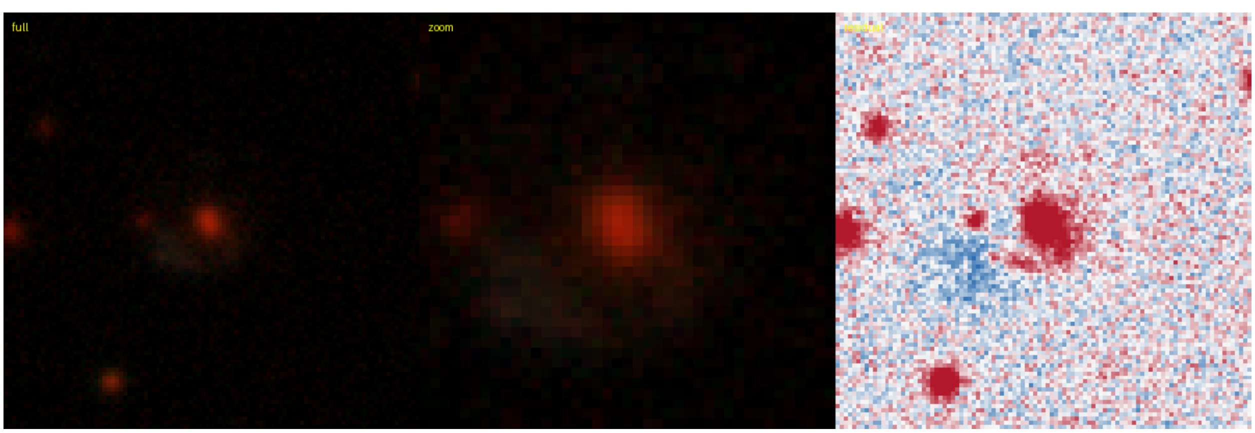



**Figure 105:** **#86** `s_405220_1292` — RA 136.4655, Dec +13.0307; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.42, `v3blend8` 0.89. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

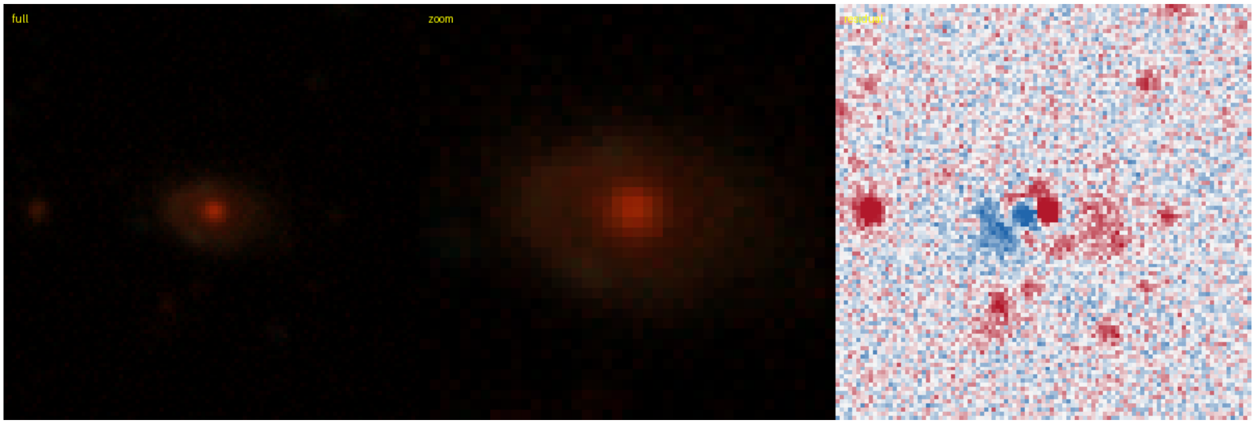



**Figure 106:** **#90** `s_188312_3322` — RA 33.2537, Dec -25.3945; grade **B**, tier-1 $p_{\rm lens}$ 0.40, `v3blend8` 0.76. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

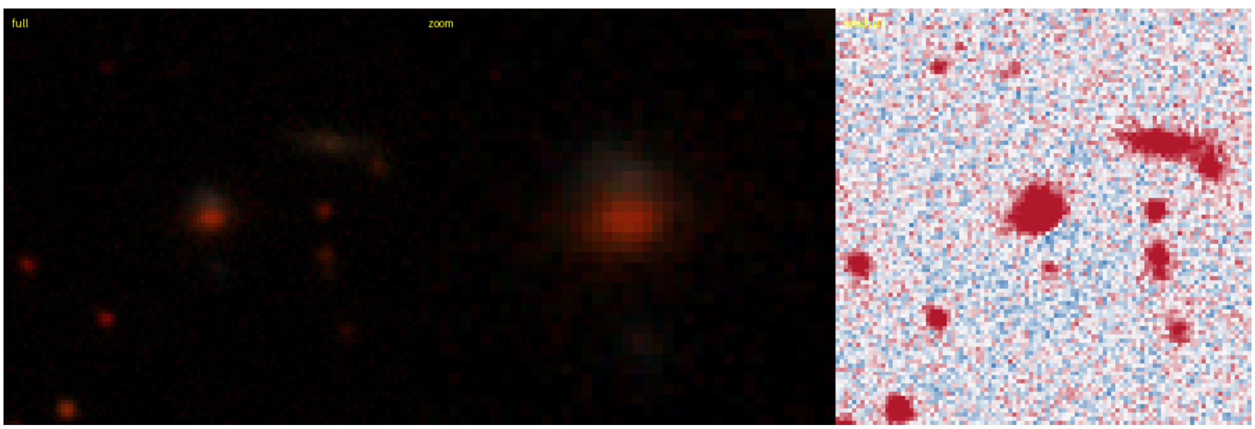



**Figure 107:** **#91** `s_120781_3317` — RA 156.8525, Dec -39.6093; grade **B**, tier-1 $p_{\rm lens}$ 0.40, `v3blend8` 0.83. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

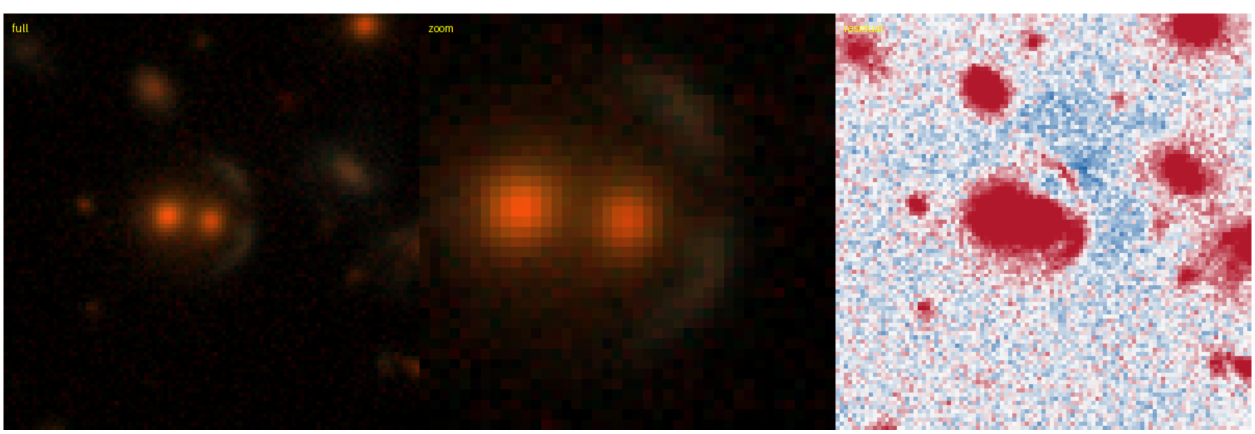



**Figure 108:** **#92** `s_266620_918` — RA 172.1320, Dec -11.3108; grade **B**, tier-1 $p_{\rm lens}$ 0.40, `v3blend8` 0.80. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

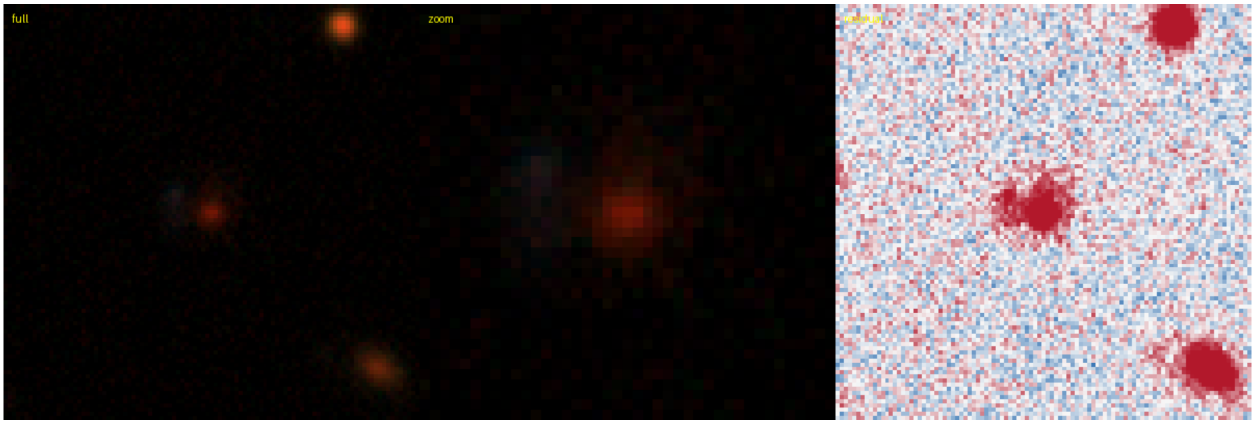



**Figure 109:** **#93** `s_410935_2097` — RA 164.1204, Dec +14.0606; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.38, `v3blend8` 0.83. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

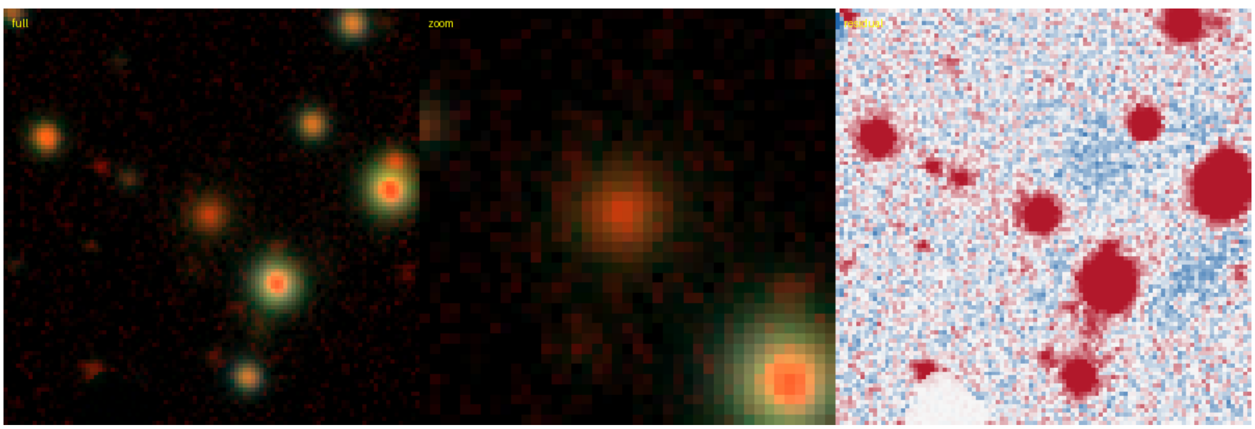



**Figure 110:** **#94** `s_32666_26056` — RA 248.6090, Dec -64.5569; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.38, `v3blend8` 0.73. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

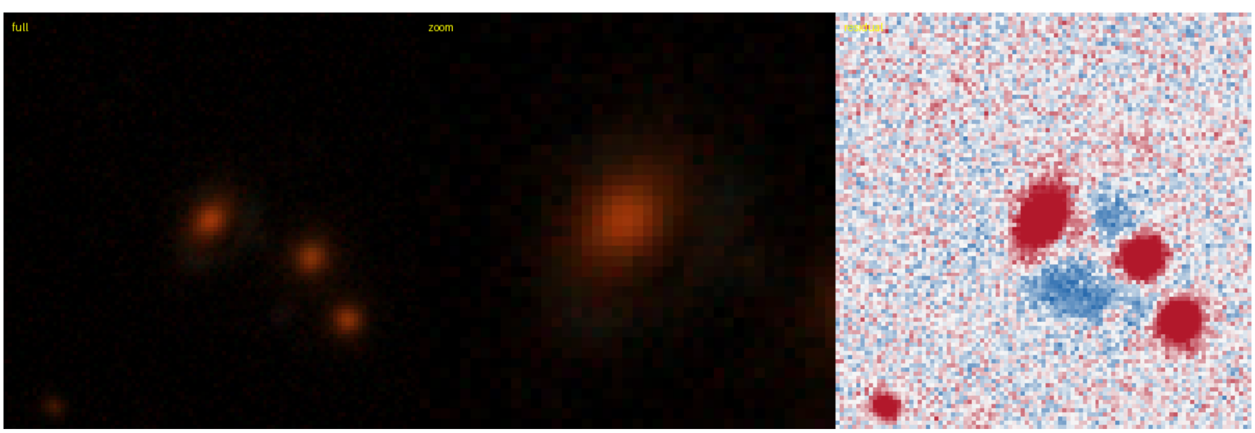



**Figure 111:** **#95** `s_244220_4253` — RA 184.3042, Dec -15.2610; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.38, `v3blend8` 0.80. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

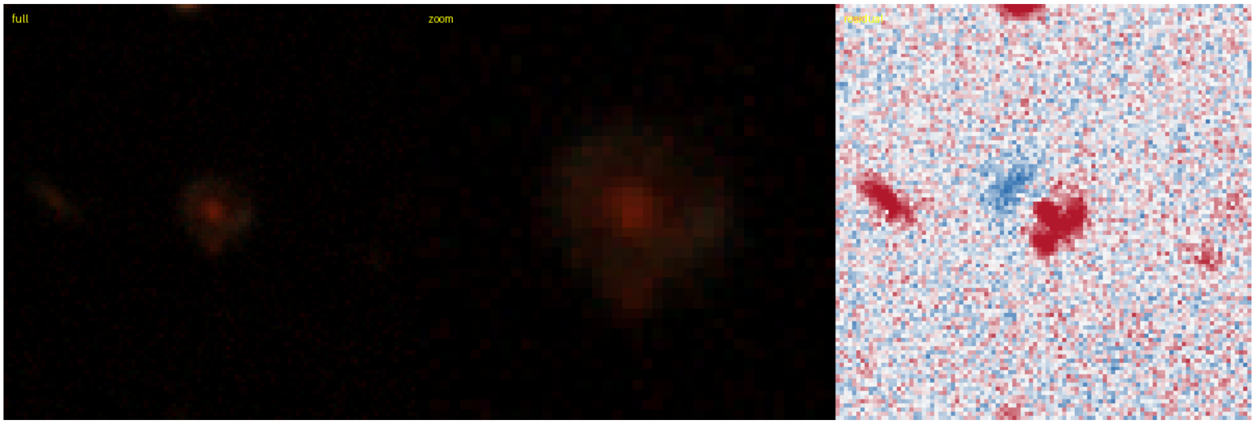



**Figure 112: #96** `s_119499_8020` — RA 101.8093, Dec -39.8359; grade **B**, tier-1 $p_{\rm lens}$ 0.38, `v3blend8` 0.75. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

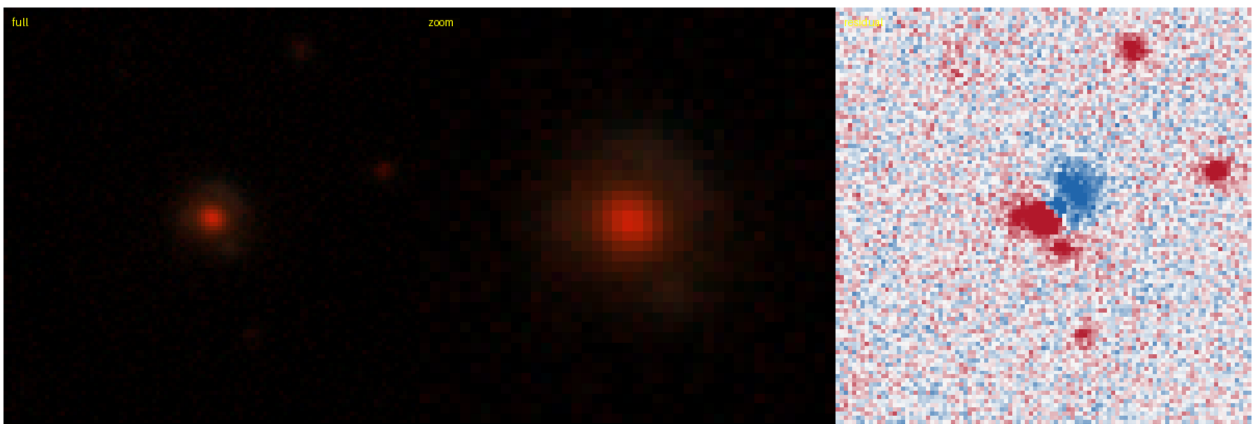



**Figure 113: #98** `s_205319_6061` — RA 359.9772, Dec -22.4596; grade **B**, tier-1 $p_{\rm lens}$ 0.38, `v3blend8` 0.88. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

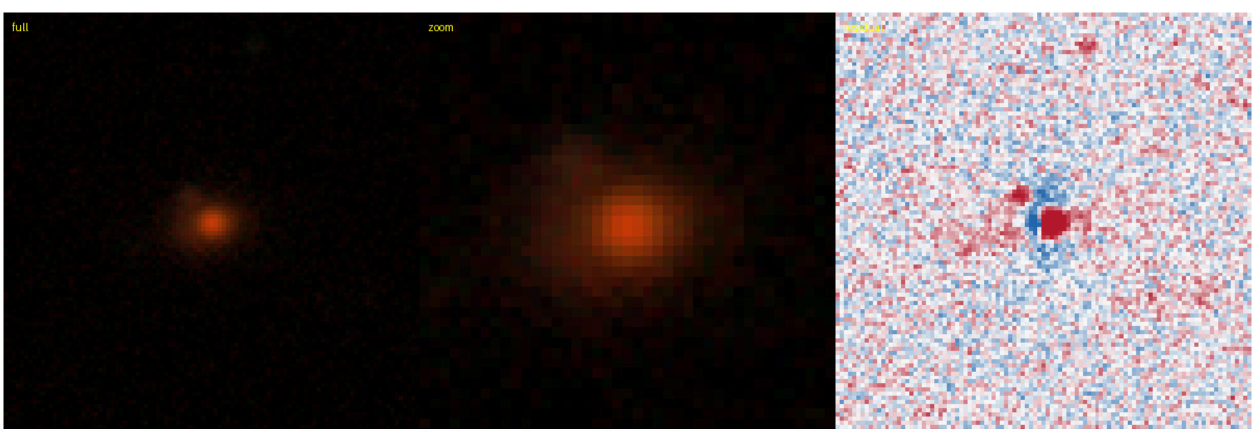



**Figure 114: #100** `s_364281_4525` — RA 207.3133, Dec +5.6308; grade **B**, tier-1 $p_{\rm lens}$ 0.35, `v3blend8` 0.90. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

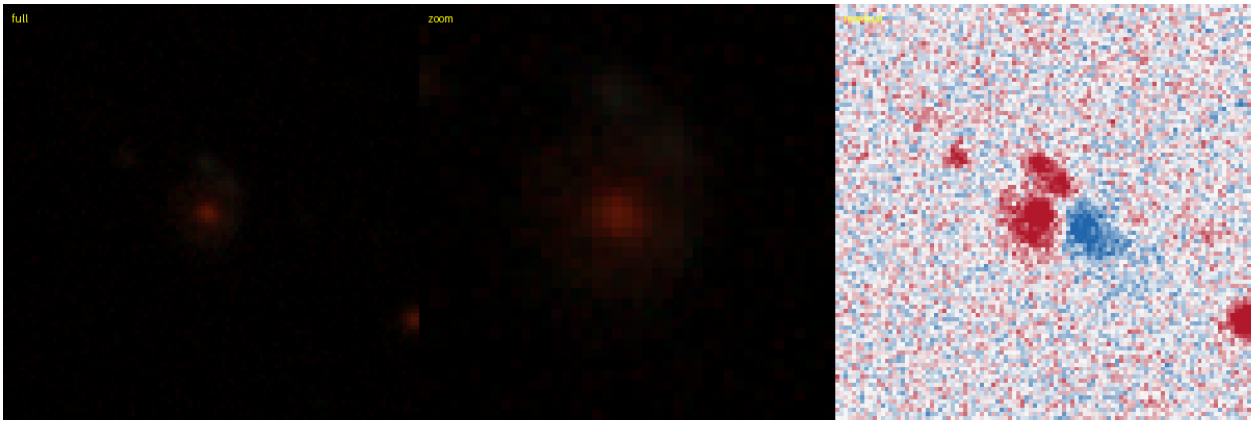



**Figure 115:** **#101** `s_250611_2161` — RA 33.7898, Dec -14.1185; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.35, `v3blend8` 0.77. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

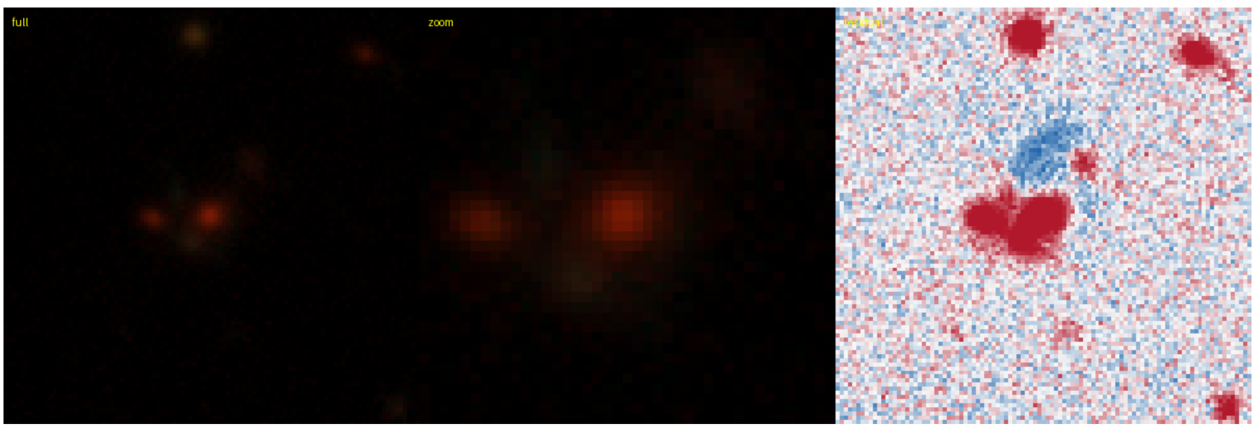



**Figure 116:** **#102** `s_212207_5081` — RA 50.6597, Dec -20.9567; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.35, `v3blend8` 0.84. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

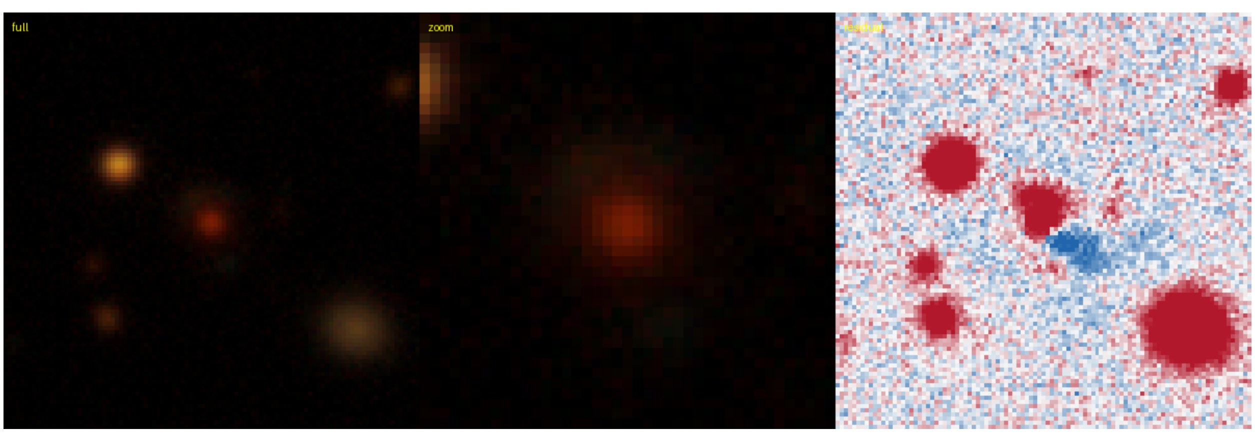



**Figure 117:** **#103** `s_128091_3374` — RA 332.6355, Dec -37.9263; grade **B**, tier-1 $p_{\mathrm{lens}}$ 0.35, `v3blend8` 0.85. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

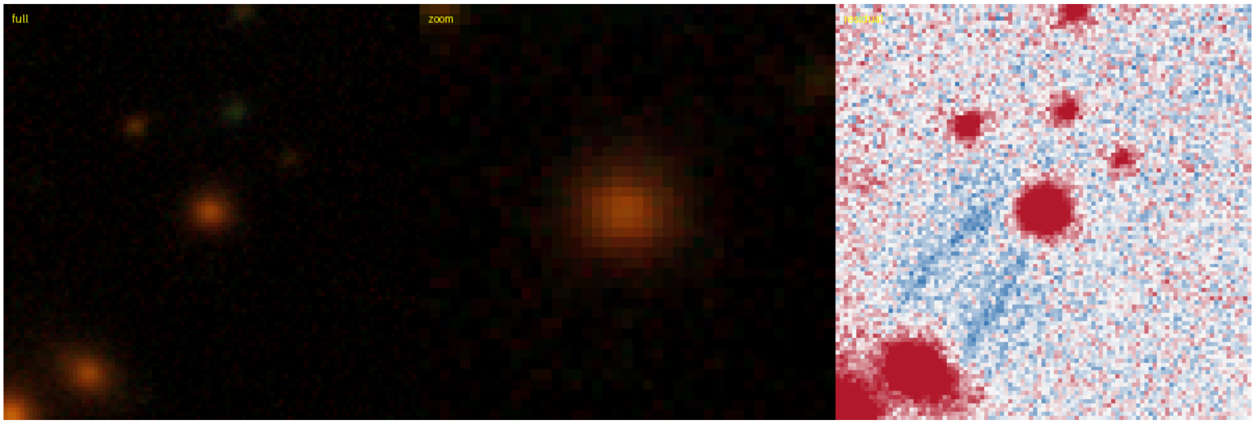


**Figure 118: #108** `s_369977_4434` — RA 198.5174, Dec +6.6887; grade **B**, tier-1 $p_{\rm lens}$ 0.28, `v3blend8` 0.75. **NEW** — new to the DESI lens-finders, the DES/AGEL/CASSOWARY searches, and the NED/SIMBAD literature.

**Table 5:** The 24 confirmed tier-2 grade-A/B systems tracked across the three residual arms (grade / $p_{\mathrm{lens}}$). **All 9/9 new-class campaign discoveries survive as A/B under the honest residual**; 13/15 SuGOHI cross-matches do likewise. $^{\downarrow}$ marks the two SuGOHI lenses the honest grader scores below A/B (one already lost to nondeterminism in the legacy-now control); both are independently-confirmed lenses, i.e. conservative grading, not campaign candidates.

| name | class | prior | legacy-now | new ($\chi$) |
|---|---|---|---|---|
| `s_341950_6798` | SuGOHI | A/0.97 | A/0.95 | A/0.96 |
| `s_340513_3688` | NEW | A/0.97 | A/0.95 | A/0.97 |
| `s_328891_7514` | SuGOHI | A/0.97 | A/0.93 | A/0.93 |
| `s_331753_3591` | SuGOHI | A/0.95 | A/0.97 | A/0.97 |
| `s_336034_1750` | SuGOHI | A/0.95 | A/0.97 | A/0.97 |
| `s_336034_1788` | SuGOHI | A/0.95 | A/0.95 | A/0.95 |
| `s_318982_1451` | SuGOHI | A/0.93 | A/0.93 | A/0.93 |
| `s_327526_9059` | NEW | A/0.92 | A/0.88 | A/0.80 |
| `s_351303_461` | NEW | A/0.92 | A/0.85 | A/0.88 |
| `s_345385_1515` | NEW | A/0.92 | A/0.90 | A/0.92 |
| `s_323786_3896` | SuGOHI | A/0.91 | D/0.04 | A/0.88 |
| `s_327403_2882` | SuGOHI | A/0.91 | A/0.90 | A/0.92 |
| `s_340971_2860` | NEW | A/0.91 | A/0.88 | A/0.85 |
| `s_345385_1514` | NEW | A/0.87 | A/0.88 | A/0.87 |
| `s_351084_1943`$^{\downarrow}$ | SuGOHI | A/0.87 | A/0.83 | C/0.22 |
| `s_332668_5796` | SuGOHI | A/0.87 | B/0.72 | A/0.83 |
| `s_326089_1785` | NEW | A/0.82 | D/0.04 | B/0.65 |
| `s_325163_2981` | NEW | A/0.82 | A/0.82 | A/0.82 |
| `s_300291_2747` | SuGOHI | A/0.82 | A/0.75 | A/0.87 |
| `s_325489_4248` | SuGOHI | B/0.68 | A/0.80 | B/0.58 |
| `s_329032_842` | SuGOHI | B/0.65 | B/0.58 | A/0.78 |
| `s_336801_4965` | SuGOHI | B/0.65 | D/0.05 | B/0.65 |
| `s_318043_1037` | NEW | B/0.45 | B/0.40 | B/0.48 |
| `s_346996_4322`$^{\downarrow}$ | SuGOHI | B/0.40 | D/0.03 | C/0.08 |

# C Residual-robustness A/B: the confirmed catalogue across residual arms

Table 5 tracks all 24 confirmed tier-2 grade-A/B systems (§3: the 9 new-class detections of Table 2 plus the 15 SuGOHI cross-matches) across the three arms of the paired residual re-run (§3, "Residual robustness"): the stored legacy vetting run, a legacy-residual rerun (the LLM-nondeterminism control), and the new honest-$\chi$ run. All 9 new-class campaign discoveries survive as A/B under the honest residual. Reproduced by `compare_residual_ab.py` from the three cascade parquets.